DANE W AI PRAKTYCZNIE

Tokens

Tokens

You can think of tokens as pieces of words used for natural language processing. For English text, 1 token is approximately 4 characters or 0.75 words.

As a point of reference, the collected works of Shakespeare are about 900,000 words or 1.2M tokens.

Understanding tokens and possibilities

Tokens:

I have an orange cat named Butterscotch.

I have an orange cat named Butterscotch.

Horses are my favorite

animal	49.65%
animals	42.58%
\n	3.49%
!	0.91%

Probabilities:

IF TEMPERATURE IS 0

Horses are my favorite animal
Horses are my favorite animal
Horses are my favorite animal
Horses are my favorite animal

IF TEMPERATURE IS 1

Horses are my favorite animal
Horses are my favorite animals
Horses are my favorite !
Horses are my favorite animal

Prompt Instruction

Suggest three names for an animal that is a superhero.

Animal: Cat
Names: Captain Sharpclaw, Agent Fluffball, The Incredible Feline
Animal: Dog
Names: Ruff the Protector, Wonder Canine, Sir Barks-a-Lot
Animal: Horse
Names:

Completion Temperature 0 (always the same)

Mighty Equine, The Great Galloper, Thunderhoof

Completion Temperature 1 (often different)

Blaze the Miracle Mare, Pegasus the Winged Warrior, Secretariat the Superhorse

Completion Temperature 1 (often different)

Blaze of Glory, Sterling Silver, Thunderbolt

[Playground - OpenAI API]

Tokens

2,762

Characters

7556

2.73

11 pięter, ponad dwa tysiące miejsc noclegowych, no i pięć gwiazdek - hotel Gołębiewski w Pobierowie miał być perłą w koronie imperium stworzonego przez zmarłego ponad rok temu Tadeusza Gołębiewskiego. Na razie to niezniszczone marzenie. Na budowie wykryto samowolę budowlaną. Sprawa trafiła do sądu. Zapadł już pierwszy wyrok.

Postawienie największego hotelu w Polsce było dla Tadeusza Gołębiewskiego dziełem życia. "Wiem, jak to

Tokens

2,542

Characters

11541

4.54

Former House Speaker Kevin McCarthy is resigning from Congress and will leave at the end of this year, he announced in a Wall Street Journal op-ed on Wednesday — a highly anticipated decision that comes two months after his unprecedented ouster from the speakership.

McCarthy’s decision will narrow the House GOP’s already historically slim majority, which just last week got smaller after the expulsion of ex-Rep. George Santos of New York.

Image input tokens

- **Image token cost** depends on detail level and image size.
- **Low detail**: allows the API to return faster responses for scenarios that don't require high image resolution analysis. The tokens consumed for low detail images are **flat rate**:
  - GPT-4o/Turbo: 85 tokens/image
  - GPT-4o mini: 2833 tokens/image
- **High detail:** image tokens are calculated based on the image's dimensions. The calculation involves the following steps:
  - Image resizing: The image is resized to fit within a 2048 x 2048 pixel square. If the shortest side is larger than 2048 pixels, the image is further resized so that the shortest side is 2048 pixels long. The aspect ratio is preserved during resizing.
  - Tile calculation: Once resized, the image is divided into 512 x 512 pixel tiles. Any partial tiles are rounded up to a full tile. The number of tiles determines the total token cost.
  - Token calculation:
    - GPT-4o and GPT-4 Turbo with Vision: Each 512 x 512 pixel tile costs 170 tokens. An extra 85 base tokens are added to the total.
    - GPT-4o mini: Each 512 x 512 pixel tile costs 5667 tokens. An extra 2833 base tokens are added to the total.

Image input tokens

- **Example: 2048 x 4096 image (high detail)**:
  - The image is initially resized to 1024 x 2048 pixels to fit within the 2048 x 2048 pixel square.
  - The image is further resized to 768 x 1536 pixels to ensure the shortest side is a maximum of 768 pixels long.
  - The image is divided into 2 x 3 tiles, each 512 x 512 pixels.
  - **Final calculation**:
    - For GPT-4o and GPT-4 Turbo with Vision, the total token cost is 6 tiles x 170 tokens per tile + 85 base tokens = 1105 tokens.
    - For GPT-4o mini, the total token cost is 6 tiles x 5667 tokens per tile + 2833 base tokens = 36835 tokens.

Image generation tokens

- GPT-image-1 generates images by first producing specialized image tokens. Both latency and eventual cost are proportional to the number of tokens required to render an image. The number of tokens generated depends on image dimensions and quality:

Quality	Square (1024×1024)	Portrait (1024×1536)	landscape (1536×1024)
Low	272 tokens	408 tokens	400 tokens
Medium	1056 tokens	1584 tokens	1568 tokens
High	4160 tokens	6240 tokens	6208 tokens

Embeddings

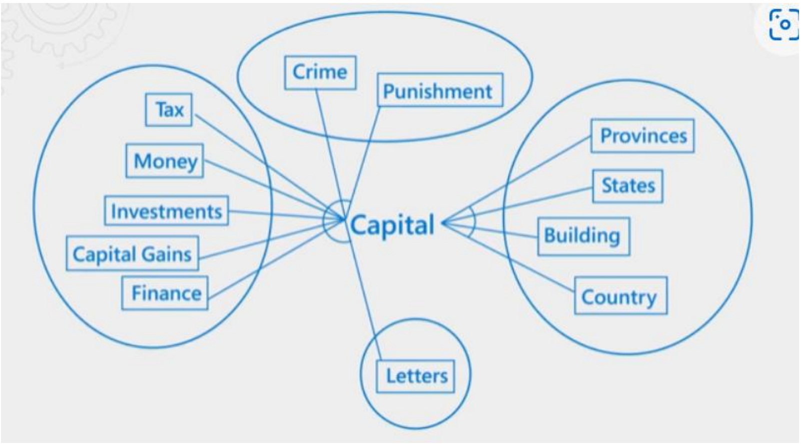
Challenges with exact term match

Query: “vehicle”

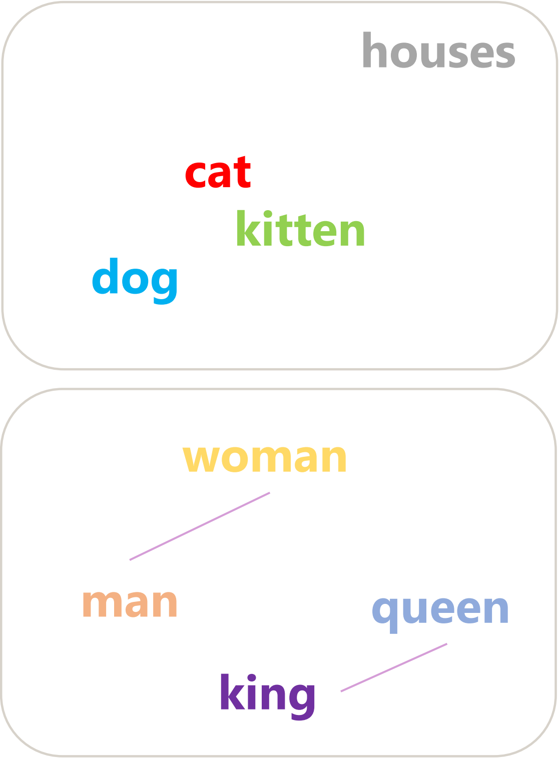
Document 1: “vehicle”
Document 2: “vehicles”
Document 3: “car”
Document 4: “sedan”

What is embedding

Word	Living Being	Feline	Human	Gender	Royalty	Verb
cat	0.6	0.9	0.1	0.4	-0.7	-0.2
kitten	0.5	0.8	-0.1	0.2	-0.6	-0.5
dog	0.7	-0.1	0.4	0.3	-0.4	-0.1
houses	-0.8	-0.4	-0.5	0.1	-0.9	0.3
man	0.6	-0.2	0.8	0.9	-0.1	0.9
woman	0.7	0.3	0.9	-0.7	0.1	-0.5
king	0.5	-0.4	0.7	0.8	0.9	-0.7
queen	0.8	-0.1	0.8	-0.9	0.8	-0.5



Dimensionality reduction of word embeddings from 7D to 2D



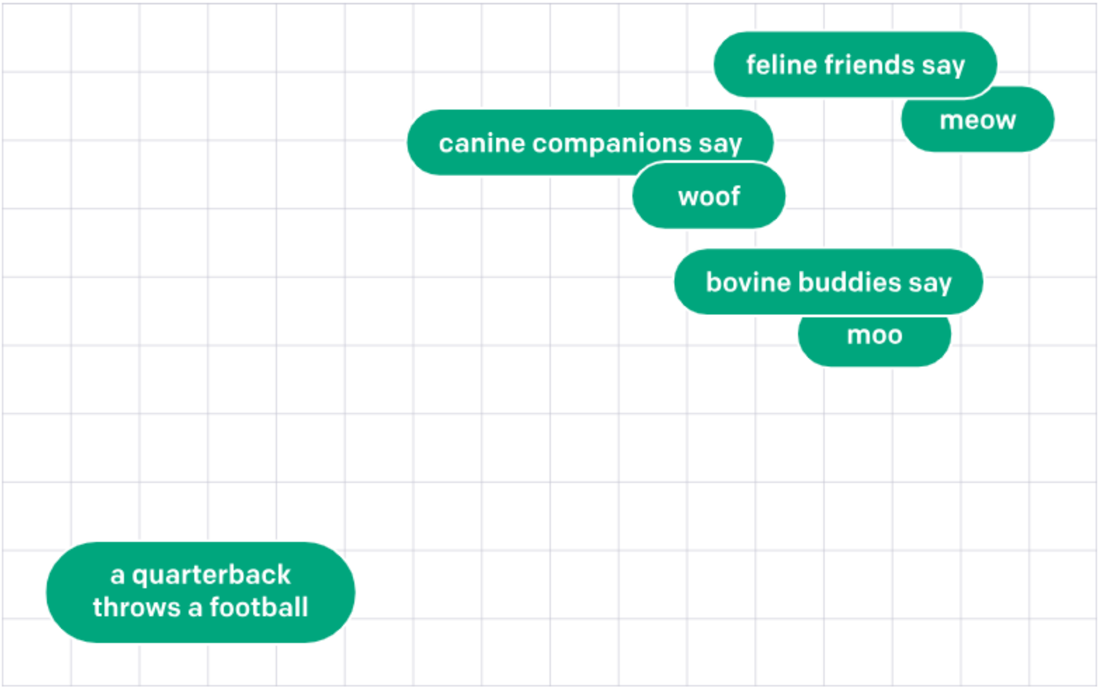
Word

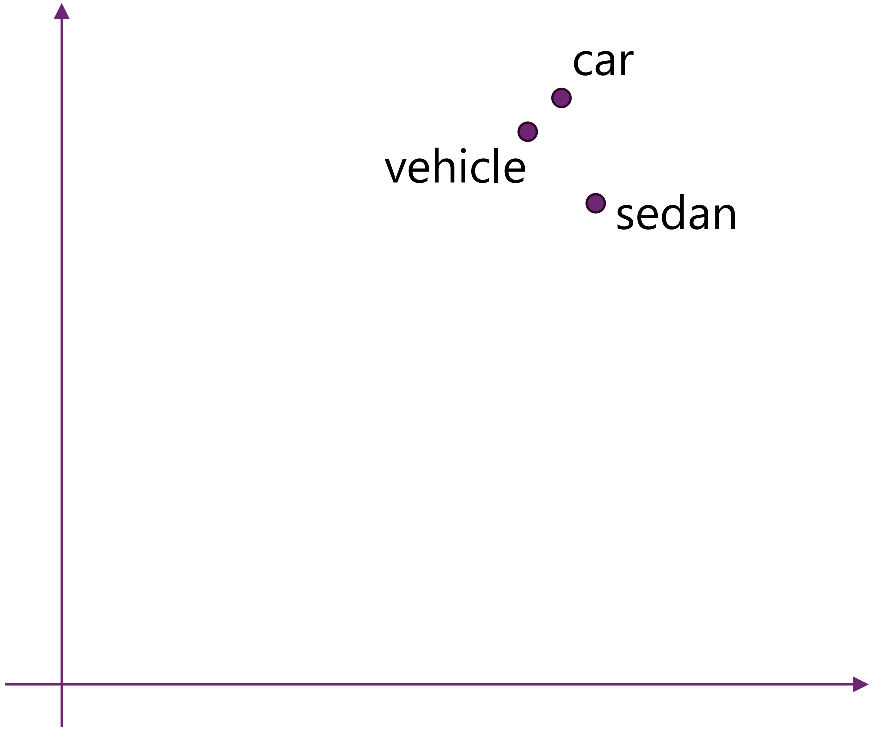
Word embedding

Dimensionality Reduction

Visualization word in 2D

Similarity search





Query: “vehicle”

Document 1: “vehicle”
Document 2: “vehicles”
Document 3: “car”
Document 4: “sedan”

Cosine similarity

Definition [edit]

The cosine of two non-zero vectors can be derived by using the Euclidean dot product formula:

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \, \|\mathbf{B}\| \cos \theta$$

Given two n-dimensional vectors of attributes, **A** and **B**, the cosine similarity, $\cos(\theta)$, is represented using a dot product and magnitude as

$$\text{cosine similarity} = S_C(A, B) := \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum\limits_{i=1}^{n} A_i B_i}{\sqrt{\sum\limits_{i=1}^{n} A_i^2} \cdot \sqrt{\sum\limits_{i=1}^{n} B_i^2}},$$

where A_i and B_i are the ith components of vectors $\mathbf{A}$ and $\mathbf{B}$, respectively.

Cosine similarity - Wikipedia

Search for: COVID | Card Results | ClinicalTrials.gov

Search Results

Card View | Table View

Viewing 1-10 out of 4,484 studies

Showing results for: **COVID**

+ Synonyms of conditions or disease (12)

None Selected

RSS

UNKNOWN STATUS

NCT04460443

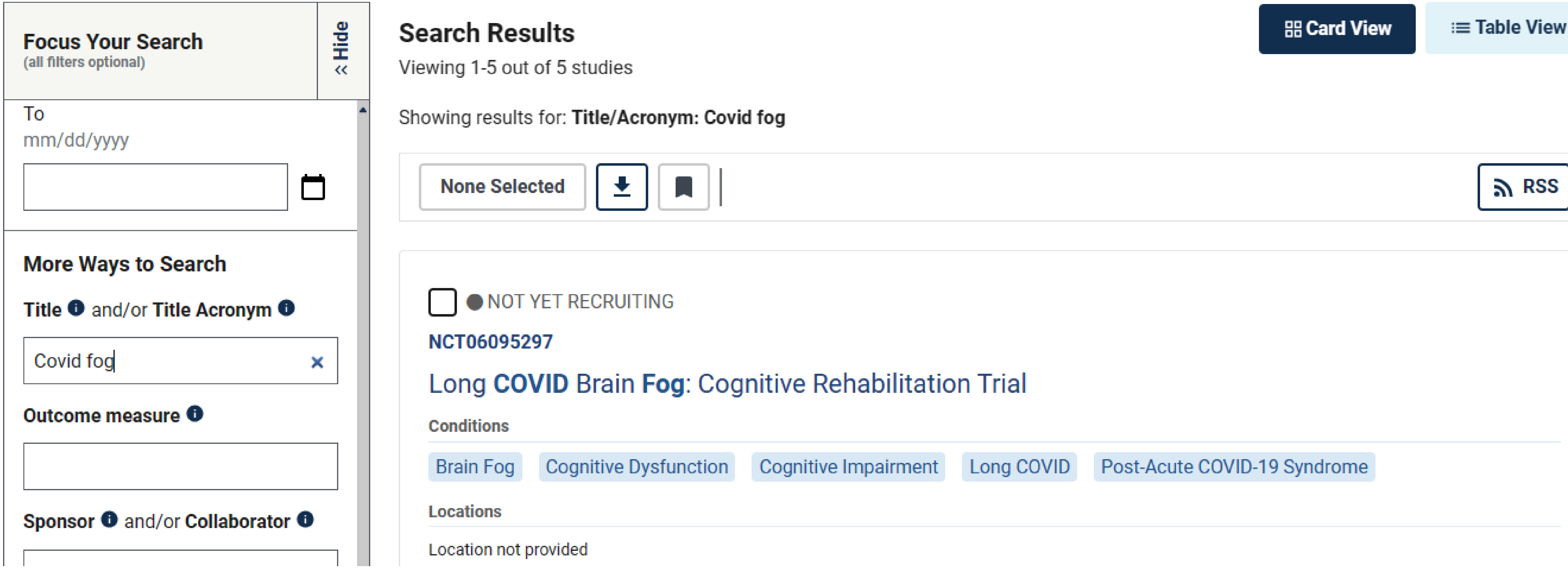
Sofosbuvir in Treatment of **COVID** 19

Conditions

COVID

Locations

COVID Fog


Focus Your Search
(all filters optional)
Hide
To
mm/dd/yyyy
More Ways to Search
Title and/or Title Acronym
Covid fog
Outcome measure
Sponsor and/or Collaborator
Search Results
Viewing 1-5 out of 5 studies
Card View
Table View
Showing results for: Title/Acronym: Covid fog
None Selected
RSS
NOT YET RECRUITING
NCT06095297
Long COVID Brain Fog: Cognitive Rehabilitation Trial
Conditions
Brain Fog
Cognitive Dysfunction
Cognitive Impairment
Long COVID
Post-Acute COVID-19 Syndrome
Locations
Location not provided

Vector search

`COVID fog`

Last 2 from top 50

nctId: NCT05758558

Title: Cognitive, Psychological, and Physical Functioning in Long-COVID Patients With Different Levels of Fatigue.

Score: 0.82114977

Content: Post COVID-19 usually occurs 3 months from the onset of COVID-19 with symptoms that last for at least 2 months and cannot be explained by an alternative diagnosis. Patients report a range of disabling symptoms such as **fatigue**, shortness of breath, cognitive impairment, memory loss and **mental health and employment issues**. This clinical heterogeneity complicates the identification of the appropriate needs and care.

The aim of this cross-sectional study is to identify subgroups (clusters) of post COVID patients based on clinical symptoms, demographic characteristics, levels of fatigue and physical, cognitive and psychological functioning of the individuals.

nctId: NCT06012552

Title: Randomized, Double-blind, Placebo-controlled Trial of the Efficacy and Safety of Tianeptine in the Treatment of Covid Fog Symptoms in Patients After COVID-19 With the Study of the Pathophysiology of the Phenomenon Using Positron Emission Tomography, Biochemical, Immunological and Electrophysiological Parameters.

Score: 0.8211213

Content: COVID-19 is associated with a high risk of complications from the central nervous system. Syndrome **of cognitive disorders- in terms of memory**, attention or executive functions among COVID-19 convalescents is often called brain fog (covid fog - CF). CF leads to psychomotor retardation and chronic fatigue syndrome, resulting in poor functioning and low quality of life. CF may affect up to 81% of patients after COVID-19.

Prevalence of CF may be even greater among patients with severe forms of COVID-19. In the preliminary assessment authors found that 83% of COVID-19 inpatients had at least mild cognitive impairment. Moreover, SARS-CoV-2 infection is associated with higher incidence of depression and anxiety disorders. CF pathogenesis is not fully understood. There exist no strict diagnostic criteria for it, as well as no therapeutic recommendations. Health care systems of many countries, including Poland, lack therapeutic programs addressed at patients with CF. Tianeptine may be a drug with potentially beneficial effects in CF. Neuroprotective, antidepressive, sleep-improving and anxiolytic properties of tianeptine allow it to choose as a candidate for CF amelioration. There is also data supporting the thesis that patients with CF may benefit from short-term group therapy. It has been proven to improve quality of life, reduce stress, and improve cognitive function in non-MC cognitive disorders.

Expected research results: A database will be created from the collected clinical, laboratory and additional data. Statistical models will be created to predict: the duration of disorders, response to therapy, the final result of treatment. Among the markers of CNS damage, those which correlates with the patient's condition will be selected.

The study will allow to estimate the prevalence of CF in the population. PET-CT and auditory evoked potentials also will be used to expand knowledge in the field of CF.

Based on the existing data, an improvement is expected in all investigated participants as a result of rehabilitation and psychotherapy.

Additional improvement is expected in the tianeptine group. Improvement will be defined as: reduction in the severity of anxiety and depression disorders, reduction in the severity of cognitive disorders, improvement in the quality of life. The results will be used to develop a new diagnostic and therapeutic pathway and a comprehensive intervention program in CF.

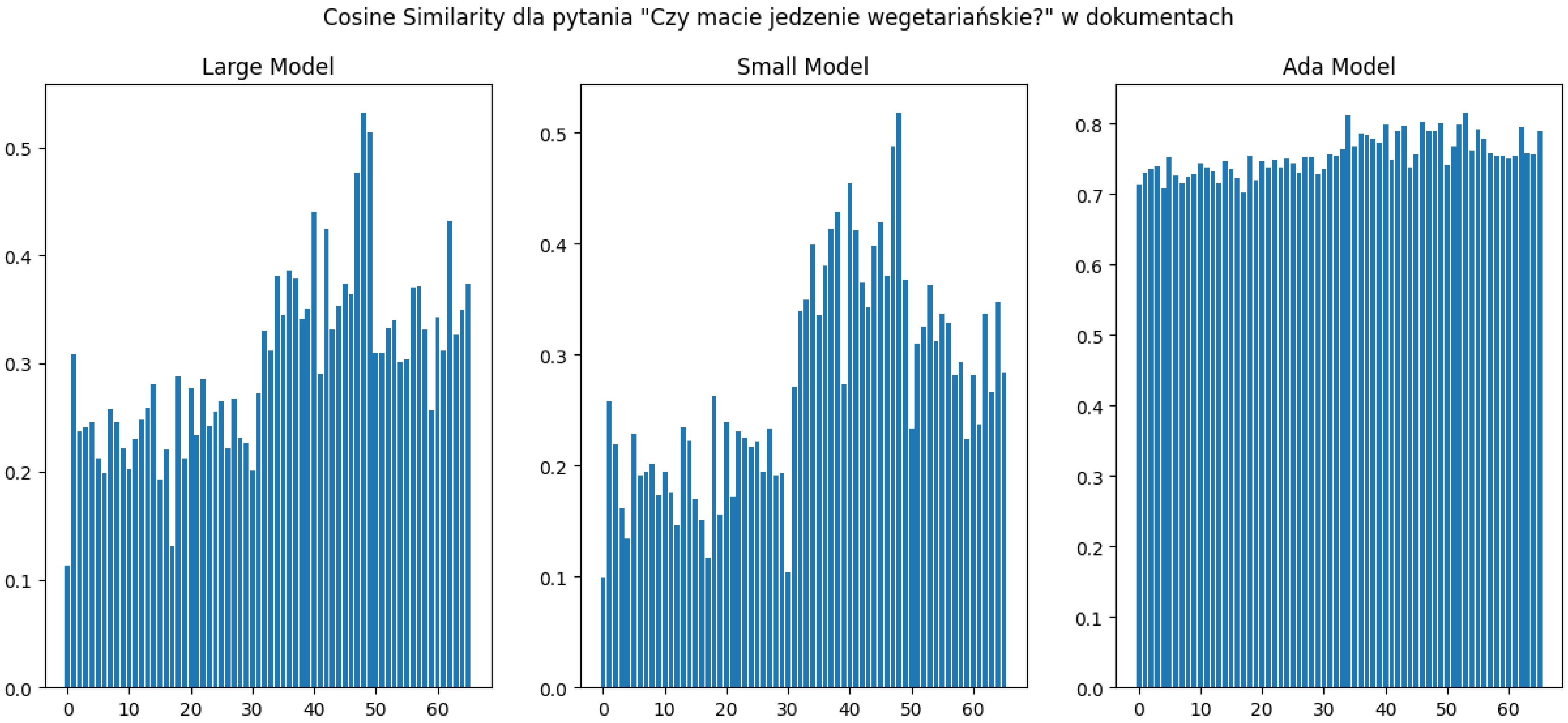
Vector search - Azure AI

Embeddings models

	ada v2	text-embedding-3-small		text-embedding-3-large		
Embedding size	1536	512	1536	256	1024	3072
Average MTEB score	61.0	61.6	62.3	62.0	64.1	64.6

[MTEB Leaderboard - a Hugging Face Space by mteb]

Embedding:


Cosine Similarity dla pytania "Czy macie jedzenie wegetariańskie?" w dokumentach
Large Model
Small Model
Ada Model
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
10
20
30
40
50
60

Word Similarity for 1000 words in English

Word	text-embedding-ada-002	text-embedding-3-small	text-embedding-3-large
time	clock, moment, hour	moment, during, now	clock, moment, period
person	people, subject, thing	people, woman, human	people, human, self
year	age, month, day	month, age, season	month, day, week
way	how, vary, far	there, thus, how	road, path, walk
day	noon, month, hour	month, week, year	month, noon, year
wood	forest, tree, yard	steel, stone, forest	forest, tree, yard
life	nature, world, live	live, born, death	live, love, fun
raft	boat, ship, rail	rail, throw, row	rail, boat, ride
hand	finger, touch, hold	finger, horse, touch	held, finger, hold
part	pair, section, party	fraction, half, segment	half, pair, section
child	children, baby, parent	children, baby, kind	children, baby, boy
chocolate	milk, sugar, food	sugar, milk, food	milk, sugar, brown
car	road, engine, truck	care, door, carry	card, truck, care
book	paper, sheet, page	record, board, dictionary	read, dictionary, write
house	home, room, office	home, property, room	home, room, property
water	rain, liquid, wash	liquid, rain, wash	liquid, air, energy
food	eat, feed, bread	eat, bread, feed	eat, fruit, feed
money	capital, cost, pay	pay, bank, cost	dollar, pay, energy
music	song, instrument, art	song, sound, art	radio, dance, song
love	joy, life, live	friend, wish, our	like, life, live

Word Similarity for 1000 words in Polish

Słowo	text-embedding-ada-002	text-embedding-3-small	text-embedding-3-large
czas	moment, czy, tam	moment, aż, ty	czy, chwila, moment
osoba	pracownik, ludzie, człowiek	ktoś, ludzie, ludność	ktoś, sobą, człowiek
rok	raz, lek, jak	raz, gra, ruch	skała, rola, bo
droga	drogi, rola, zgoda	drogi, substancja, rzecz	drogi, metoda, rada
dzień	tydzień, miesiąc, dziś	tydzień, dzisiaj, niedziela	dziś, święto, miesiąc
drewno	drewniany, ziarno, drzewo	drewniany, drzewo, ziarno	drewniany, drzewo, materiał
życie	żyć, działalność, śmierć	żyć, żywy, szczęście	żyć, miłość, żywy
tratwa	trawa, potrawa, walka	trawa, trzeba, trudny	statek, rzeka, podróż
ręka	palec, ramię, ząb	woda, rzeka, nóż	ramię, prawy, broń
część	połowa, kawałek, większość	połowa, dział, całość	dział, kawałek, partia
dziecko	chłopiec, chłopak, dziewczyna	chłopiec, chłopak, dziewczyna	chłopiec, dziewczyna, chłopak
czekolada	mleko, cukier, cecha	ciasto, chleb, jabłko	cukier, ciasto, słodki
samochód	pojazd, maszyna, obiekt	pojazd, silnik, autobus	pojazd, samolot, silnik
książka	księga, język, tytuł	księga, słownik, ciasto	księga, tytuł, artykuł
dom	pod, od, sam	od, do, domowy	do, sto, pod
woda	wodny, drogi, piwo	węgiel, atomowa, płyn	wodny, płyn, rzeka
jedzenie	jeść, kuchnia, leczenie	jeść, potrawa, chleb	jeść, obiad, mięso
pieniądze	pieniądz, długi, cienki	pieniądz, długi, dodatek	pieniądz, drobny, złoty
muzyka	muzyczny, sztuka, taniec	muzyczny, język, dźwięk	muzyczny, sztuka, utwór
miłość	miły, imię, miękki	kochać, miły, lubić	kochać, szczęście, życie

Azure OpenAI API

Applications submit prompts to deployed models. Responses are completions.

REST API endpoints:

- **Embeddings** - model takes input and returns a vector representation of that input
- **ChatCompletion** - model takes input in the form of a chat conversation (where roles are specified with the message they send), and the next chat completion is generated

Using the Azure OpenAI REST API

Embedding Endpoint

https://*endpoint*.openai.azure.com/openai/deployments/*deployment*/**embeddings**

```
{
  "input": "The food was delicious and
            the waiter was very
            friendly..."
}
```

```
{
  "object": "list",
  "data": [
    {
      "object": "embedding",
      "embedding": [
        0.0172990688066482523,
        ....
        0.0134544348834753042,
      ],
      "index": 0
    }
  ],
  "model": "text-embedding-ada:002"
}
```

Using the Azure OpenAI REST API

ChatCompletion Endpoint

https://*endpoint*.openai.azure.com/openai/deployments/*deployment*/**chat/completions**


```
{
  "messages":[
    {"role": "system",
     "content": "You are an assistant
      that teaches people about AI."},
    {"role": "user",
     "content": "Does Azure OpenAI
          support multiple languages?"},
    {"role": "assistant",
     "content": "Yes, Azure OpenAI
          supports several languages."},
    {"role": "user",
     "content": "Do other Cognitive
         Services support translation?"}
  ]
}
```



```
{
    "id": “unique_id", "object": "chat.completion",
    "created": 1679001781, "model": "gpt-35-turbo",
    "usage": { "prompt_tokens": 95,
               "completion_tokens": 84, "total_tokens": 179},
    "choices": [
          {"message":
                {  "role": "assistant",
                    "content": "Yes, other Azure Cognitive
                        Services also support translation…"},
            "finish_reason": "stop",
            "index": 0}
    ]
}
```

Using Azure OpenAI SDKs – Python

```
import openai

# Configure service settings
openai.api_type = "azure"
openai.api_base = your_endpoint
openai.api_version = "20XX-XX-XX“ # target version
openai.api_key = your_key

# Send request to Azure OpenAI model
completion = openai.ChatCompletion.create(
    engine=your_model_deployment,
    messages=[
        {"role": "system", "content": "You are a helpful AI bot."},
        {"role": "user", "content": “What is Azure OpenAI?"},
    ]
)
print(completion['choices'][0]['message']['content'])
```

OpenAI vs Azure SDK

Sample Code

You can use the following code to start integrating your current prompt and settings into your application.

https://aif-agents-sec.openai.azure.com/openai/v1/chat/completions | python | OpenAI SDK

Key authentication

```python
from openai import OpenAI

endpoint = "https://aif-agents-sec.openai.azure.com/openai/v1/"
deployment_name = "gpt-4.1"
api_key = "<your-api-key>"

client = OpenAI(
    base_url=endpoint,
    api_key=api_key
)

completion = client.chat.completions.create(
    model=deployment_name,
    messages=[
        {
            "role": "user",
            "content": "What is the capital of France?",
        }
    ],
)

print(completion.choices[0].message)

```

https://aif-agents-sec.openai.azure.com/openai/deployments/gpt-4.1/chat/completions?api-version=2025-01-01-preview | python | Azure OpenAI SDK

Entra ID authentication | Key authentication

Open in VS Code

```python
import os
import base64
from openai import AzureOpenAI
from azure.identity import DefaultAzureCredential, get_bearer_token_provider

endpoint = os.getenv("ENDPOINT_URL", "https://aif-agents-sec.openai.azure.com/")
deployment = os.getenv("DEPLOYMENT_NAME", "gpt-4.1")

# Initialize Azure OpenAI client with Entra ID authentication
token_provider = get_bearer_token_provider(
    DefaultAzureCredential(),
    "https://cognitiveservices.azure.com/.default"
)

client = AzureOpenAI(
    azure_endpoint=endpoint,
    azure_ad_token_provider=token_provider,
    api_version="2025-01-01-preview",
)


# IMAGE_PATH = "YOUR_IMAGE_PATH"
# encoded_image = base64.b64encode(open(IMAGE_PATH, 'rb').read()).decode('ascii')
chat_prompt = [
    {
        "role": "system",
        "content": [
            {
                "type": "text",
                "text": "You are an AI assistant that helps people find information."
            }
        ]
    }
]

# Include speech result if speech is enabled
messages = chat_prompt

completion = client.chat.completions.create(
    model=deployment,
    messages=messages,
    max_tokens=6553,
    temperature=0.7,
    top_p=0.95,
    frequency_penalty=0,
    presence_penalty=0,
    stop=None,
    stream=False
)

print(completion.to_json())

```

Building AI Apps

Introduction

- Streamlit is a free and open-source framework to rapidly build and share beautiful machine learning and data science web apps.
- No front-end (html, js, css) experience or knowledge is required.
- You don't have to spend days or months developing a web app; instead, you can construct a stunning machine learning or data science app in a matter of hours or even minutes.
- It is compatible with many Python libraries, including pandas, matplotlib, seaborn, plotly, Keras, PyTorch, and SymPy (latex).
- Less code is needed to create amazing web apps.

Benefits

Python-scripting: Streamlit embraces Python, making it easy for developers to create web apps with less code.

Real-time visualization: Streamlit updates dashboards instantly as data changes, useful for monitoring live data feeds or tracking KPIs.

Interactive dashboards: Streamlit allows for interactive web apps, not just static images.

Ease of use: Streamlit is designed to be user-friendly.

Page Elements

Text Elements

Title: Display text in title formatting.

Header: Display text in header formatting.

Caption: Display text in small font.

Divider: Display a horizontal rule.

Latex: Display mathematical expressions formatted as LaTeX.

Text: Write text without Markdown or HTML parsing.

Html: Insert HTML into your app.

Data Elements

Data frame: Display a data frame as an interactive table.

Data editor: Allows you to edit data frames and many other data structures in a table-like UI.

Table: Display a static table.

Metric: Display a metric in big bold font, with an optional indicator of how the metric changed.

JSON: Display an object or string as a pretty-printed, interactive JSON string.

Chart

Chart Elements

- area_chart: Display an area chart.
- bar_chart: Display an bar chart.
- line_chart: Display text in small font.
- map: Display a map with a scatterplot overlaid onto it.

Input

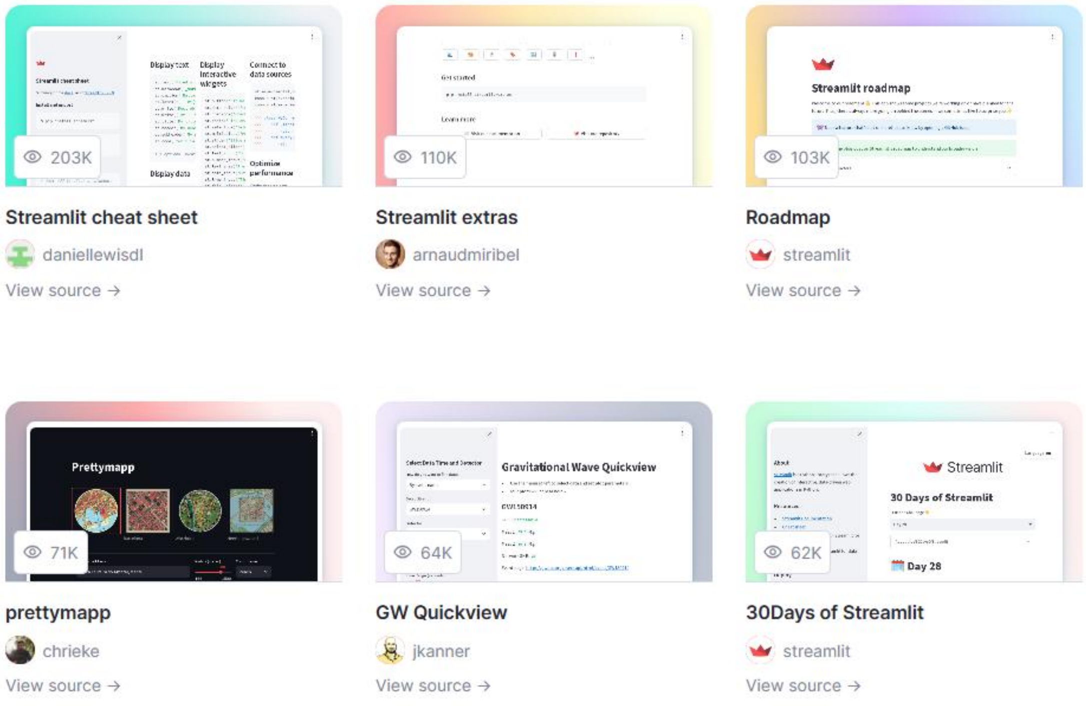
Input Elements

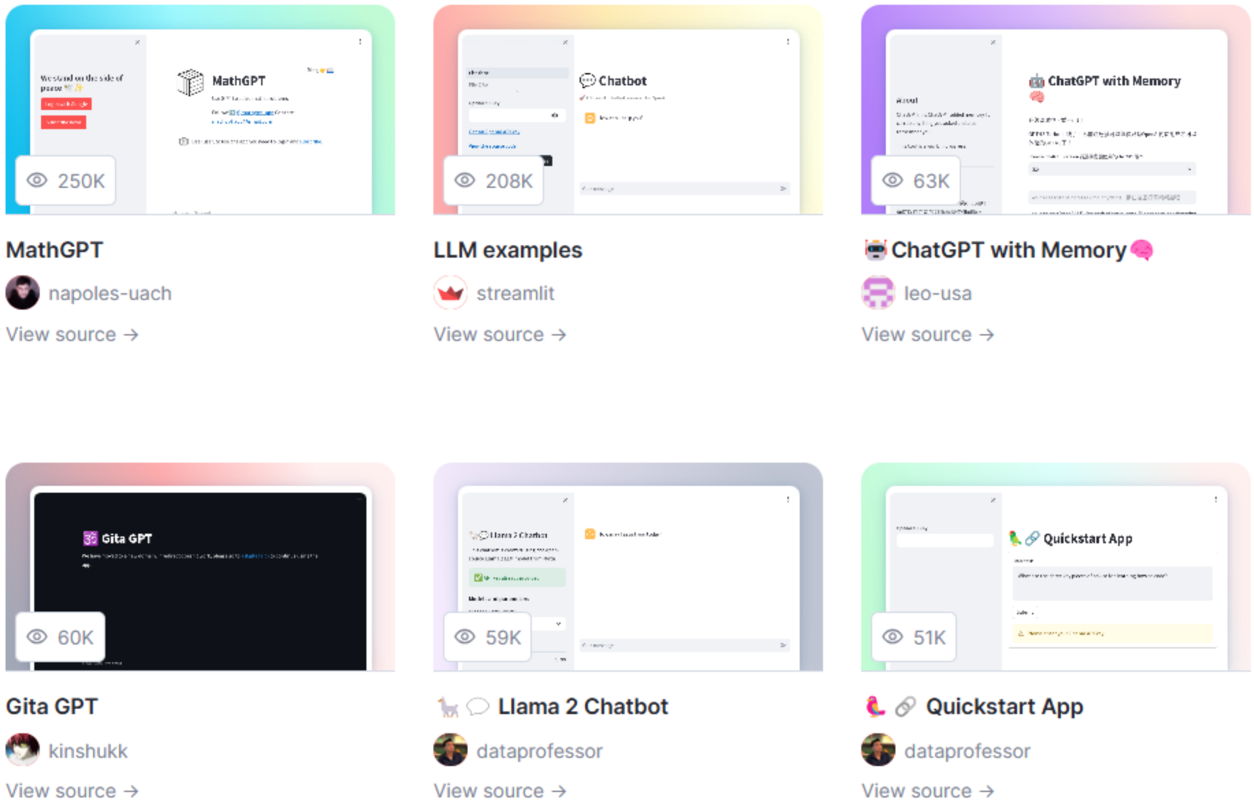
- Button: Display a button widget.
- checkbox: Display a checkbox widget.
- color_picker: Display a color picker widget.
- Radio: Display a radio button widget.
- Selectbox: Drop down widget.
- select_slider: Display a slider widget to select items from a list.
- number_input: Display a numeric input widget.
- Date and time Input: Display a date and time input widget.
- camera_input: Display a widget that returns pictures from the user's webcam.
- file_uploader: Display a file uploader widget.

Page Elements

Examples Gallery

[App Gallery • Streamlit]()





Installation

Streamlit documentation

- Python 3.7 – Python 3.12
- Using a virtual environment is always recommended (pipenv, poetry, venv…)
- Install Streamlit

  pip install streamlit
- Test the installation

  streamlit hello
- Launch your own application

  streamlit run your_script.py [-- script args]

  Or

  python -m streamlit run your_script.py

JSON response

Retrieval Augmented Generation

W jaki sposób poszukujesz informacji?

Wprowadzanie wiedzy domenowej do LLM

Prompt engineering

Uczenie się w kontekście

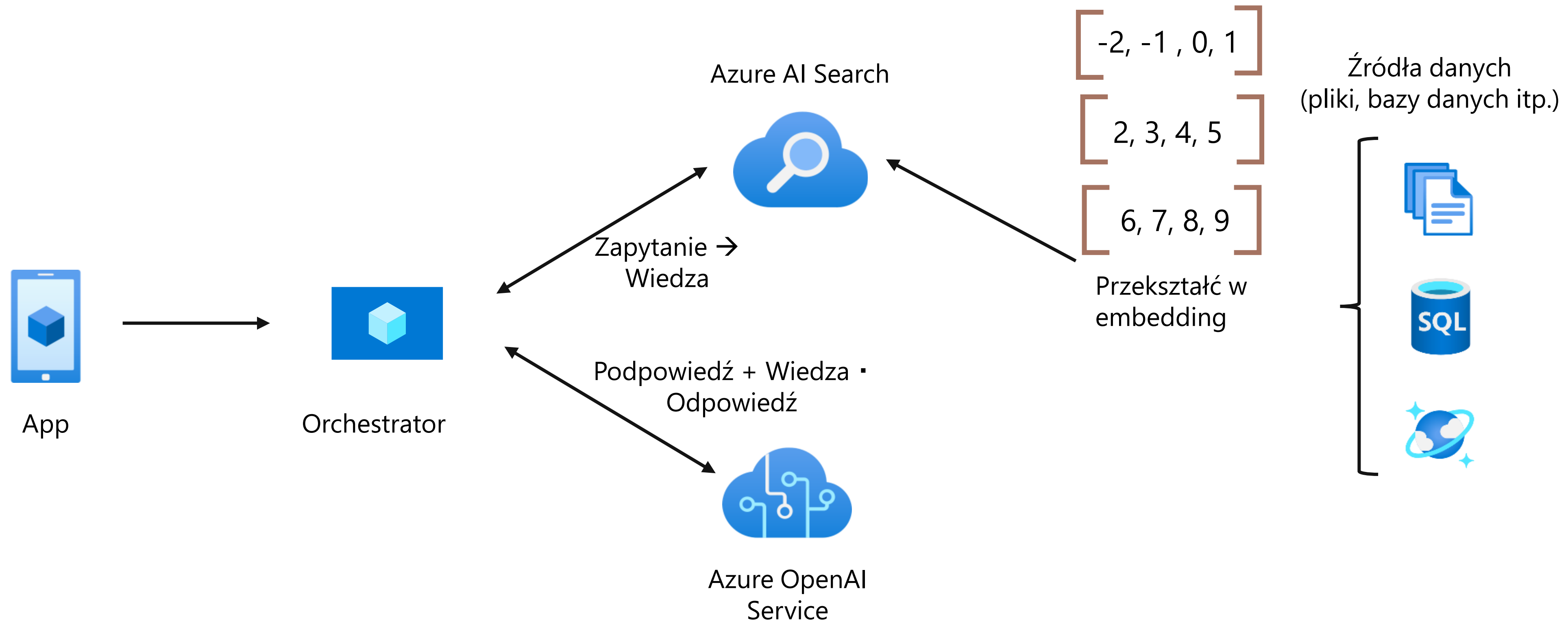
Fine tuning

Zdobądź nowe umiejętności

Retrieval augmentation

Poznaj nowe fakty

Retrieval Augmented Generation



Przykład Retrieval Augmented Generation: przenieś dane do promptu

Wprowadzanie, które zapewnia ramy dotyczące tego, jak model powinien się zachowywać

Prompt

Jesteś inteligentnym asystentem, który pomaga pracownikom firmy Contoso, Inc. w przypadku pytań dotyczących planu opieki zdrowotnej, a także podręcznika dla pracowników. Odpowiedz na poniższe pytanie, korzystając wyłącznie z danych podanych w poniższych źródłach.

Pytanie: Czy mój plan zdrowotny obejmuje coroczne badania wzroku?

Pytanie zadane przez użytkownika, na które należy odpowiedzieć

Źródła:

1. Northwind Health Plus oferuje pokrycie kosztów badań wzroku, okularów i soczewek kontaktowych, a także badań dentystycznych, czyszczenia i wypełnień.
2. Northwind Standard oferuje pokrycie tylko w przypadku badań wzroku i okularów.
3. Oba plany oferują pokrycie kosztów usług okulistycznych i dentystycznych.

Źródła, z których udzielono odpowiedzi na pytanie

```
Odpowiedź

Na podstawie podanych informacji,
jeśli Twój plan jest z Northwind
Standard, obejmuje badania wzroku.
```

Ulepszanie RAG dzięki zaawansowanym funkcjom wyszukiwania

Inwestycja w najnowocześniejszą technologię wyszukiwania w celu uzyskania lepszych wyników

R ← Jakość **retrievera** ma kluczowe znaczenie!

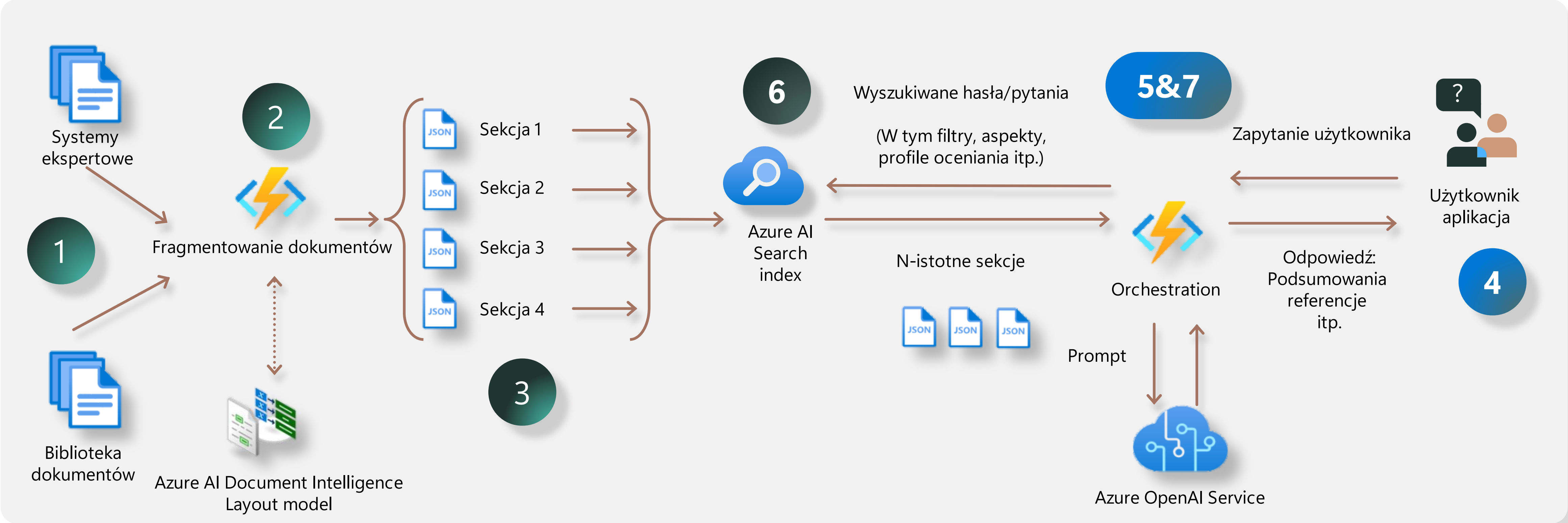
A

G

Usługa Azure AI Search dokłada wszelkich starań, aby zapewnić NAJLEPSZE rozwiązanie do pobierania za pośrednictwem:

- Możliwości wyszukiwania wektorowego
- Wyszukiwanie hybrydowe
- Zaawansowane filtrowanie
- Bezpieczeństwo dokumentów
- Przeszeregowanie/optymalizacja L2
- Wbudowane dzielenie na fragmenty
- Automatyczna wektoryzacja
- I wiele więcej!

Anatomia RAG



1. Pozyskiwanie danych

Różne formaty danych i system ewidencji

2. Chunking

Jaka jest najlepsza strategia chunkingu?

3. Indeksowanie

Czy powinienem używać osadzania wektorów, przekształcania danych, mapowań?

4. Interfejs użytkownika

Chatbot do pytań i odpowiedzi udostępniony użytkownikom końcowym

5. Orkiestracja

Koordynacja komunikacji i promptowanie — propmt o treść wyszukiwania

6. Pobieranie danych

Czy powinienem stosować podejście wektorowe, semantyczne, słowa kluczowe czy hybrydowe?

7. Orkiestracja

Koordynacja komunikacji: tworzenie odpowiedzi użytkownika na podstawie pobranych danych

Niezawodne pobieranie aplikacji RAG

- Odpowiedzi tylko tak dobre, jak pobrane dane
- Wyzwania związane z wyszukiwaniem słów kluczowych
  - "Luka w słownictwie"
  - Gorzej z pytaniami w języku naturalnym
- Wyszukiwanie oparte na wektorach działa dobrze z językiem naturalnym
  - Odporny na różnice w sposobie wyrażania pojęć (dobór słów, morfologia, specyfika itp.)

Pytanie:

"Czy mój **plan zdrowotny** obejmuje **coroczne** badania **wzroku**?"

Wyszukiwanie słów kluczowych nie pasuje:

"Northwind Standard oferuje pokrycie kosztów tylko w przypadku badań wzroku i okularów".

"Northwind Health Plus oferuje pokrycie kosztów badań wzroku, okularów i soczewek kontaktowych, a także badań dentystycznych, czyszczenia i wypełnień".

Dlaczego pobieranie jest ważne?

Co się stanie, jeśli wyniki, których oczekuje użytkownik, nie zostaną zwrócone jako część najważniejszych źródeł użytych do udzielenia odpowiedzi na pytanie?

Wprowadzanie, które zapewnia ramy dotyczące tego, jak model powinien się zachowywać

Prompt

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Źródła, z których udzielono odpowiedzi na pytanie

Źródło, które nie zostało wybrane, ponieważ najtrafniejsze wyniki nie znalazły się na górze:

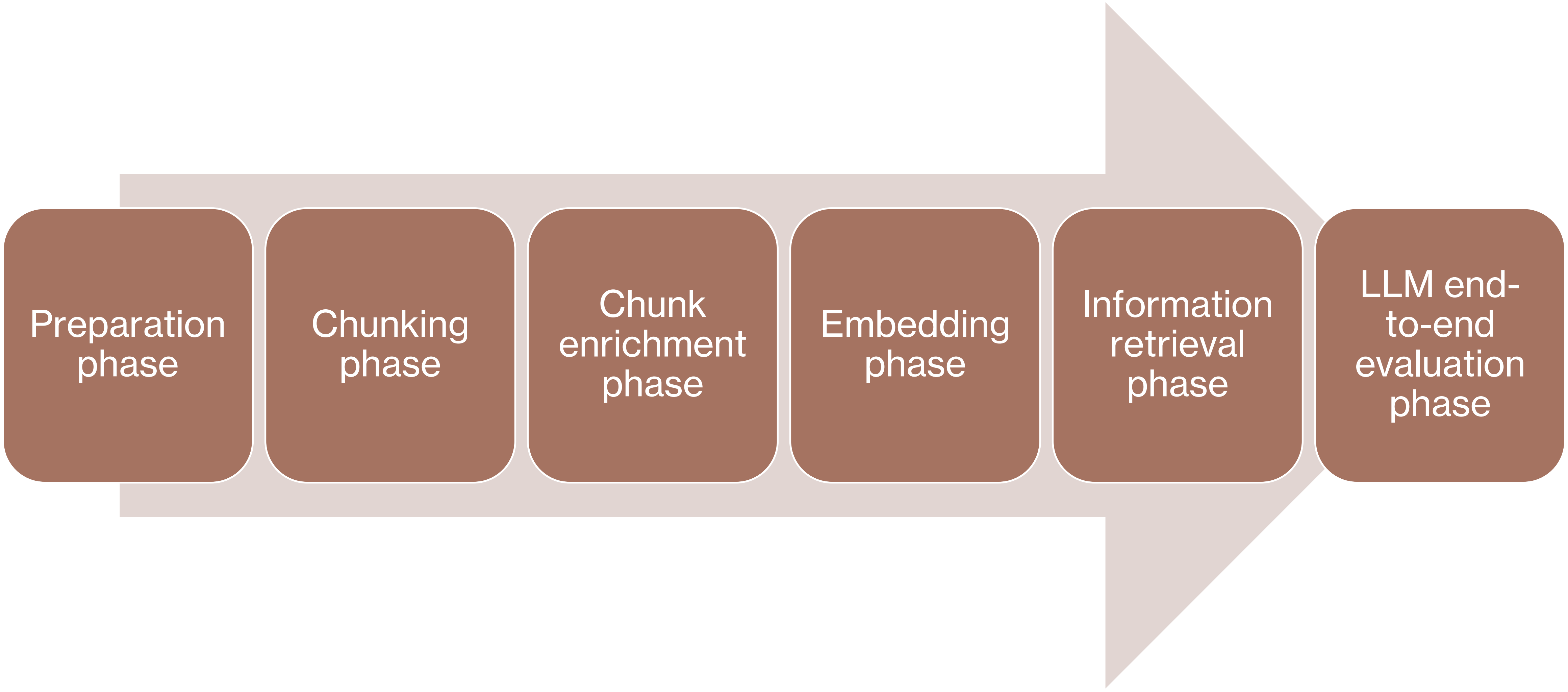
3. Northwind Health Plus oferuje pokrycie kosztów badań wzroku, okularów i soczewek kontaktowych, a także badań dentystycznych, czyszczenia i wypełnień.

Odpowiedni wynik, który nie został podany w ramach najlepszych źródeł

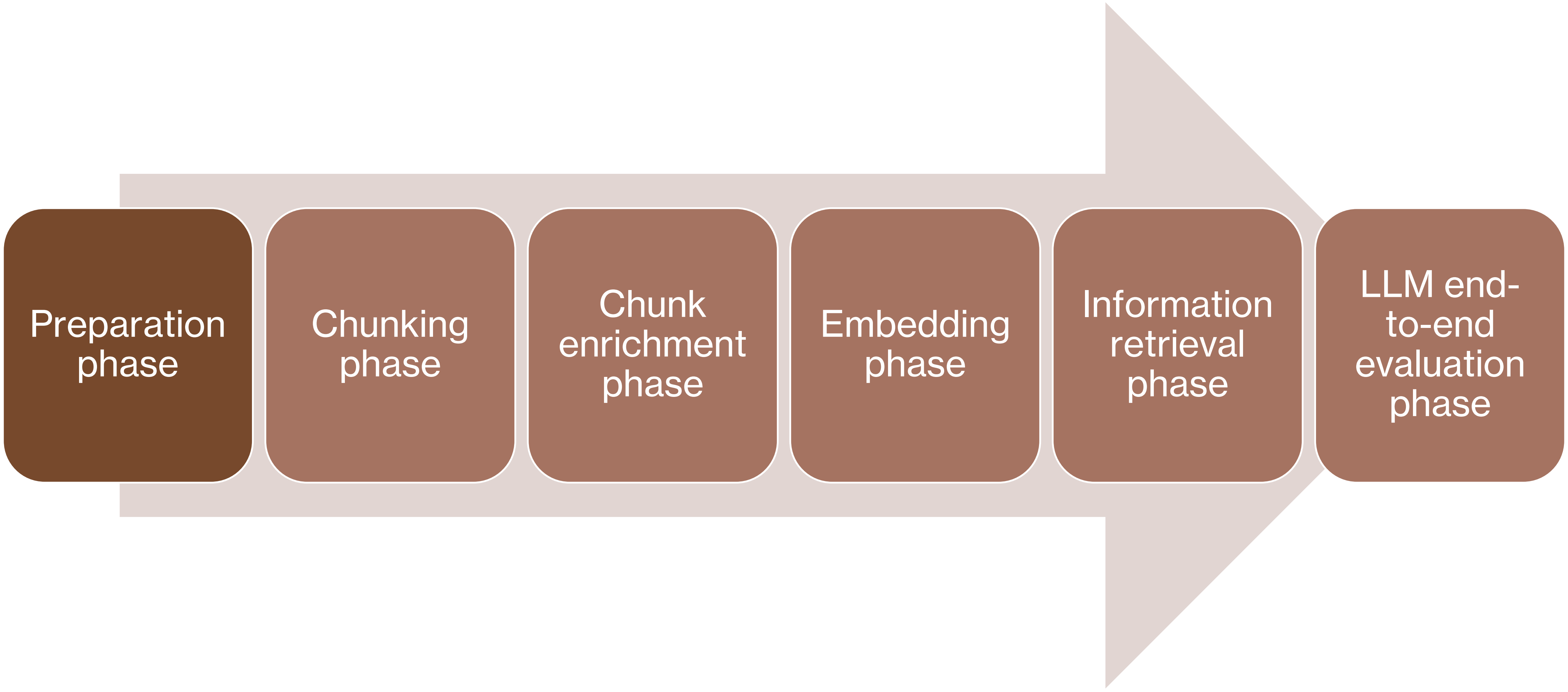
```
Odpowiedź

Na podstawie podanych informacji,
jeśli Twój plan jest z Northwind
Standard, obejmuje badania wzroku.
```

Jeśli retriever nie zwraca najtrafniejszych wyników, odpowiedź nie jest kompletna

RAG data pipeline flow



RAG data pipeline flow



Określenie domeny rozwiązania

Sprecyzowanie wymagań biznesowych i zakresu rozwiązania.

Zdefiniowanie typów pytań, na które system ma odpowiadać.

Wskazanie źródłowych danych i dokumentów potrzebnych do odpowiedzi na te pytania.

Ta faza ma wpływ na dalszą strategię modelowania embeddingów.

Analiza dokumentów

Klasyfikacja dokumentów

- Wyróżnienie głównych i podrzędnych klas dokumentów (np. ubezpieczenia zdrowotne, samochodowe).
- Upewnienie się, że każda klasa jest odpowiednio reprezentowana w testach.

Typy dokumentów

- Identyfikacja formatów plików (PDF, DOCX, HTML, Markdown).
- Sprawdzenie wymagań technicznych dotyczących przetwarzania każdego formatu.

Ograniczenia bezpieczeństwa

- Określenie potrzeby uwierzytelniania i autoryzacji dostępu.
- Zapewnienie, że kod ma dostęp do dokumentów i osadzonych mediów.

Struktura dokumentów

- Analiza układu, długości, obecności multimediów, tabel, wykresów.
- Identyfikacja elementów do pominięcia (stopki, przypisy, znaki wodne, itd.).
- Określenie koniecznych działań wstępnych (czyszczenie, ekstrakcja mediów, formatowanie).

Dobór strategii chunkowania i przetwarzania

Strategia podziału dokumentów (chunking)

- Uwzględnienie struktury dokumentu (np. kolumny, nagłówki, akapity, listy).
- Rozróżnienie elementów istotnych i nieistotnych dla modelu.

Przetwarzanie obrazów i mediów

- Określenie wymagań co do rozdzielczości, obecności tekstu w obrazach, powiązania obrazu z tekstem.
- Decyzja, które elementy multimedialne należy uwzględnić w indeksie.

Zgromadzenie reprezentatywnych dokumentów testowych

- Dokumenty muszą być zgodne z wymaganiami biznesowymi i odpowiadać na zebrane pytania.
- Dokumenty powinny reprezentować wszystkie typy i warianty dokumentów używanych w rozwiązaniu.
- Wysoka jakość fizyczna i merytoryczna dokumentów (brak błędów, czytelność).
- Zalecane: minimum dwa dokumenty na każdą klasę/typ.
- Preferowane dokumenty rzeczywiste, z anonimizacją danych osobowych; dopuszczalne uzupełnienie o dane syntetyczne.

Zbieranie zapytań testowych

Równoległe zbieranie pytań testowych i dokumentów.

Zapytania powinny być:

- Reprezentatywne dla rzeczywistych potrzeb użytkowników.
- Odpowiedziane na podstawie zgromadzonych dokumentów.
- Uwzględniać także przypadki, gdy dokumenty nie zawierają odpowiedzi.

Tworzenie zapytań syntetycznych na podstawie fragmentów dokumentów (chunków) – możliwe automatyzowanie przy użyciu modeli językowych.

Weryfikacja pytań i odpowiedzi przez eksperta dziedzinowego (SME).

Uwzględnienie zapytań do treści osadzonych (obrazy, tabele, wykresy, multimedia).

Możliwość wykorzystania rzeczywistych pytań klientów (helpdesk, FAQ).

Wyjście z fazy przygotowania

Zestaw pytań testowych, powiązanych fragmentów kontekstu (tekst z dokumentów), oraz poprawnych odpowiedzi.

Gotowy zbiór dokumentów testowych odpowiadających zdefiniowanej domenie.

Podstawy do dalszych etapów budowy i testowania rozwiązania RAG.

Dobre praktyki

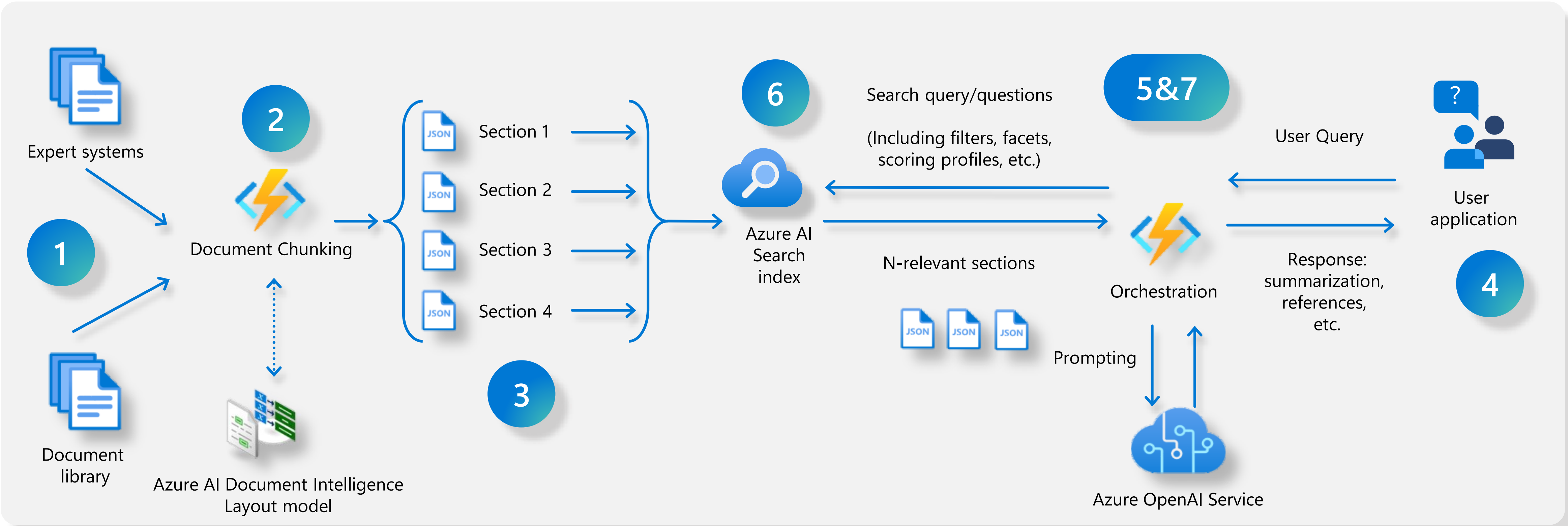
Regularna aktualizacja i przegląd pytań oraz dokumentów, by odzwierciedlały zmiany w domenie.

Utrzymywanie wysokiej jakości dokumentów i reprezentatywności zbioru testowego.

Zapewnienie zgodności z wymaganiami bezpieczeństwa i dostępności danych oraz mediów.

RAG: Deep Dive

Anatomy of RAG



1. Data ingestion

Different data formats and system of records

2. Chunking

What is the best Chunking strategy?

3. Indexing

Shall I use vector embeddings data transformation, mappings?

4. User interface

Chatbot for Q&A surfaced to end users

5. Orchestration

Communication coordination and prompting—Prompt to get retriever query

6. Data retrieving

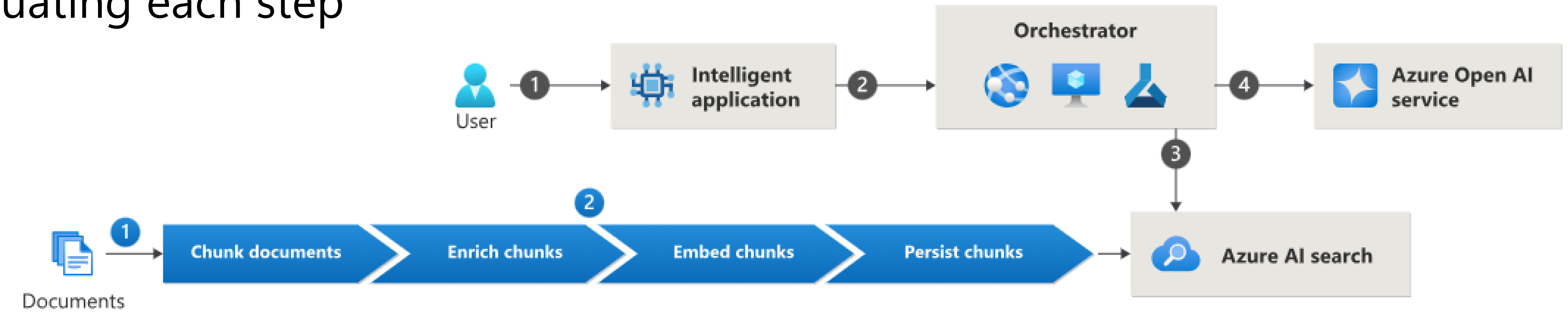
Shall I use vector, semantic, keyword or hybrid approach?

7. Orchestration

Communication coordination: create user response based on retrieve data and send to User app

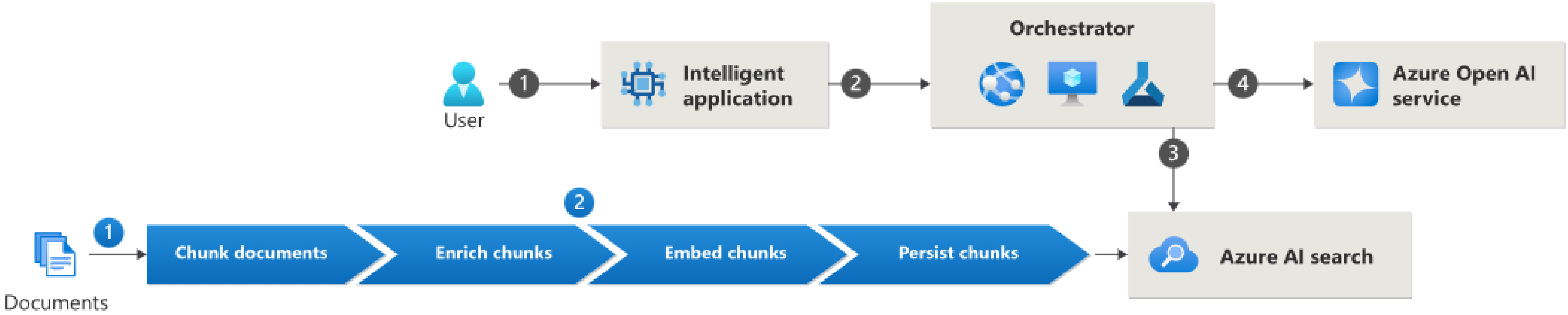
RAG considerations

- Determining what test documents and queries to use during evaluation
- Choosing a chunking strategy
- Determining what and how you should enrich the chunks
- Choosing the right embedding model
- Determining how to configure the search index
- Determining what searches you want to perform: vector, full text, hybrid, manual multiple
- Evaluating each step



RAG application flow

- The user issues a query in an intelligent application user interface.
- The intelligent application makes an API call to an orchestrator. The orchestrator can be implemented with tools or platforms like Semantic Kernel, Azure Machine Learning prompt flow, or LangChain.
- The orchestrator determines what search to perform on Azure AI Search and issues the query.
- The orchestrator packages the top N results from the query, packages them as context within a prompt, along with the query, and sends the prompt to the large language model. The orchestrator returns the response to the intelligent application for the user to read.



Data ingestion

Consider:

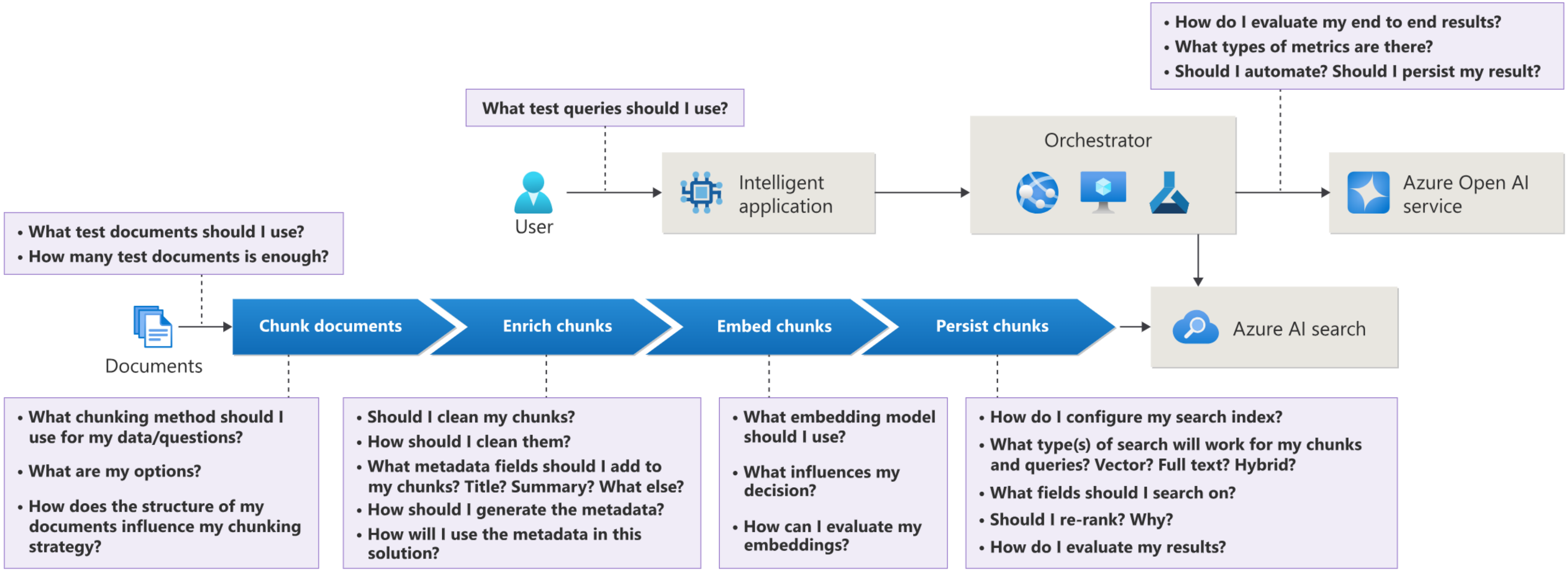
- Data sources
- Data preparation
- Understanding your data
- Ingestion mechanisms (Pull/Push)
- Data AI enrichment
- Chunking
- Vectorization
- Incremental indexing
- Deletion policies
- Scheduling
- Security
- Data and service limits
- Scalability
- Cost
- Error handling and logging
- Compliance

RAG data pipeline flow

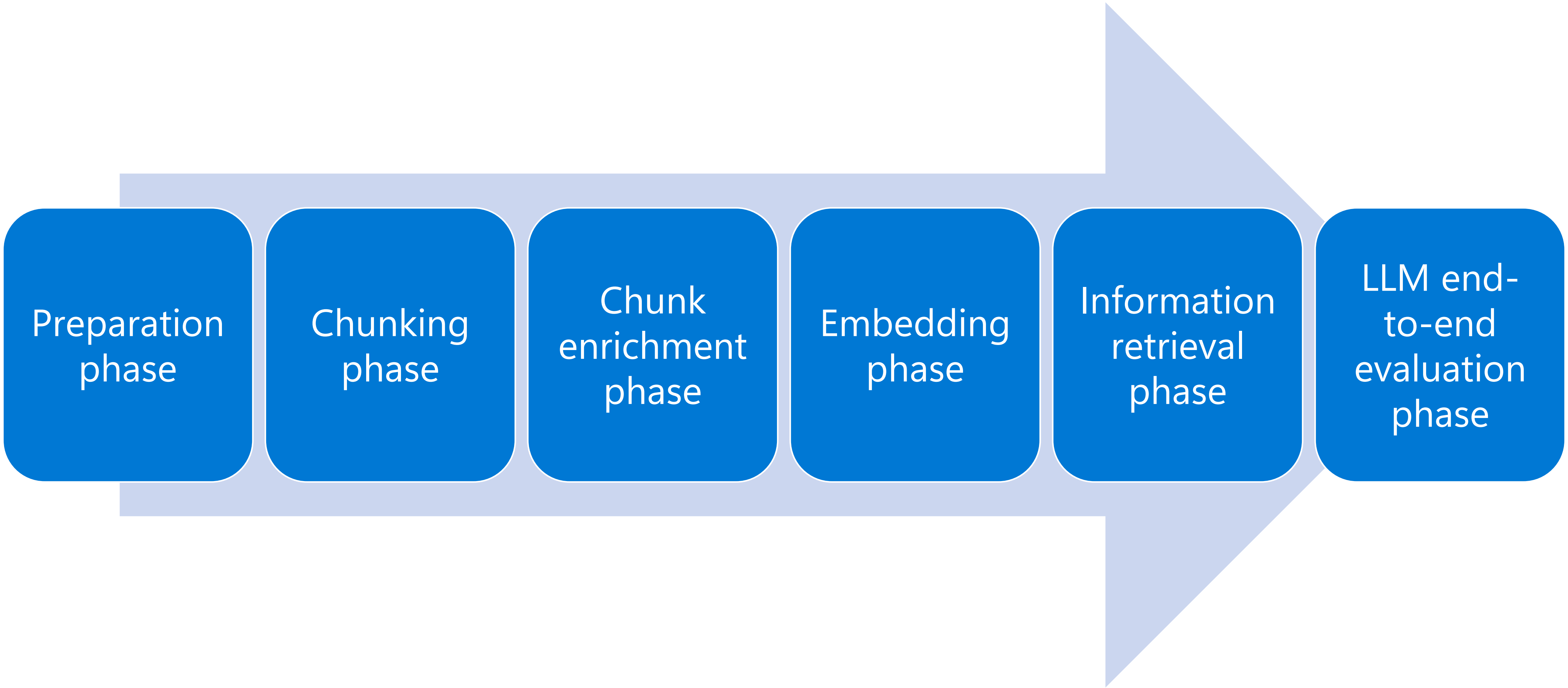
- Documents are either pushed or pulled into a data pipeline.
- The data pipeline processes each document individually with the following steps:
  - Chunk document - Break down the document into semantically relevant parts that ideally have a single idea or concept.
  - Enrich chunks - Adds metadata fields created from the content in the chunks to discrete fields, such as title, summary, and keywords.
  - Embed chunks - Uses an embedding model to vectorize the chunk and any other metadata fields that are used for vector searches.
  - Persists chunks - Stores the chunks in the search index.

RAG design and evaluation considerations

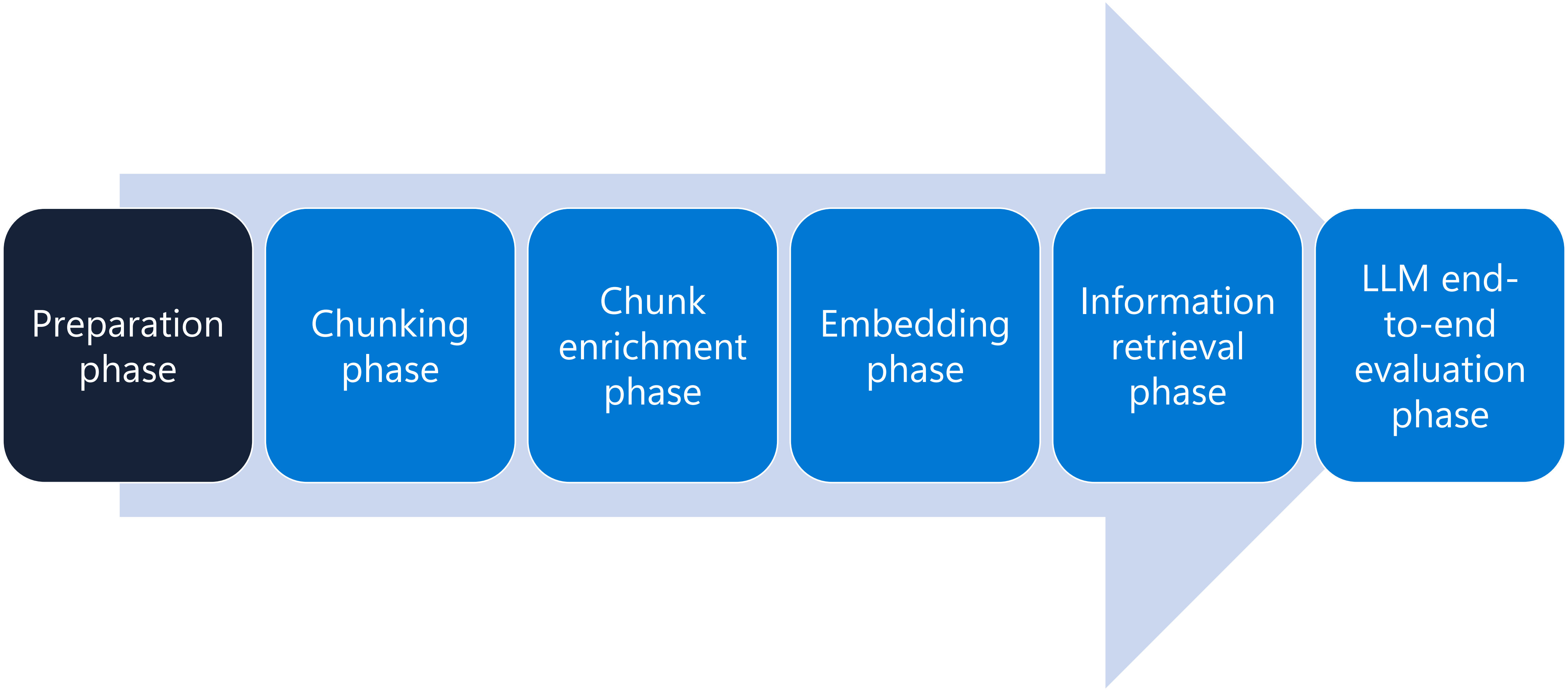
- There are a variety of implementation decisions you must make when designing your RAG solution. The following figure illustrates some of those decisions.



RAG data pipeline flow



RAG data pipeline flow



Preparation

- Determine solution domain
  - The first step in this process is to clearly define the business requirements for the solution or the use case. These requirements help determine what kind of questions the solution intends to address and what source data or documents help address those questions. In later stages, the solution domain helps inform your embedding model strategy.

- Gather representative test documents
  - In this step, you're gathering documents that are the best representation of the documents that you use in your production solution. The documents must address the defined use case and be able to answer the questions gathered in the question gathering parallel phase.

Considerations

- Consider these areas when evaluating potential representative test documents:
- Pertinence - The documents must meet the business requirements of the conversational application. For example, if you're building a chat bot tasked with helping customers perform banking operations, the documents should match that requirement, such as documents showing how to open or close a bank account. The documents must be able to address the test questions that are being gathered in the parallel step. If the documents don't have the information relevant to the questions, it cannot produce a valid response.
- Representative - The documents should be representative of the different types of documents that your solution will use. For example, a car insurance document is different to a health insurance or life insurance document. Suppose the use case requires the solution to support all three types, and you only had two car insurance documents, your solution would perform poorly for both health and life insurance. You should have at least 2 for each variation.
- Physical document quality - The documents need to be in a usable shape. Scanned images, for example, might not allow you to extract usable information.
- Document content quality - The documents must have high content quality. There should not be misspellings or grammatical errors. Large language models don't perform well if you provide them with poor quality content.

Test document guidance

- Prefer real documents over synthetic. Real documents must go through a cleaning process to remove personally identifiable information (PII).
- Consider augmenting your documents with synthetic data to ensure you're handling all kinds of scenarios.
- If you must use synthetic data, do your best to make it as close to real data as possible.
- Make sure that the documents can address the questions that are being gathered.
- You should have at least two documents for each document variant.
- You can use large language models or other tools to help evaluate the document quality.

Gather test queries

- Gather test query output
  - Query - The question, representing a legitimate user's potential prompt.
  - Context - A collection of all the actual text in the documents that address the query. For each bit of context, you should include the page and the actual text.
  - Answer - A valid response to the query. The response be content directly from the documents or it might be rephrased from one or more pieces of context.
- Creating synthetic queries
  - It's often challenging for the subject matter experts (SMEs) for a particular domain to put together a comprehensive list of questions for the use case. One solution to this challenge is to generate synthetic questions from the representative test documents that were gathered.
- Unaddressed queries
  - It's important to gather queries that the documents don't address, along with queries that are addressed.

Gather test queries: Creating synthetic queries

- Chunk the documents - Break down the documents into chunks. This chunking step isn't using the chunking strategy for your overall solution. It's a one-off step that will be used for generating synthetic queries. The chunking can be done manually if the number of documents is reasonable.

```
Please read the following CONTEXT and generate two question and answer json objects in an array based on the CONTEXT provided. The questions should require deep reading comprehension, logical inference, deduction, and connecting ideas across the text. Avoid simplistic retrieval or pattern matching questions. Instead, focus on questions that test the ability to reason about the text in complex ways, draw subtle conclusions, and combine multiple pieces of information to arrive at an answer. Ensure that the questions are relevant, specific, and cover the key points of the CONTEXT.  Provide concise answers to each question, directly quoting the text from provided context. Provide the array output in strict JSON format as shown in output format. Ensure that the generated JSON is 100 percent structurally correct, with proper nesting, comma placement, and quotation marks. There should not be any comma after last element in the array.

Output format:[  { "question": "Question 1","answer": "Answer 1" }, { "question": "Question 2", "answer": "Answer 2" } ]

CONTEXT:
```

- Generate queries per chunk - For each chunk, generate queries either manually or using a large language model. When using a large language model, we generally start by generating two queries per chunk. The large language model can also be used to create the answer. The following example shows a prompt that generates questions and answers for a chunk.
- Verify output - Verify that the questions are pertinent to the use case and that the answers address the question. This verification should be performed by a SME.

Gather test queries

- Unadddressed queries
  - It's important to gather queries that the documents don't address, along with queries that are addressed. When you test your solution, particularly when you test the large language model, you need to determine how the solution should respond to queries it doesn't have sufficient context to answer. Approaches to responding to queries you can't address include:
    - Responding that you don't know
    - Responding that you don't know and providing a link where the user might find more information
- Gather test queries guidance
  - Determine whether there is a system that contains real customer questions that you can use. For example, if you're building a chat bot to answer customer questions, you might be able to use customer questions from your help desk, FAQs, or ticketing system.
  - The customer or SME for the use case should act as a quality gate to determine whether or not the gathered documents, the associated test queries, and the answers to the queries from the documents are comprehensive, representative, and are correct.
  - Reviewing the body of questions and answers should be done periodically to ensure that they continue to accurately reflect the source documents.

Document analysis

- The goal of document analysis is to determine the following three things:
  - What in the document you want to ignore or exclude
  - What in the document you want to capture in chunks
  - How you want to chunk the document

Document analysis - common questions you can ask when analyzing a document

- Does the document contain a table of content?
- Does the document contain images?
  - Are they high resolution images?
  - What kind of data do you have on images?
  - Are there captions for the images?
  - Is there text embedded in the images?
- Does the document have charts with numbers?
- Does the document contain tables?
  - Are the tables complex (nested tables) or noncomplex?
  - Are there captions for the tables?
- Is there multi-column data or multi column paragraphs? You don't want to parse multi-column content as though it were a single column.
- How many paragraphs are there? How long are the paragraphs? Are the paragraphs roughly equal length?
- What languages, language variant, or dialects are in the documents?
- Does the document contain Unicode characters?
- How are numbers formatted? Are they using commas or decimals?
- Are there headers and footers? Do you need them?
- Are there copyrights or disclaimers? Do you need them?
- What in the document is uniform and what isn't uniform?
- Is there a header structure where semantic meaning can be extracted?
- Are there footnotes or endnotes?
- Are there watermarks?
- Are there annotations or comments (for example, in PDFs or Word documents)
- Are there other types of embedded media like videos or audio?
- Are there any mathematical equations/scientific notations in the document?
- Are there bullets or meaningful indentations?

PDF in LLM

PDFs: The Dominant Data Format

- Enterprise and personal docs
- PDFs are a universal format for document exchange
- PDFs used in contracts,invoices, medical records, etc..
- PDFs maintain the same formatting across different platforms
- PDFs can be encrypted and signed digitally to secure sensitive information
- PDF became an open standard in 2008, promoting widespread use and innovation

LLM

- Understanding and Generating Text: LLMs are AI systems that can understand and generate human language. They are designed to mimic human-like text understanding and generation based on the data they were trained on.
- Training on Large Datasets: LLMs are trained on extensive sets of data, which enables them to recognize patterns, nuances, and complexities in language. This training involves the use of large neural networks and deep learning techniques.
- Next-Word Prediction Engines: Fundamentally, LLMs are next-word prediction engines. They process natural language inputs and predict the next word based on the context they've been given.
- Wide Range of Tasks: LLMs can perform a variety of tasks, including inferring from context, generating coherent and contextually relevant responses, translating languages, summarizing text, answering questions, and even assisting in creative writing or code generation tasks.
- Text Input Requirement: LLMs require text input to operate. They use this input to understand the context and generate appropriate responses.
- Limitations in processing long documents

Challenges of extracting text from PDF

- Complex structure - Nested elements, varying data types.
- No standard layout - Different standards and versions.
- Text extraction challenges - Encodings, fonts, formatting.
- Table and image handling - Accurate recognition and data capture.
- Error handling and validation - Ensuring data integrity.

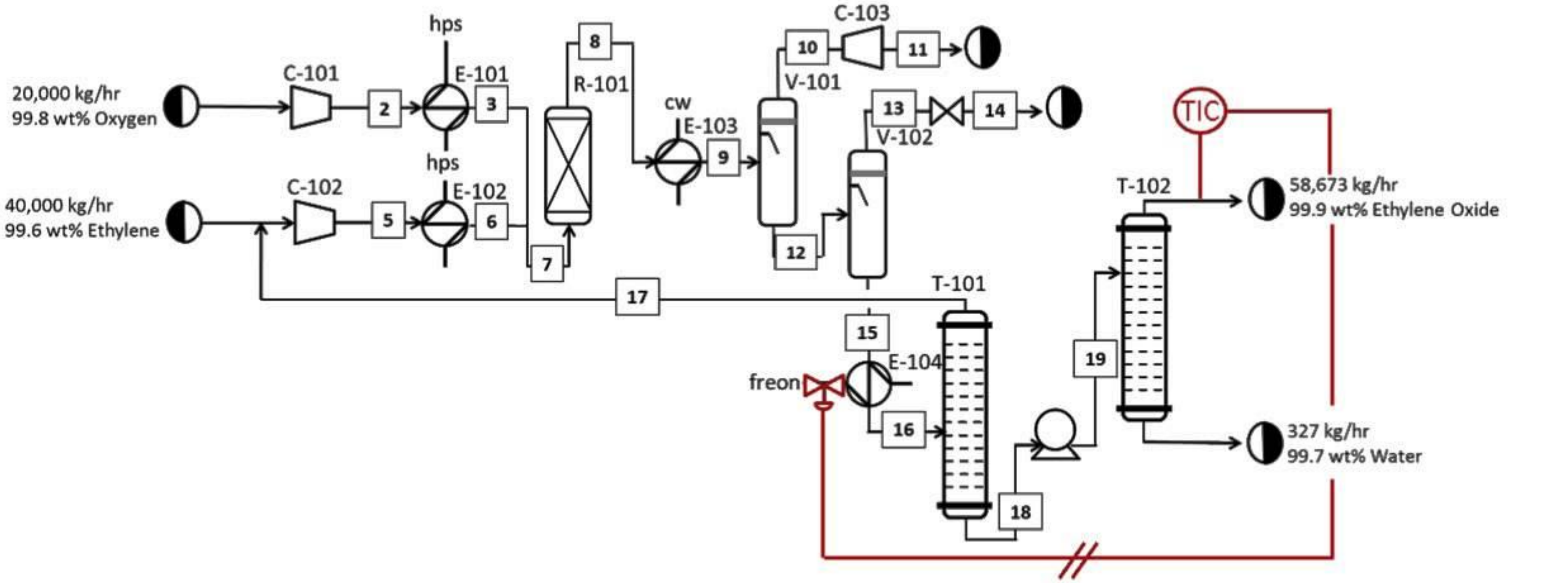
Use PDF in LLM

- Convert PDFs to plain text for easier parsing.
- Use machine learning model strained on diverse PDF layouts.
- Develop custom parsers tailored to specific PDF structures.
- Leverage OCR techniques to extract text from images within PDFs.
- Employ LLMs to understand and structure extracted PDF content.


PICTURES!
1:1.4 D
AF
50mm

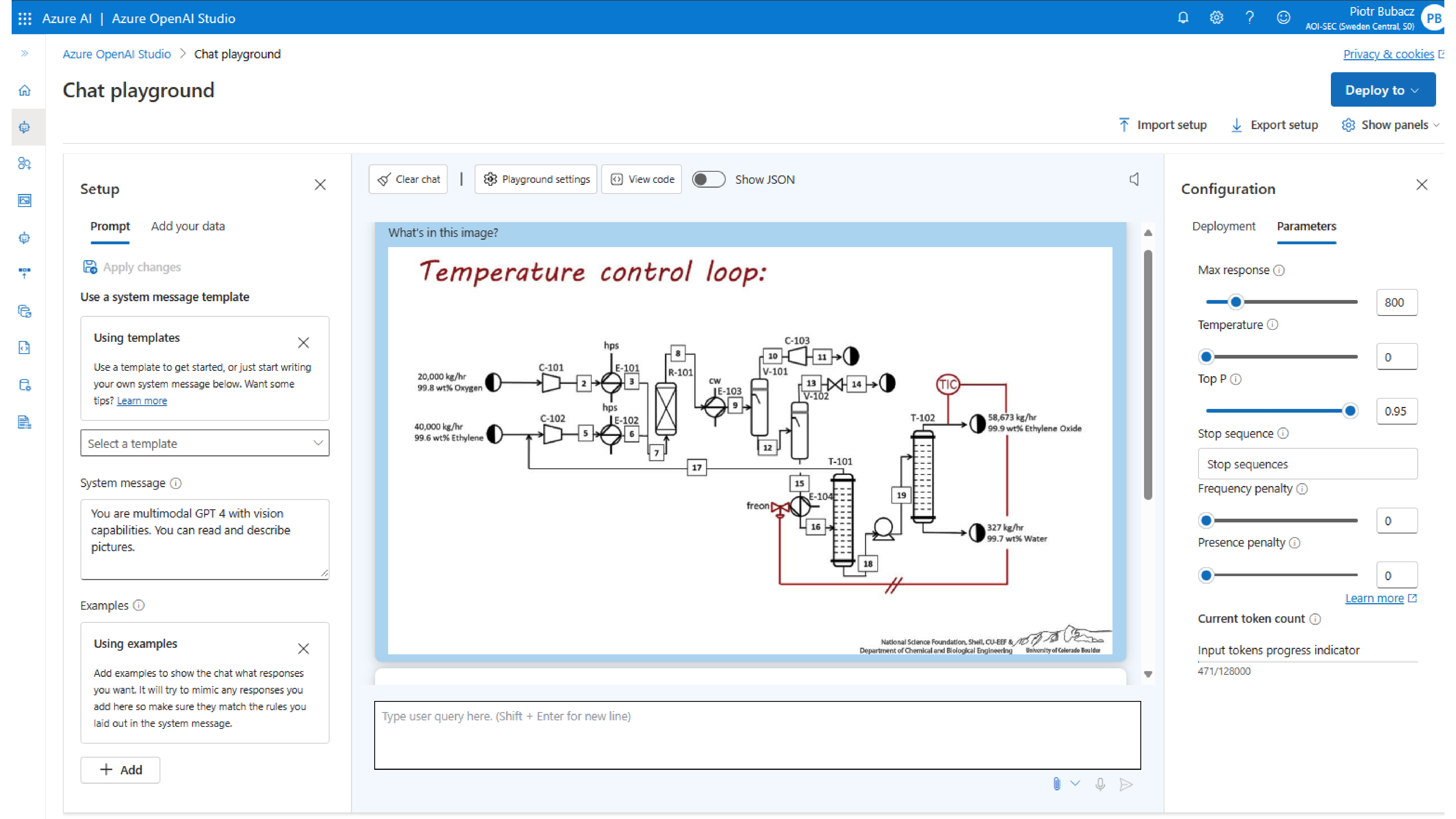
I have a picture!

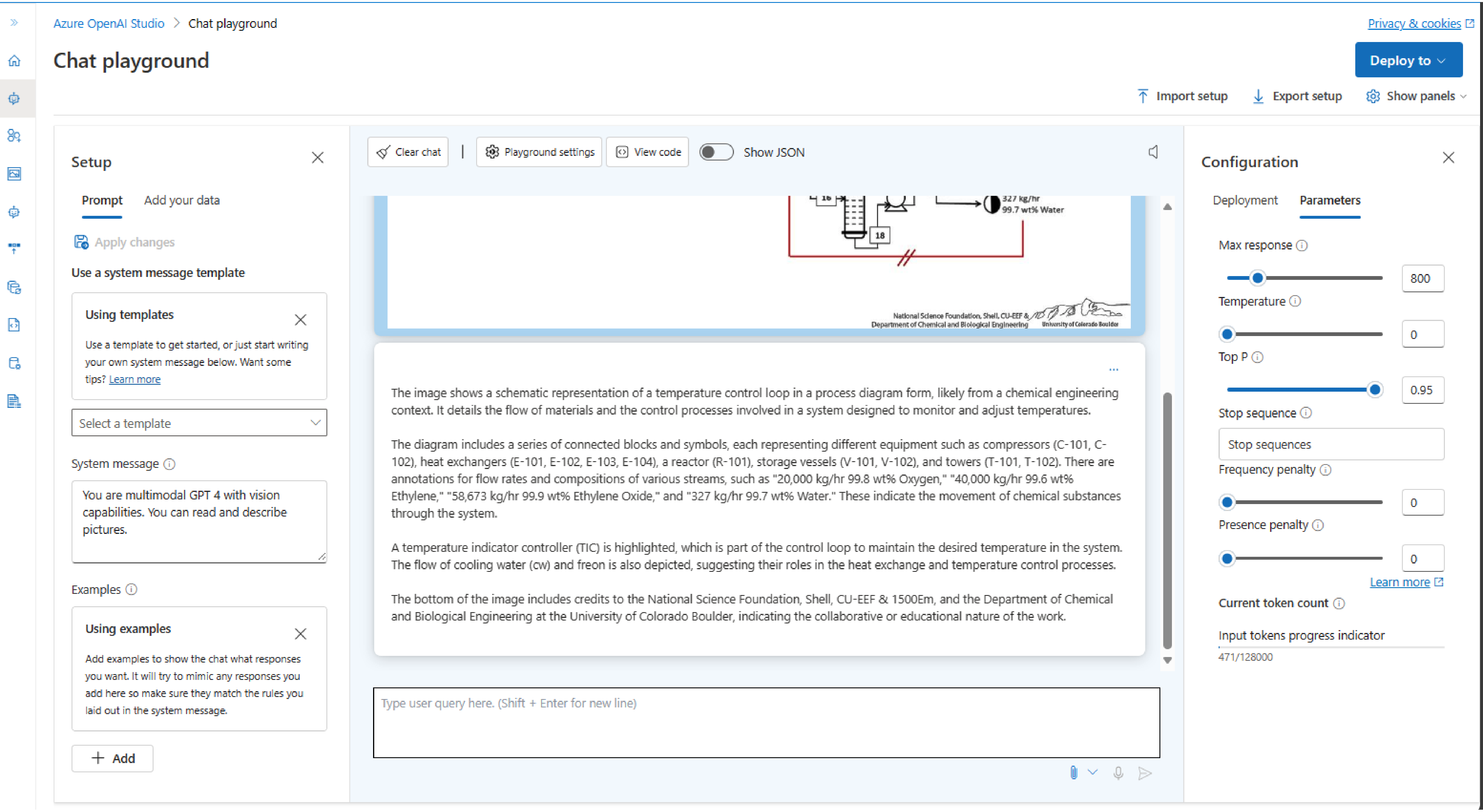
Temperature control loop:



What's in this image?

National Science Foundation, Shell, CU-EEF & Department of Chemical and Biological Engineering University of Colorado Boulder


Azure AI | Azure OpenAI Studio
Piotr Bubacz
AOI-SEC (Sweden Central, S0)
Azure OpenAI Studio > Chat playground
Privacy & cookies
Chat playground
Deploy to
Import setup
Export setup
Show panels
Setup
Prompt
Add your data
Apply changes
Use a system message template
Using templates
Use a template to get started, or just start writing your own system message below. Want some tips? Learn more
Select a template
System message
You are multimodal GPT 4 with vision capabilities. You can read and describe pictures.
Examples
Using examples
Add examples to show the chat what responses you want. It will try to mimic any responses you add here so make sure they match the rules you laid out in the system message.
Add
Clear chat
Playground settings
View code
Show JSON
What's in this image?
Temperature control loop:
20,000 kg/hr
99.8 wt% Oxygen
40,000 kg/hr
99.6 wt% Ethylene
58,673 kg/hr
99.9 wt% Ethylene Oxide
327 kg/hr
99.7 wt% Water
TIC
freon
National Science Foundation, Shell, CU-EEF &
Department of Chemical and Biological Engineering
University of Colorado Boulder
Type user query here. (Shift + Enter for new line)
Configuration
Deployment
Parameters
Max response
800
Temperature
0
Top P
0.95
Stop sequence
Stop sequences
Frequency penalty
0
Presence penalty
0
Learn more
Current token count
Input tokens progress indicator
471/128000


Azure OpenAI Studio > Chat playground
Privacy & cookies
Chat playground
Deploy to
Import setup
Export setup
Show panels
Setup
Prompt
Add your data
Apply changes
Use a system message template
Using templates
Use a template to get started, or just start writing your own system message below. Want some tips? Learn more
Select a template
System message
You are multimodal GPT 4 with vision capabilities. You can read and describe pictures.
Examples
Using examples
Add examples to show the chat what responses you want. It will try to mimic any responses you add here so make sure they match the rules you laid out in the system message.
Add
Clear chat
Playground settings
View code
Show JSON
327 kg/hr
99.7 wt% Water
16
18
National Science Foundation, Shell, CU-EEF &
Department of Chemical and Biological Engineering
University of Colorado Boulder
The image shows a schematic representation of a temperature control loop in a process diagram form, likely from a chemical engineering context. It details the flow of materials and the control processes involved in a system designed to monitor and adjust temperatures.
The diagram includes a series of connected blocks and symbols, each representing different equipment such as compressors (C-101, C-102), heat exchangers (E-101, E-102, E-103, E-104), a reactor (R-101), storage vessels (V-101, V-102), and towers (T-101, T-102). There are annotations for flow rates and compositions of various streams, such as "20,000 kg/hr 99.8 wt% Oxygen," "40,000 kg/hr 99.6 wt% Ethylene," "58,673 kg/hr 99.9 wt% Ethylene Oxide," and "327 kg/hr 99.7 wt% Water." These indicate the movement of chemical substances through the system.
A temperature indicator controller (TIC) is highlighted, which is part of the control loop to maintain the desired temperature in the system. The flow of cooling water (cw) and freon is also depicted, suggesting their roles in the heat exchange and temperature control processes.
The bottom of the image includes credits to the National Science Foundation, Shell, CU-EEF & 1500Em, and the Department of Chemical and Biological Engineering at the University of Colorado Boulder, indicating the collaborative or educational nature of the work.
Type user query here. (Shift + Enter for new line)
Configuration
Deployment
Parameters
Max response
800
Temperature
0
Top P
0.95
Stop sequence
Stop sequences
Frequency penalty
0
Presence penalty
0
Learn more
Current token count
Input tokens progress indicator
471/128000

AI GPT 4o description:

...

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Can we do more?

Please provide a detailed technical description of the system depicted in the image

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...

The image depicts a detailed process flow diagram (PFD) for a temperature control loop, commonly used in chemical engineering to illustrate the flow of chemical substances and the equipment involved in the process. Here's a technical breakdown of the system:

1. **Compressors (C-101 and C-102):** These are likely used to increase the pressure of the chemical feedstocks entering the system. C-101 handles a stream of oxygen at 20,000 kg/hr with a purity of 99.8 wt%, while C-102 handles ethylene at 40,000 kg/hr with a purity of 99.6 wt%.
2. **Heat Exchangers (E-101, E-102, E-103, E-104):** These units are used to transfer heat between streams. E-101 and E-102 are probably used to control the temperature of the feedstocks by either heating or cooling them before they enter the reactor. E-103 and E-104 might be used for cooling the products and by-products after the reaction.
3. **Reactor (R-101):** This is where the chemical reaction takes place. The reactor is fed by the oxygen and ethylene streams, where they react to form ethylene oxide and other by-products.
4. **Separation Vessels (V-101 and V-102):** These vessels are used to separate the products from the unreacted feed and by-products. V-101 might be a flash drum used to separate gases from liquids, while V-102 could be a decanter or another type of separator.
5. **Distillation Towers (T-101 and T-102):** These towers are used for further separation of the components based on their boiling points. T-101 could be separating lighter components from heavier ones, while T-102 is likely purifying the main product, ethylene oxide, to a high purity of 99.9 wt%.
6. **Cooling Water (cw) and Freon Lines:** These lines indicate the flow of cooling agents used in the heat exchangers and possibly in the condensers associated with the distillation towers.
7. **Temperature Indicator Controller (TIC):** This is a control system that monitors and adjusts the temperature within the system, likely in the reactor or one of the distillation towers, to maintain optimal reaction conditions and product quality.
8. **Flow Rates and Purity Indicators:** The diagram includes specific flow rates and purity percentages for various streams, indicating the scale of the operation and the efficiency of the separation processes.
9. **Flow Direction Indicators:** The arrows show the direction of the process flow through the system.
10. **Valves and Control Points:** Various points in the system, such as number 15, indicate valves or control points that can be adjusted to control the flow rates, pressures, and compositions of the streams.

The diagram is a high-level representation of a chemical process and is typically used for process design, analysis, and optimization. The inclusion of the National Science Foundation, Shell, CU-EEF & 1500Em, and the Department of Chemical and Biological Engineering at the University of Colorado Boulder suggests that this PFD might be part of an educational module, research project, or a collaborative industry-academic study.

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GPT-4 vision limitations

Rotation: The model may misinterpret rotated / upside-down text or images.

Spatial reasoning: The model struggles with tasks requiring precise spatial localization, such as identifying chess positions.

Metadata and resizing: The model doesn't process original file names or metadata, and images are resized before analysis, affecting their original dimensions.

Counting: May give approximate counts for objects in images.

Visual elements: The model may struggle to understand graphs or text where colors or styles like solid, dashed, or dotted lines vary.

Images in prompt: max 20 MB per image. Formats: png, jpeg, gif and webp format. no limit defined (<10)

Vision Limitations - OpenAI API

Classification

- PDF contains multiple sub-documents requiring classification and separation
- Challenging to distinguish classes based on layout only

NexGen

Innovation drives progress

Mr Joe Bloggs
73 Regal Way
LEEDS
West Yorkshire
LS1 5AB

3rd June 2024

Policy number
GB20246717948

Manage your account
NexGen.com/my-account

Hello Mr Bloggs,

Welcome to your new car insurance

Thanks for choosing us. Your new car insurance is effective from 3rd June 2024. Please check the summary table below and let us know if there's anything incorrect.

Details	Option
HYUNDAI IONIQ 5 PREMIUM 73 KWH RWD (2024) VS24DMC	
Class Of Use	Business – class 1
Purchase Date	03/06/2024
Costs of insurance	£532.19 per year or £44.35 per month via Direct Debit

Your new documents

You'll find your new documents in the app or MyAccount.

- Schedule of insurance
- Statement of insurance
- Certificate of insurance
- Cover summary

We're here to help

If you have any questions about this, there are lots of ways you can get in touch. Please visit the Help Centre on our website for details.

Thanks,

Jim Taylor
Operations Director

NexGen

Document issued: 3rd June 2024
Reason for issue: New contract
Policy number: GB20246717948

Your statement of insurance

Please read and check the information shown in this document carefully. It represents the answers given during the initial quotation process, and any changes that have been made to the policy since then. It forms part of your contract for motor insurance.

If any of the details are incorrect or have changed, please contact us immediately.

You can do this:

- Through MyAccount
- By calling 0123 321 1234

We'll tell you if this changes your premium, terms or insurer.

Under the Consumer Insurance (Disclosure and Representations) Act 2012, you have a duty to take reasonable care to answer all questions fully and accurately. If you volunteer information over and above that requested, you must do so honestly and carefully. If you don't answer all questions fully and accurately, it could invalidate your insurance cover and result in all, or part of a claim not being paid.

Policyholder details	
Policyholder	Mr Joe Bloggs
Email address	Joe.Bloggs@me.com
Have any drivers on this policy ever had an insurance policy declined, cancelled, voided or had any special terms imposed?	No
Total vehicles in household	1
Total number of drivers in household	1
Your address	73 Regal Way, LEEDS, West Yorkshire, LS1 5AB
Years at address	4
Homeowner	Yes
Children under 16	No
Vehicle key at address overnight	Yes
Kept overnight	Locked Garage

Insured drivers	
Name	Joe Bloggs
Date of birth	05/01/1990
Marital status	Married-Common Law
Permanent resident in the UK since	Since birth
Employment status	Employed
Primary occupation	Software Engineer
Industry	Computers – Software
Driving licence number	BLOGGS901050JJ1AB
Medical conditions reportable to the DVLA/DVANI	No
Use of other vehicle	No
Unspent non-motoring convictions	No

Your policy details	
Cover type	Comprehensive
Class of use	Business – class 1
Your no claims discount	You told us you have 9 or more years of no claims discount, but we haven't confirmed this with the previous insurer. If you had a non-recoverable claim in this policy period, we'll adjust your no claims discount at renewal.
How no claims discount was earned	Private car discount

Incident history

For all drivers named in this policy, we need to know of any incident, claim or damage involving any motor vehicle in the past five years, including windscreen damage. This applies whether a claim was made, and regardless of blame.

None disclosed

Conviction history

For all drivers named in this policy, we need to know of any driving related convictions, endorsements, fixed penalties, disqualifications or bans in the past five years.

Driver	Date	Type	Licence points	Disqualification
Joe Bloggs	-	-	-	-

Insured vehicle details	
Registration number	VS24DMC
Annual mileage	5,000
Make & model	Hyundai IONIQ 5 Premium 73 kWh RWD
Body type	HATCHBACK
Year registered	2024
Transmission	Automatic
Engine size	0
Number of seats	5
Number of doors	5
Date purchased	05/2024
Imported	No
Left hand drive	No
Declared value	£40,000.00
Registered keeper & legal owner	Yes
Security devices or immobiliser	Factory fitted Alarm + Immobiliser
Tracker fitted	No
Modifications	No

What's a modification?

A modification is any alteration to your car from the manufacturer's standard specification. This includes, but isn't limited to:

- Changes to the bodywork, such as spoilers or body kit
- Changes to suspension or brakes
- Alloy wheels
- Audio/entertainment system
- Changes affecting performance, such as to the engine management system or exhaust system.

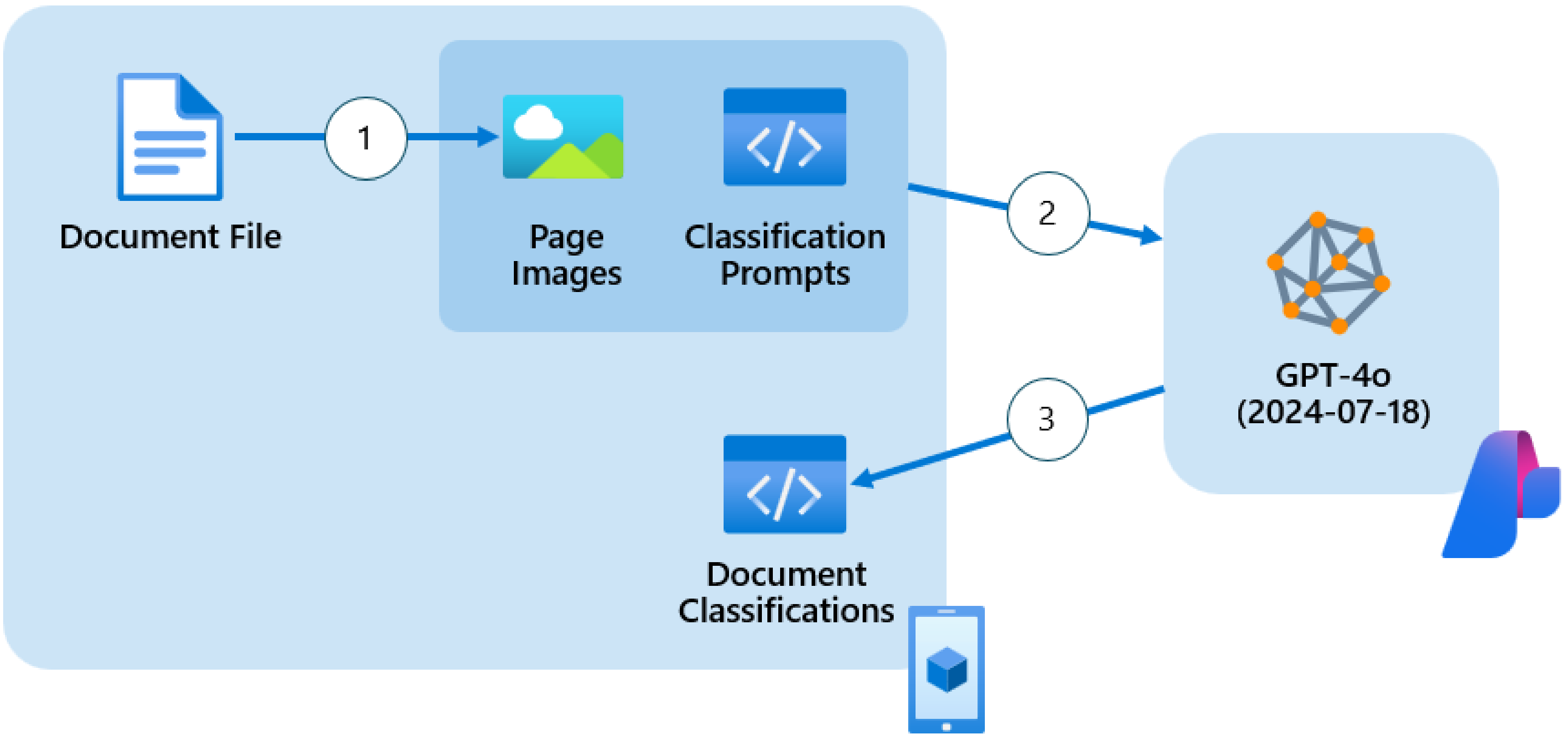
You must also update your Policy if you intend to alter or modify your Car/s from the manufacturer's standard specification

If you don't tell us about a modification, we may cancel your Policy from its start date, apply additional premium or add new terms to your Policy. If you make a claim your insurer may reject the claim or only provide partial payment for it.

We may need to check your vehicle for modifications and accessories at the time of your claim.

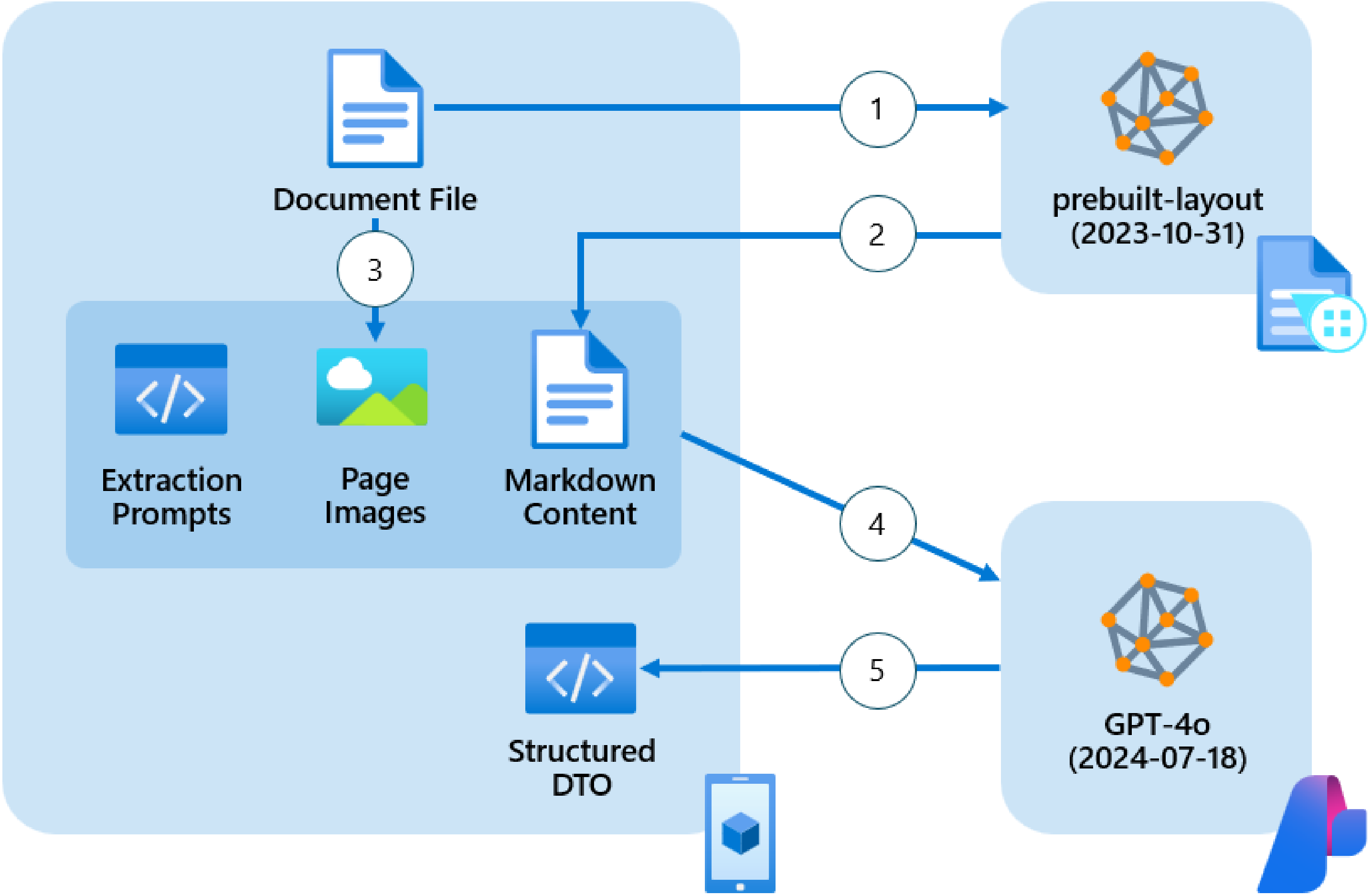
Please note: optional extras fitted at the point of manufacture aren't classed as modifications.

Classification - Azure OpenAI GPT-4o with Vision



classification/openai.ipynb

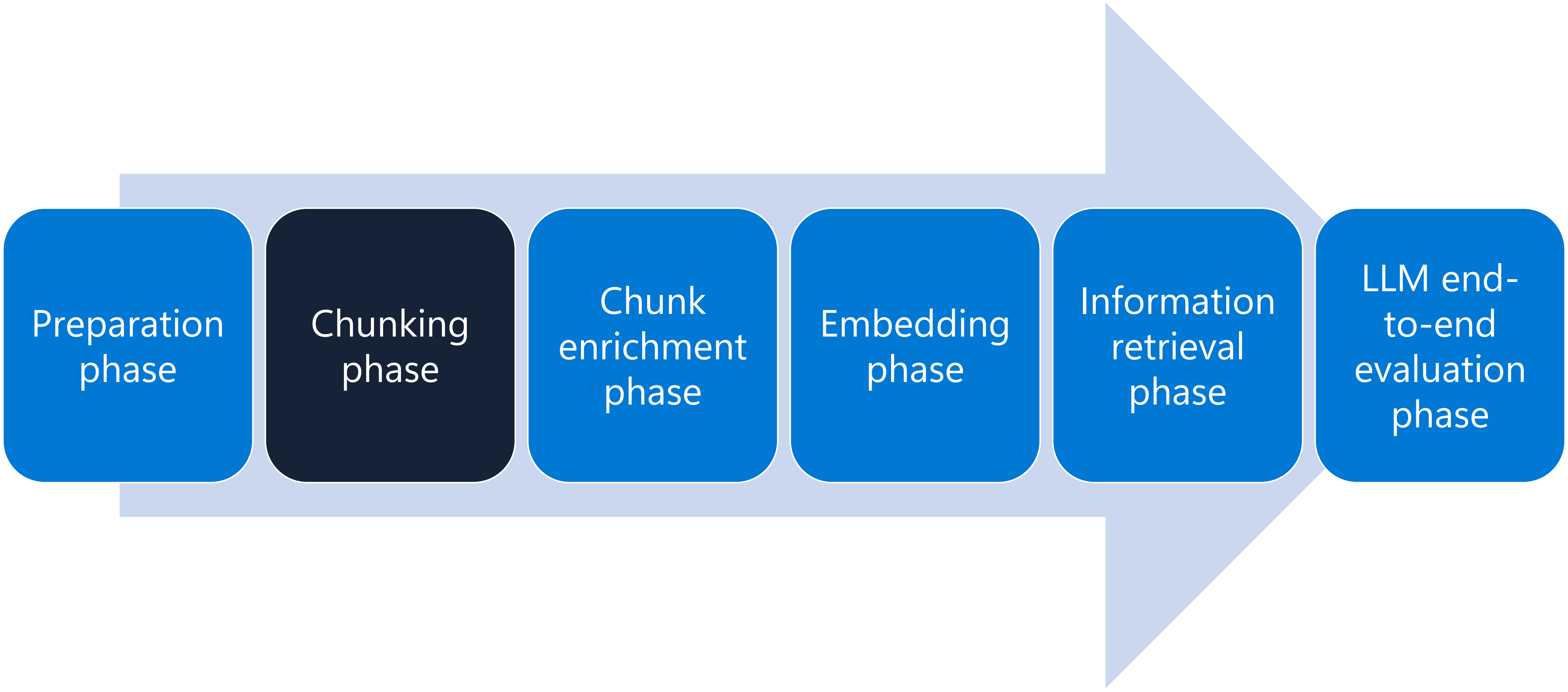
Data Extraction - Comprehensive Azure AI Document Intelligence + Azure OpenAI GPT-4o with Vision



extraction/vision-based/comprehensive

Document Intelligence
Content Understanding

RAG data pipeline flow



Your document Chunking strategy matters

Chunking solves 3 problems for generative AI applications:

1. Allows multiple retrieved documents to be passed to the LLM within its context window limit
2. Provides a mechanism for the most relevant passages of a given document to be ranked first
3. Vector search has a per-model limit to how much content can be embedded into each vector

Retrieval Configuration	**Single vector per document** *[Recall@50]*	**Chunked documents** *[Recall@50]*
Queries whose answer is in long documents	28.2	**45.7**
Queries whose answer is deep into a document	28.7	**51.4**

Chunk boundary strategy	**Recall@50**
512 tokens, break at token boundary	40.9
512 tokens, preserve sentence boundaries	42.4
512 tokens with 10% overlapping chunks	43.1
512 tokens with 25% overlapping chunks	**43.9**

Retrieval Configuration	**Recall@50**
512 input tokens per vector	**42.4**
1024 input tokens per vector	37.5
4096 input tokens per vector	36.4
8191 input tokens per vector	34.9

Chunking economics

- When determining your overall chunking strategy, you must consider your budget along with your quality and throughput requirements for your text corpus. There are engineering costs for the design and implementation of each unique chunking implementation and per-document processing costs that differ depending upon the approach.
- The following are factors to consider when looking at the cost of your overall solution:
  - Number of unique chunking implementations - Each unique implementation has both an engineering and maintenance cost. You need to consider the number of unique document types in your corpus and the cost vs. quality tradeoffs of unique implementations for each.
  - Per-document cost of each implementation - Some chunking approaches might lead to better quality chunks but have a higher financial and temporal cost to generate those chunks. For example, using a prebuilt model in Azure AI Document Intelligence likely has a higher per-document cost than a pure text parsing implementation, but might lead to better chunks.
  - Number of initial documents - The number of initial documents you need to process to launch your solution.
  - Number of incremental documents - The number and rate of new documents that you must process for ongoing maintenance of the system.

Chunking strategies

Chunking methods

Fixed-size Chunking	Simple and common, maintaining a balance between overlap and semantic context
Specialized Chunking	For structured content like Markdown and LaTeX
“Content-aware” Chunking	Content splitting using NLP libraries such as AI Document Intelligence Layout API
Recursive Chunking	Divides text into smaller Chunks in a hierarchical manner

→ AI Search Built-in Split Skill

→ AI Search Custom Skill

Determining the best Chunk size

- Preprocessing data for quality
- Selecting a range of potential Chunk sizes considering content nature and embedding model capabilities
- Evaluating performance of each Chunk size

🔗 Tech blog with suggested Chunking sizes depending on scenario

Chunking

Fixed-size chunking

```python
text = "..." # your text
from langchain.text_splitter import CharacterTextSplitter
text_splitter = CharacterTextSplitter(
    separator = "\n\n",
    chunk_size = 256,
    chunk_overlap  = 20
)
docs = text_splitter.create_documents([text])
```

"Content-aware" Chunking

- Sentence splitting

```python
text = "..." # your text
docs = text.split(".")
```

- Recursive Chunking

```python
text = "..." # your text
from langchain.text_splitter import RecursiveCharacterTextSplitter
text_splitter = RecursiveCharacterTextSplitter(
    # Set a really small chunk size, just to show.
    chunk_size = 256,
    chunk_overlap  = 20
)

docs = text_splitter.create_documents([text])
```

Fixed-size parsing

ChunkViz (railway.app)

Upload .txt

Splitter: Character Splitter

Chunk Size: 25

Chunk Overlap: 5

Total Characters: 3318

Number of chunks: 133

Average chunk size: 24.9

One of the most important things I didn't understand about the world when I was a child is the degree to which the returns for performance are superlinear.

Teachers and coaches implicitly told us the returns were linear. "You get out," I heard a thousand times, "what you put in." They meant well, but this is rarely true. If your product is only half as good as your competitor's, you don't get half as many customers. You get no customers, and you go out of business.

It's obviously true that the returns for performance are superlinear in business. Some think this is a flaw of capitalism, and that if we changed the rules it would stop being true. But superlinear returns for performance are a feature of the world, not an artifact of rules we've invented. We see the same pattern in fame, power, military victories, knowledge, and even benefit to humanity. In all of these, the rich get richer. [1]

You can't understand the world without understanding the concept of superlinear returns. And if you're ambitious you definitely should, because this will be the wave you surf on.

It may seem as if there are a lot of different situations with superlinear returns, but as far as I can tell they reduce to two fundamental causes: exponential growth and thresholds.

The most obvious case of superlinear returns is when you're working on something that grows exponentially. For example, growing bacterial cultures. When they grow at all, they grow exponentially. But they're tricky to grow. Which means the difference in outcome between someone who's adept at it and someone who's not is very great.

Startups can also grow exponentially, and we see the same pattern there. Some manage to achieve high growth rates. Most don't. And as a result you get qualitatively different outcomes: the companies with high growth rates tend to become immensely valuable, while the ones with lower growth rates may not even survive.

Y Combinator encourages founders to focus on growth rate rather than absolute numbers. It prevents them from being discouraged early on, when the absolute numbers are still low. It also helps them decide what to focus on: you can use growth rate as a compass to tell you how to evolve the company. But the main advantage is that by focusing on growth rate you tend to get something that grows exponentially.

YC doesn't explicitly tell founders that with growth rate "you get out what you put in," but it's not far from the truth. And if growth rate were proportional to performance, then the reward for performance p over time t would be proportional to pt.

Even after decades of thinking about this, I find that sentence startling.

Sentence-based parsing

- This straightforward approach breaks down text documents into chunks made up of complete sentences. The benefits of this approach include that it's inexpensive to implement, it has low processing cost, and it can be applied to any text-based document that is written in prose, or full sentences. A challenge with this approach is that each chunk might not capture the complete context of a thought or meaning. Often, multiple sentences must be taken together to capture the semantic meaning.

- Tools: spaCy sentence tokenizer, LangChain recursive text splitter, NLTK sentence tokenizer
- Engineering effort: Low
- Processing cost: Low
- Use cases: Unstructured documents written in prose, or full sentences, and your corpus of documents contains a prohibitive number of different document types to build individual chunking strategies for
- Examples: User-generated content like open-ended feedback from surveys, forum posts, reviews, email messages, a novel or an essay

Fixed-size advanced parsing (with overlap)

- This approach breaks up a document into chunks based on a fixed number of characters or tokens and allows for some overlap of characters between chunks. This approach has many of the same advantages and disadvantages as sentence-based parsing. An advantage this approach has over sentence-based parsing is that it's possible to get chunks with semantic meaning that spans multiple sentences.
- You must choose the fixed size of the chunks and the amount of overlap. Because the results differ for different document types, it's best to use a tool like the HuggingFace chunk visualizer to do exploratory analysis. Tools like this allow you to visualize how your documents are chunked, given your decisions. It's best practice to use BERT tokens over character counts when using fixed-sized parsing. BERT tokens are based on meaningful units of language, so they preserve more semantic information than character counts.

- Tools: LangChain recursive text splitter, Hugging Face chunk visualizer
- Engineering effort: Low
- Processing cost: Low
- Use cases: Unstructured documents written in prose or non-prose with complete or incomplete sentences. Your corpus of documents contains a prohibitive number of different document types to build individual chunking strategies for
- Examples: User-generated content like open-ended feedback from surveys, forum posts, reviews, email messages, personal or research notes or lists

Fixed-size advanced parsing

Upload .txt

Splitter: Recursive Character Text Splitter

Chunk Size: 25

Chunk Overlap: 0

Total Characters: 2658

Number of chunks: 133

Average chunk size: 20.0

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Custom code

- his approach parses documents using custom code to create chunks. This approach is most successful for text-based documents where the structure is either known or can be inferred and a high degree of control over chunk creation is required. You can use text parsing techniques like regular expressions to create chunks based on patterns within the document's structure. The goal is to create chunks that have similar size in length and chunks that have distinct content. Many programming languages provide support for regular expressions, and some have libraries or packages that offer more elegant string manipulation features.

- Tools: Python (re, regex, BeautifulSoup, lxml, html5lib, marko, unstructured), R (stringr, xml2), Julia (Gumbo.jl)
- Engineering effort: Medium
- Processing cost: Low
- Use cases: Semi-structured documents where structure can be inferred
- Examples: Patent filings, research papers, insurance policies, scripts, and screenplays

Large language model augmentation

- Large language models can be used to create chunks. Common use cases are to use a large language model, such as GPT-4, to generate textual representations of images or summaries of tables that can be used as chunks. Large language model augmentation is used with other chunking approaches such as custom code.

- Tools: Azure OpenAI, OpenAI
- Engineering effort: Medium
- Processing cost: High
- Use cases: Images, tables
- Examples: Generate text representations of tables, and images, summarize transcripts from meetings, speeches, interviews, or podcasts

Document layout analysis

- Document layout analysis libraries and services combine optical character recognition (OCR) capabilities with deep learning models to extract both the structure of documents, and text. Structural elements can include headers, footers, titles, section headings, tables and figures. The goal is to provide better semantic meaning to content contained in documents.

- Document layout analysis libraries and services expose a model that represents the content, both structural and text, of the document. You still have to write code that interacts with the model.

- Tools: Azure AI Document Intelligence document analysis models, Donut, Layout Parser
- Engineering effort: Medium
- Processing cost: Medium
- Use cases: Semi-structured documents
- Examples: News articles, web pages, resumes

Prebuilt model

- There are services, such as Azure AI Document Intelligence, that offer prebuilt models you can take advantage of for a various document types. Some models are trained for specific document types, such as the US Tax W-2 form, while others target a broader genre of document types such as an invoice.

- Tools: Azure AI Document Intelligence prebuilt models, Power Automate Intelligent Document Processing, LayoutLMv3
- Engineering effort: Low
- Processing cost: Medium/High
- Use cases: Structured documents where a prebuilt model exists
- Specific examples: Invoices, receipts, health insurance card, W-2 form

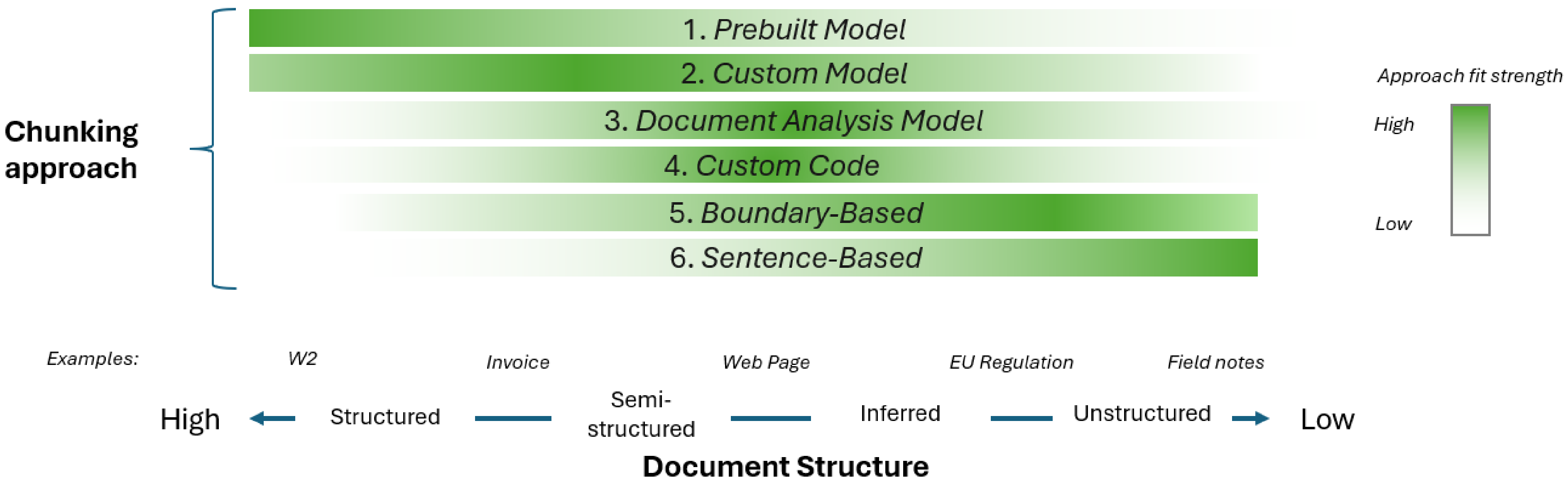
Custom model

- For highly structured documents where no prebuilt model exists, you might have to build a custom model. This approach can be effective for images or documents that are highly structured, making them difficult to use text parsing techniques.

- Tools: Azure AI Document Intelligence custom models, Tesseract
- Engineering effort: High
- Processing cost: Medium/High
- Use cases: Structured documents where a prebuilt model doesn't exist
- Examples: Automotive repair and maintenance schedules, academic transcripts, and records, technical manuals, operational procedures, maintenance guidelines

Document structure

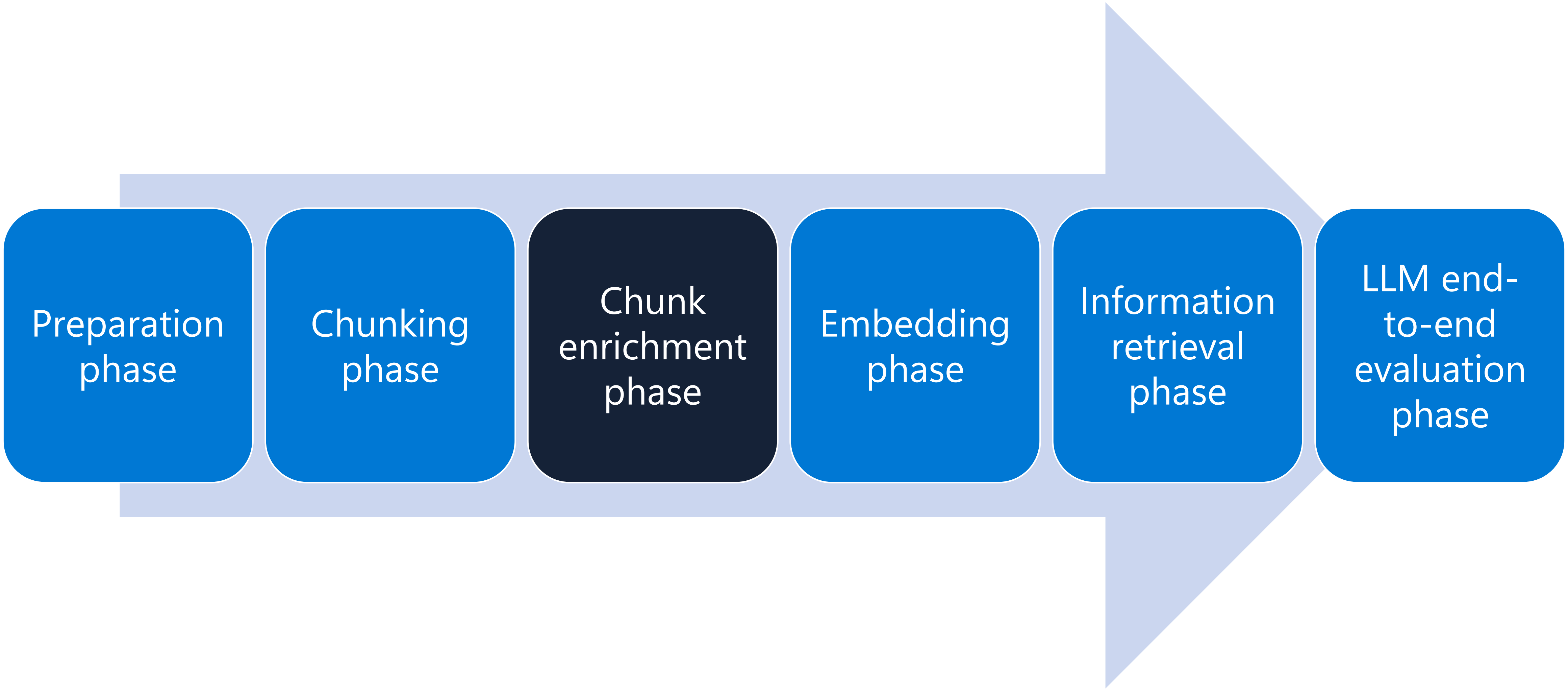
- Documents vary in the amount of structure they have. Some documents, like government forms have a complex and well-known structure, such as a W-2 U.S. tax document. At the other end of the spectrum are unstructured documents like free-form notes. The degree of structure to a document type is a good starting point for determining an effective chunking approach. While there are no hard and fast rules, this section provides you with some guidelines to follow.



Document structure

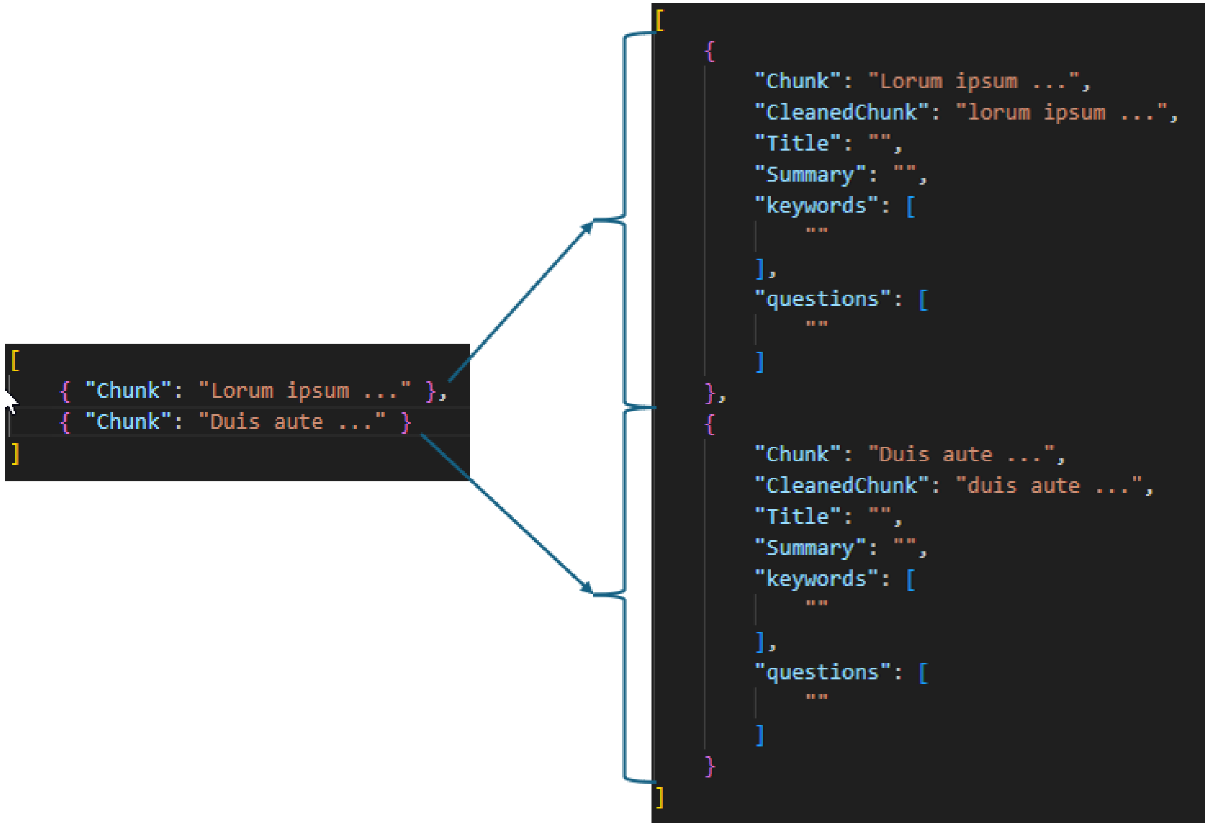
- Structured documents, sometimes referred to as fixed-format documents, have defined layouts. The data in these documents is located at fixed locations. For example, the date, or customer last name, is found in the same location in every document of the same fixed format. Examples of fixed format documents are the W-2 U.S. tax document.
- Semi-structured documents don't have a fixed format or schema, like the W-2 form, but they do offer consistency regarding format or schema. For example, all invoices aren't laid out the same, however, in general they have a consistent schema. You can expect an invoice to have an invoice number and some form of bill to and ship to name and address, among other data. A web page might not have schema consistencies, but they have similar structural or layout elements, such as body, title, H1, and p that can be used to add semantic meaning to the surrounding text.
- Inferred structure - Some documents have a structure but aren't written in markup. For these documents, the structure must be inferred. A good example is the following regulation document.
- Unstructured documents A good approach for documents with little to no structure are sentence-based or fixed-size with overlap approaches.

RAG data pipeline flow



Chunk enrichment

- Once you've broken down your documents down into a collection of chunks, the next step is to enrich each chunk by both cleaning and augmenting the chunks with metadata. Cleaning the chunks allows you to achieve better matches for semantic queries in a vector search. Adding information allows you to support searches beyond semantic searches of the chunks. Both cleaning and augmenting involve extending the schema for the chunk.



Cleaning

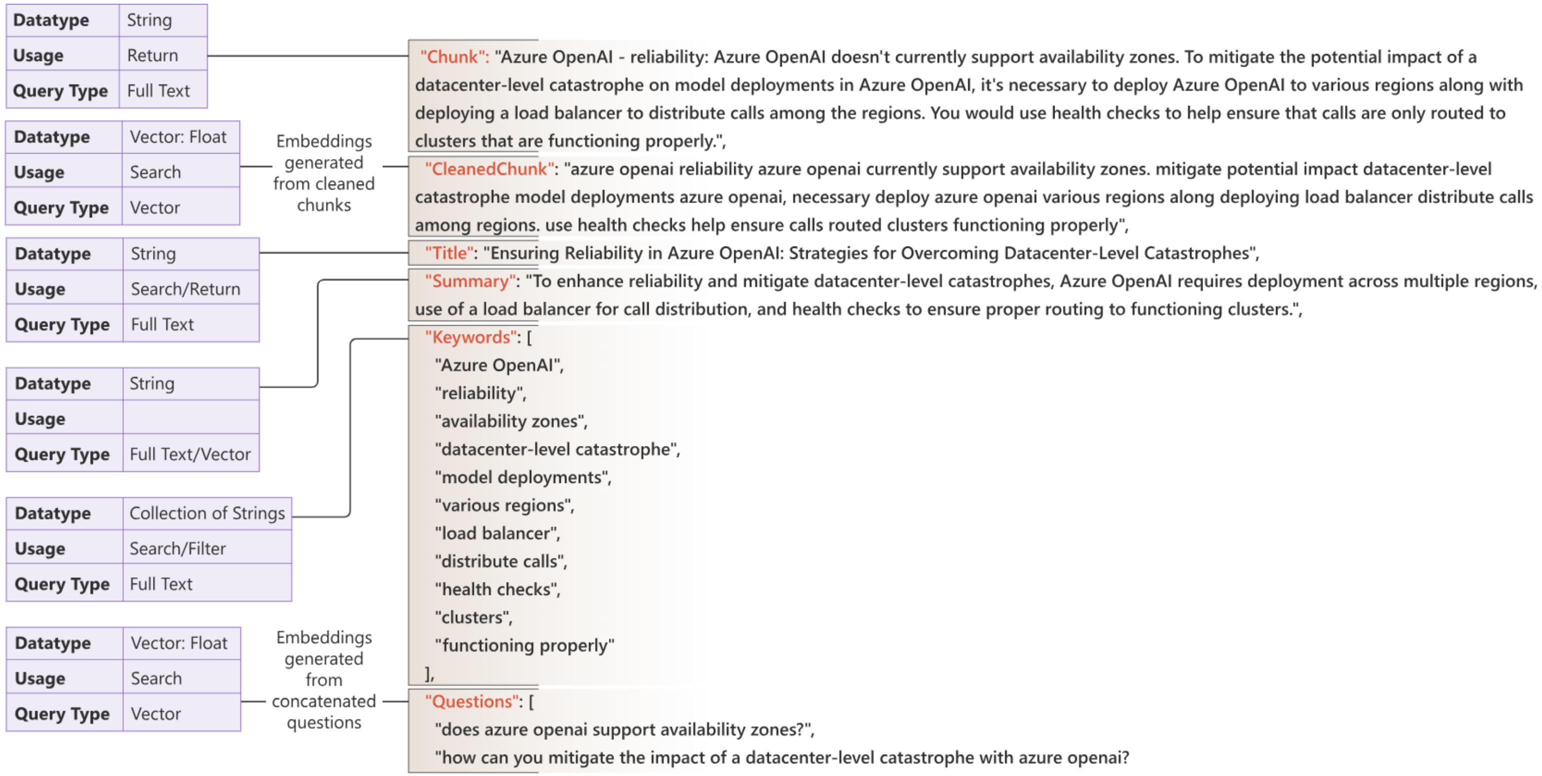
- Chunking your data supports your workload in its efforts to find the most relevant chunks, typically through vectorizing those chunks and storing them in a vector database. An optimized vector search returns only those rows in that database which have the closest semantic matches to the query. The goal of cleaning the data to support closeness matches by eliminating potential differences that aren't material to the semantics of the text. The following are some common cleaning procedures.
- **You will want to return the original, uncleaned chunk as the query result, so you will add an additional field to store the cleaned and vectorized data.**

Cleaning

- Lowercasing - Lowercasing allows words that are capitalized, such as words at the beginning of a sentence, to match with those same words within a sentence. Embeddings are typically case-sensitive meaning "Cheetah" and "cheetah" would result in a different vector for the same logical word. For example, for the embedded query: "what is faster, a cheetah or a puma?" The following embedding: "cheetahs are faster than pumas" is a closer match than embedding "Cheetahs are faster than pumas." Some lowercasing strategies lowercase all words, including proper nouns, while other strategies include just lowercasing the first words in a sentence.
- Remove stop words - Stop words are words such as "a", "an" and "the" that commonly occur in sentences. You can remove stop words to reduce the dimensionality of the resulting vector. Removing stop words would allow both "a cheetah is faster than a puma" and "the cheetah is faster than the puma" to both be vectorially equal to "cheetah faster puma." However, it's important to understand that some stop words hold semantic meaning. For example, "not" might be considered a stop word, but would hold significant semantic meaning. It's important to test to see the effect of removing stop words.
- Fix spelling mistakes - A misspelled word doesn't match with the correctly spelled word in the embedding model. For example, "cheatah" (sic) isn't the same as "cheetah" in the embedding. You should fix spelling mistakes to address this challenge.
- Remove unicode characters - Removing Unicode characters can reduce noise in your chunks and reduce dimensionality. Like stop words, some Unicode characters might contain relevant information. It's important to test to understand the impact of removing Unicode characters.
- Normalization - Normalizing the text to standards such as expanding abbreviations, converting numbers to words, and expanding contractions like "I'm" to "I am" can help increase the performance of vector searches.

Augmenting chunks

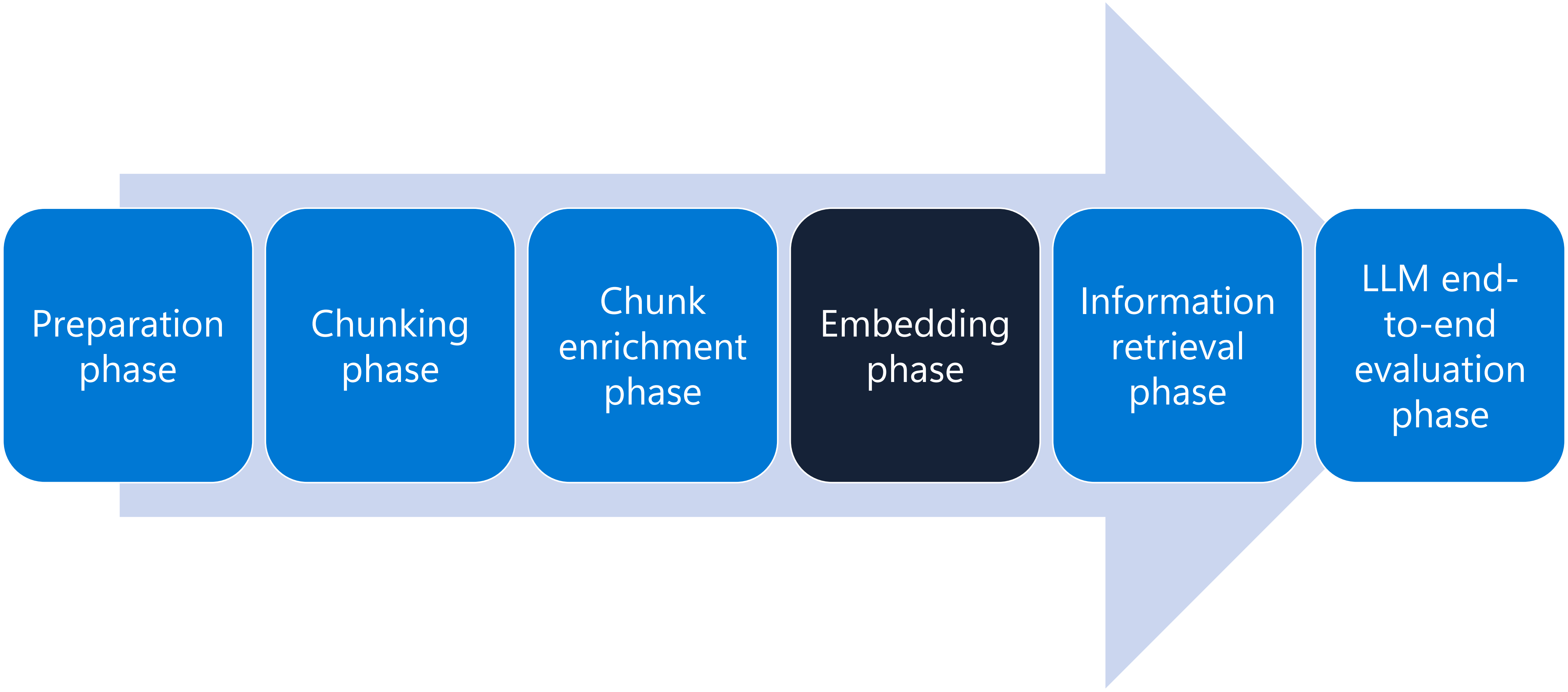
- Semantic searches against the vectorized chunks work well for some types of queries, but not as well for others. Depending upon the types of queries you need to support, you might need to augment your chunks with additional information. The additional metadata fields are all stored in the same row as your embeddings and can be used in the search solution as either filters or as part of the search.



Augmenting chunks

- The following are some common metadata fields, along with the original chunk text, some guidance about their potential uses, and tools or techniques that are commonly used to generate the metadata content.
  - **ID** - ID is a key metadata field that is used to uniquely identify a chunk. A unique ID is useful in processing to determine whether a chunk already exists in the store or not. An ID can be a hash of some key field. Tools: Hashing library
  - **Title** - A title is a useful return value for a chunk. It provides a quick summary of the content in the chunk. The summary can also be useful to query with an indexed search as it can contain keywords for matching. Tools: large language model
  - **Summary** - The summary is similar to the title in that it's a common return value and can be used in indexed searches. Summaries are generally longer than the title. Tools: large language model
  - **Rephrasing of chunk** - Rephrasing of a chunk can be helpful as a vector search field because rephrasing captures variations in language such as synonyms and paraphrasing. Tools: large language model
  - **Keywords** - Keyword searches are good for data that is noncontextual, for searching for an exact match, and when a specific term or value is important. For example, an auto manufacturer might have reviews or performance data for each of their models for multiple years. Review for product X for year 2009" is semantically like "Review for product X for 2010" and "Review for product Y for 2009." In this case, you would be better off matching on keywords for the product and year. Tools: large language model, RAKE, KeyBERT, MultiRake
  - **Entities** - Entities are specific pieces of information such as people, organizations, and locations. Like keywords, entities are good for exact match searches or when specific entities are important. Tools: SpaCy, Stanford Named Entity Recognizer (SNER, scikit-learn, Natural Language Toolkit (NLTK).
  - **Cleaned chunk text** - The cleaned chunk text. Tools: large language model
  - **Questions that the chunk can answer** - Sometimes, the query that is embedded and the chunk embedded isn't a great match. For example, the query might be small regarding the chunk size. It might be better to formulate the queries that the chunk can answer and do a vector search between the user's actual query and the preformulated queries. Tools: large language model
  - **Source** - The source of the chunk can be valuable as a return for queries. It allows the querier to cite the original source.
  - Language - The language of the chunk can be good as a filter in queries.

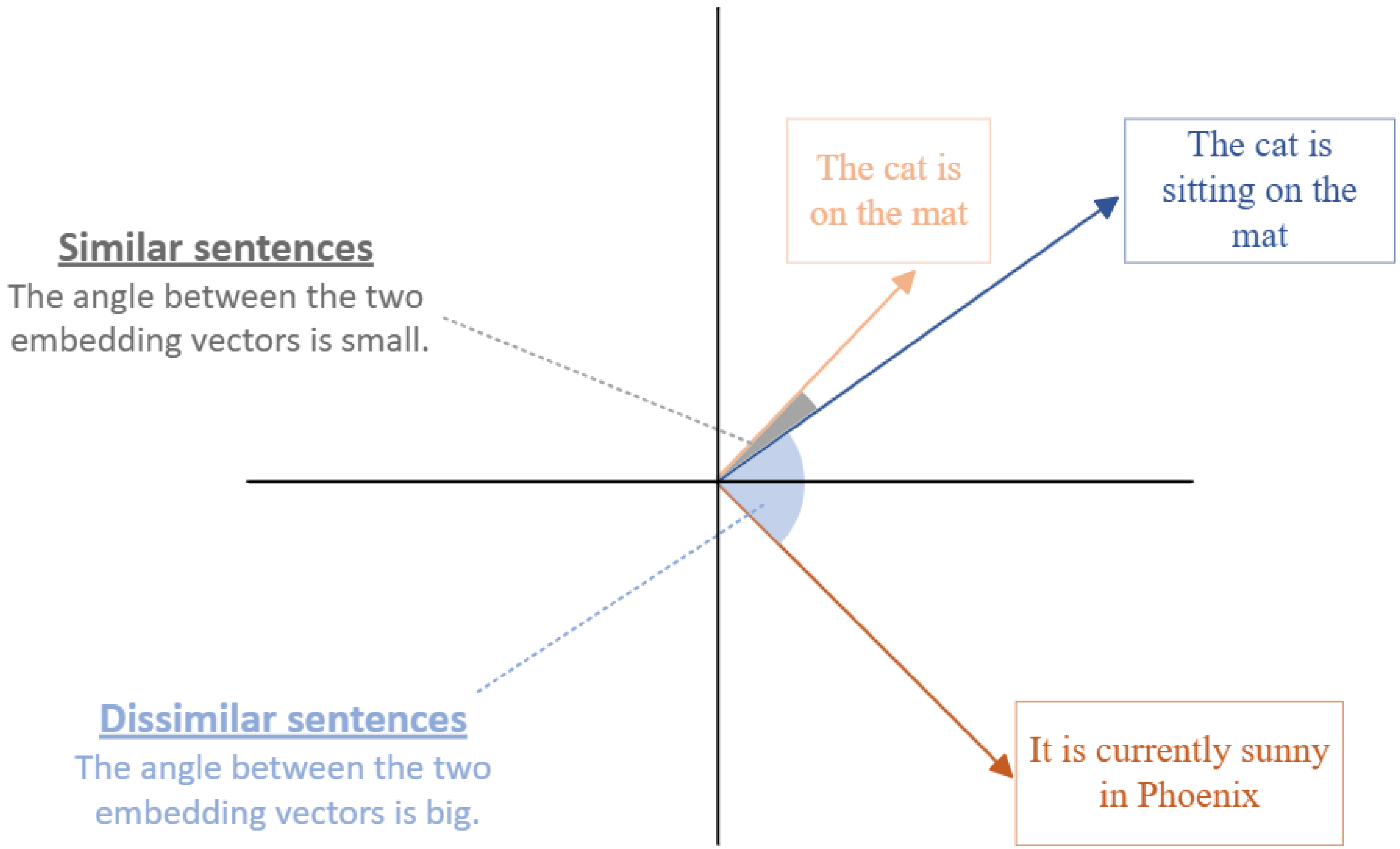
RAG data pipeline flow



Generate embeddings

- Now that you've broken your documents down into chunks and enriched the chunks, the next step is to generate embeddings for those chunks and any metadata fields over which you plan on performing vector searches.
- An embedding is a mathematical representation of an object, such as text.
- When a neural network is being trained, many representations of an object are created and each representation has connections to other objects in the network.
- An embedding is one of the representations of the object that is selected because it captures the semantic meaning of the object.

Embeddings are compared to one another using the notions of similarity and distance

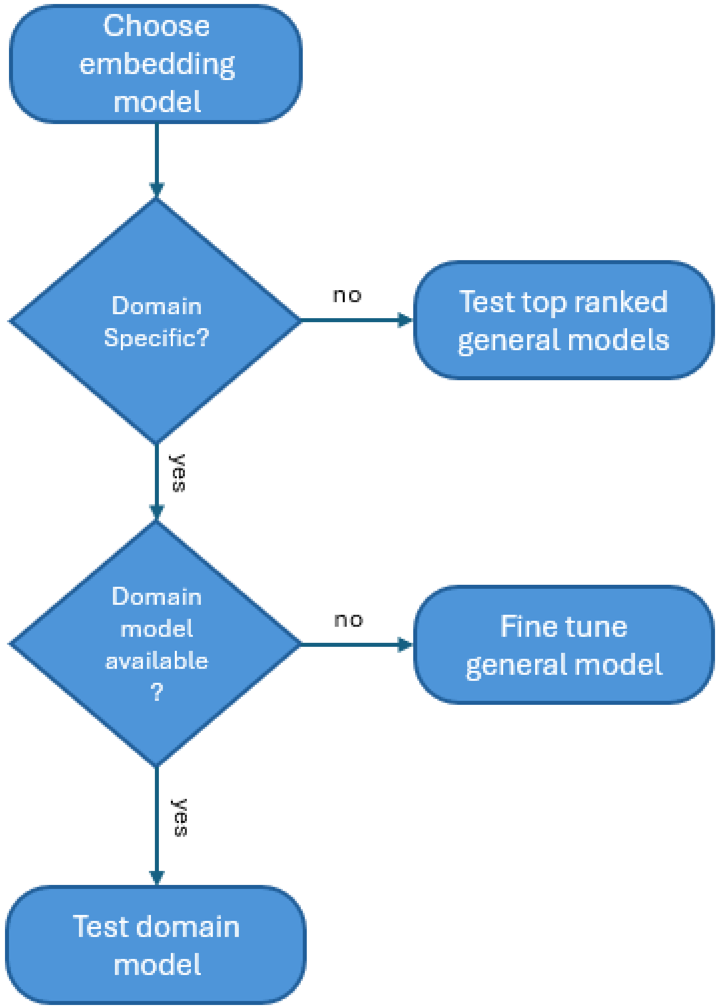


Importance of the embedding model

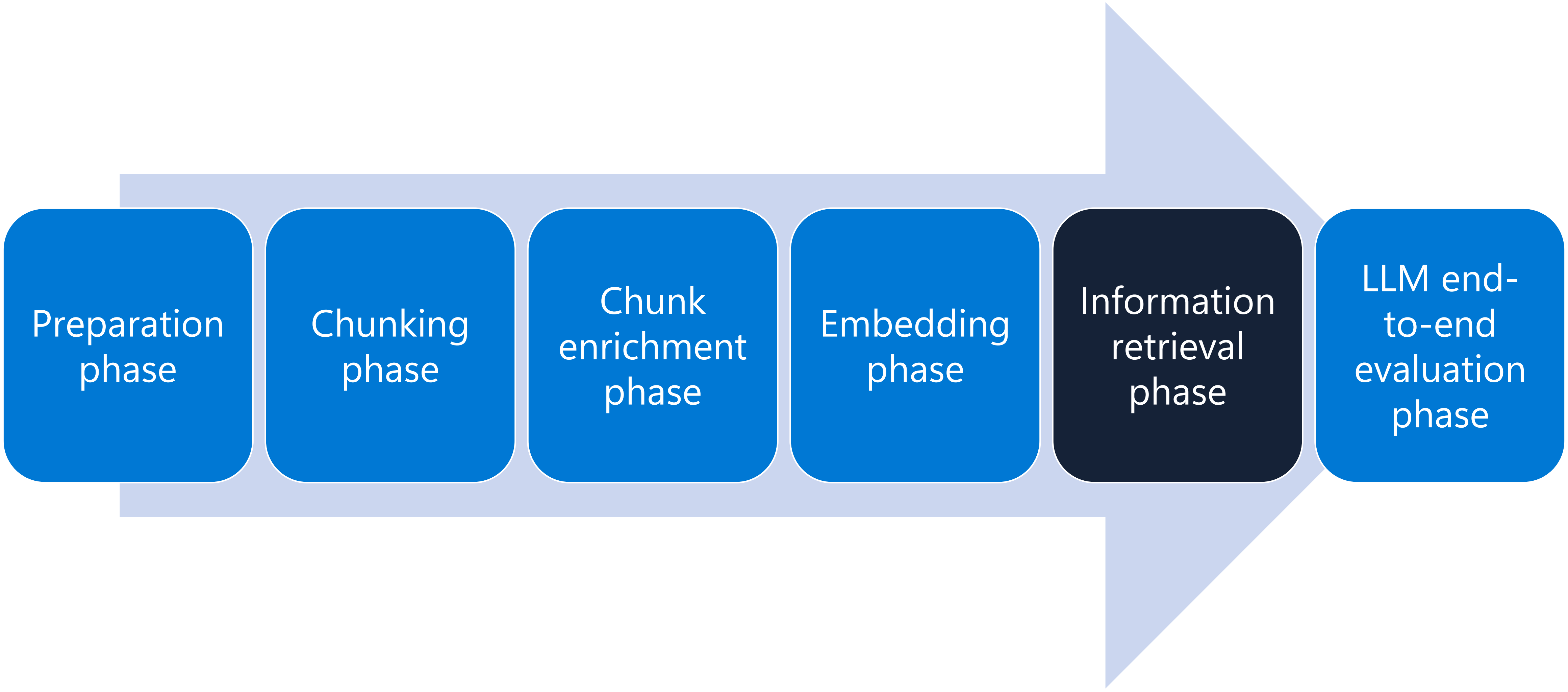
- The embedding model you choose can have a significant effect on relevancy of your vector search results. One of the key factors you must consider when choosing an embedding model is the vocabulary of the model. Every embedding model is trained with a specific vocabulary. For example, the vocabulary size of BERT is around 30,000 words.
- The vocabulary of an embedding model is important because of how embedding models treat words that aren't in their vocabulary. Even though the word isn't in its vocabulary, the model still needs to calculate a vector for it. To do this, many models break down the words into subwords, which they treat as distinct tokens or they aggregate the vectors for the subwords to create a single embedding.

Choosing an embedding model

- Determining the right embedding model for your use case is a human activity. The overlap with the embedding model's vocabulary with your data's words should be a key factor you consider when choosing your embedding model.



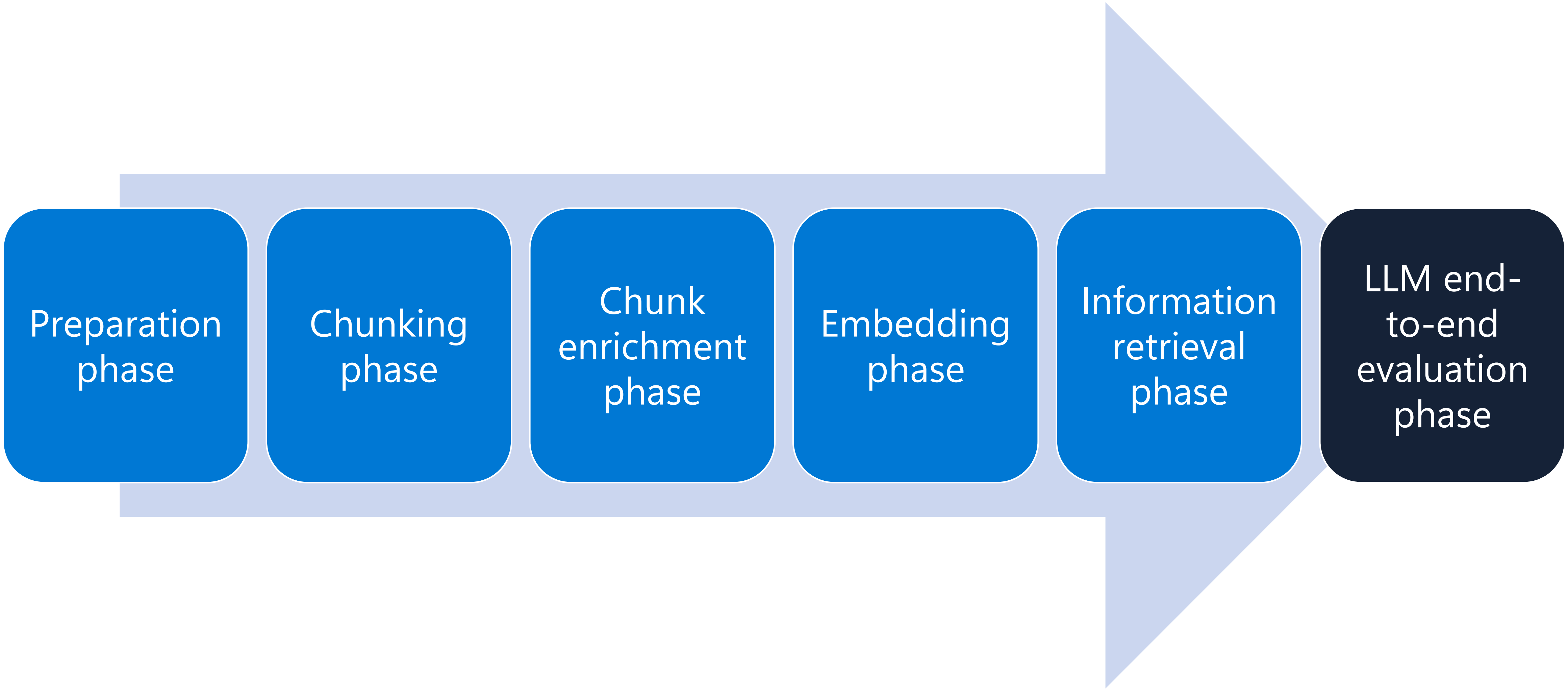
RAG data pipeline flow



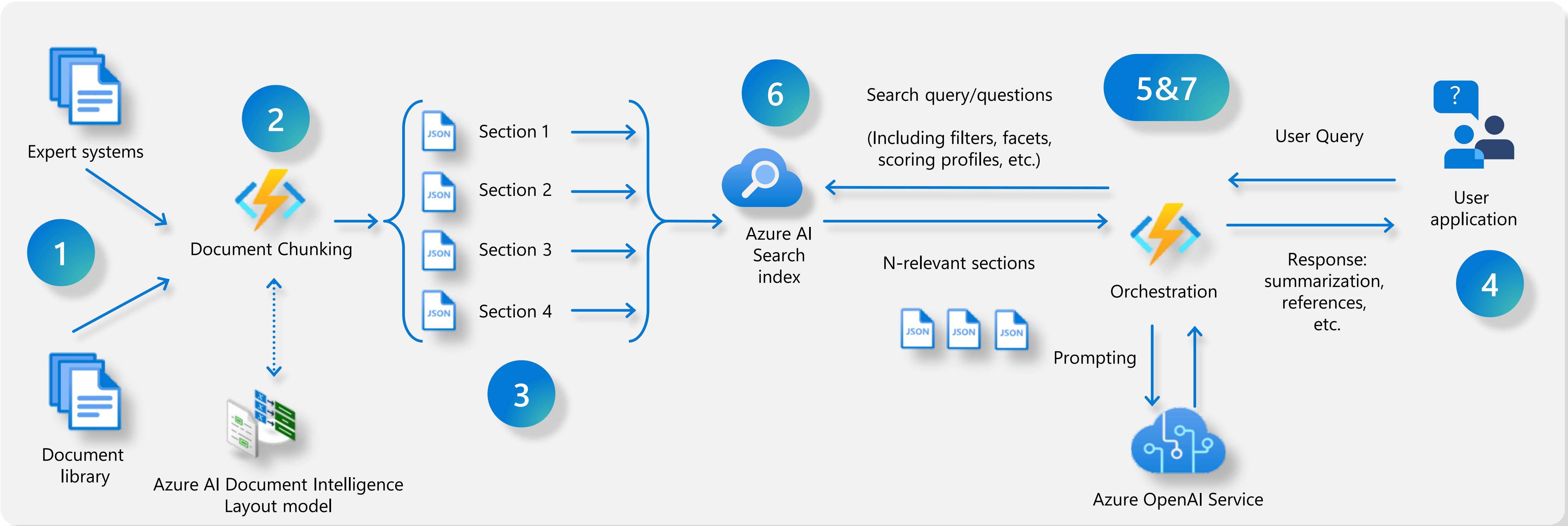
Searches

- When executing queries from your prompt orchestrator against your search store, you have many options to consider. You need to determine:
  - What type of search you're going to perform: vector or keyword or hybrid
  - Whether you're going to query against one or more columns
  - Whether you're going to manually run multiple queries, such as a keyword query and a vector search
  - Whether the query needs to be broken down into subqueries
  - Whether filtering should be used in your queries

RAG data pipeline flow



Anatomy of RAG



1. Data ingestion

Different data formats and system of records

2. Chunking

What is the best Chunking strategy?

3. Indexing

Shall I use vector embeddings data transformation, mappings?

4. User interface

Chatbot for Q&A surfaced to end users

5. Orchestration

Communication coordination and prompting—Prompt to get retriever query

6. Data retrieving

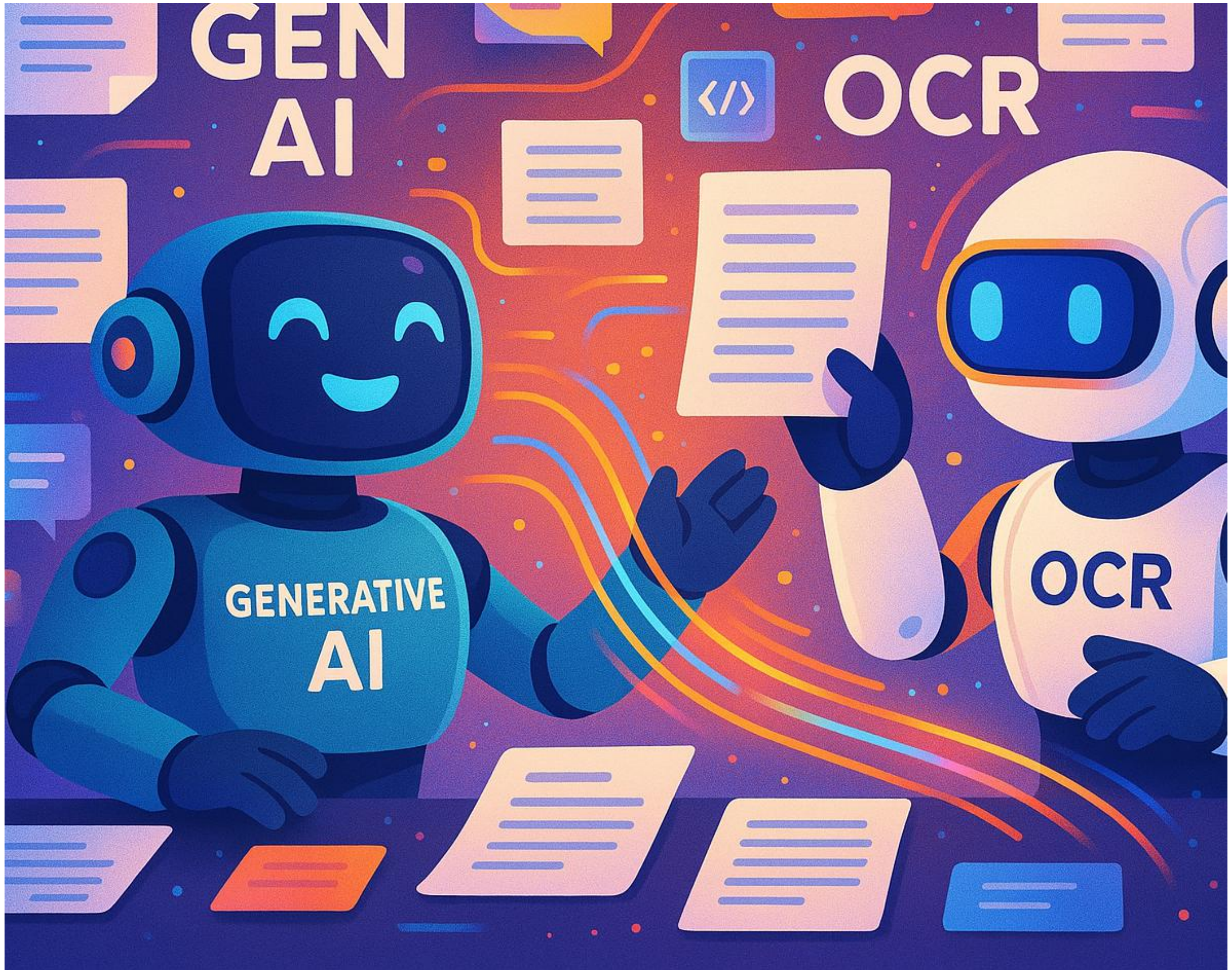
Shall I use vector, semantic, keyword or hybrid approach?

7. Orchestration

Communication coordination: create user response based on retrieve data and send to User app

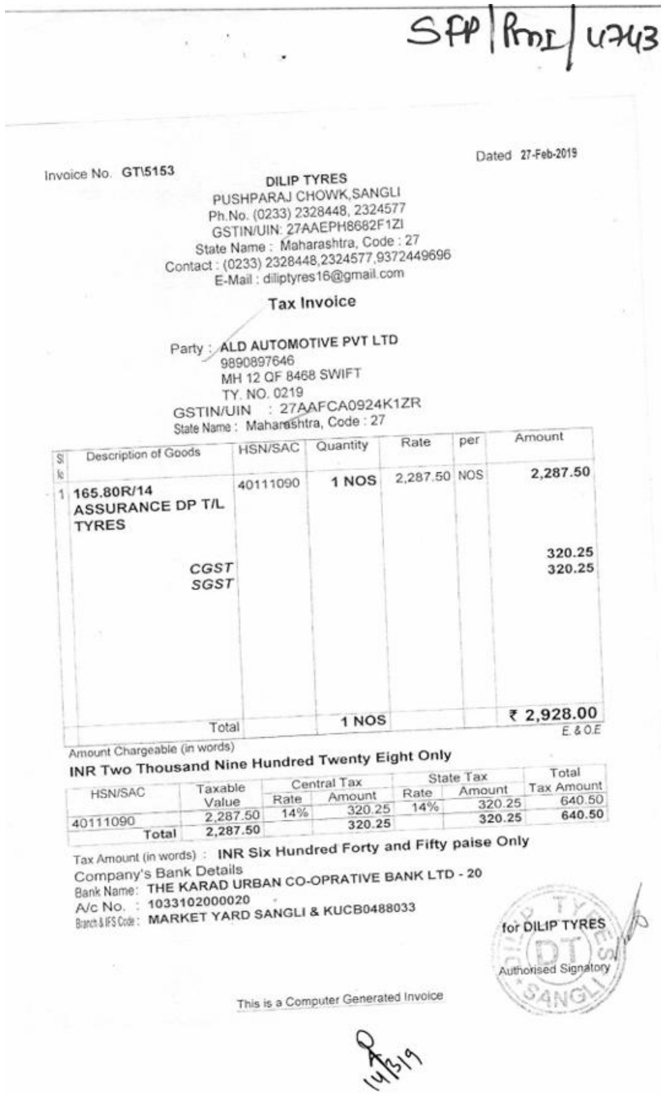
Odczytaj mnie, jeśli potrafisz:

GenAI i OCR genialny duet w świecie dokumentów



Obiekty naszego działań

[Evaluating the quality of AI document data extraction with small and large language models]



SFP | RmI | 4743

Invoice No. GT\5153

Dated 27-Feb-2019

DILIP TYRES
PUSHPARAJ CHOWK,SANGLI
Ph.No. (0233) 2328448, 2324577
GSTIN/UIN: 27AAEPH8682F1ZI
State Name : Maharashtra, Code : 27
Contact : (0233) 2328448,2324577,9372449696
E-Mail : diliptyres16@gmail.com

Tax Invoice

Party : ALD AUTOMOTIVE PVT LTD
9890897646
MH 12 QF 8468 SWIFT
TY. NO. 0219
GSTIN/UIN : 27AAFCA0924K1ZR
State Name : Maharashtra, Code : 27

Sl No	Description of Goods	HSN/SAC	Quantity	Rate	per	Amount
1	165.80R/14 ASSURANCE DP T/L TYRES	40111090	1 NOS	2,287.50	NOS	2,287.50
	CGST					320.25
	SGST					320.25
	Total		1 NOS			₹ 2,928.00

E. & O.E

Amount Chargeable (in words)
INR Two Thousand Nine Hundred Twenty Eight Only

HSN/SAC	Taxable Value	Central Tax		State Tax		Total Tax Amount
		Rate	Amount	Rate	Amount	
40111090	2,287.50	14%	320.25	14%	320.25	640.50
Total	2,287.50		320.25		320.25	640.50

Tax Amount (in words) : INR Six Hundred Forty and Fifty paise Only

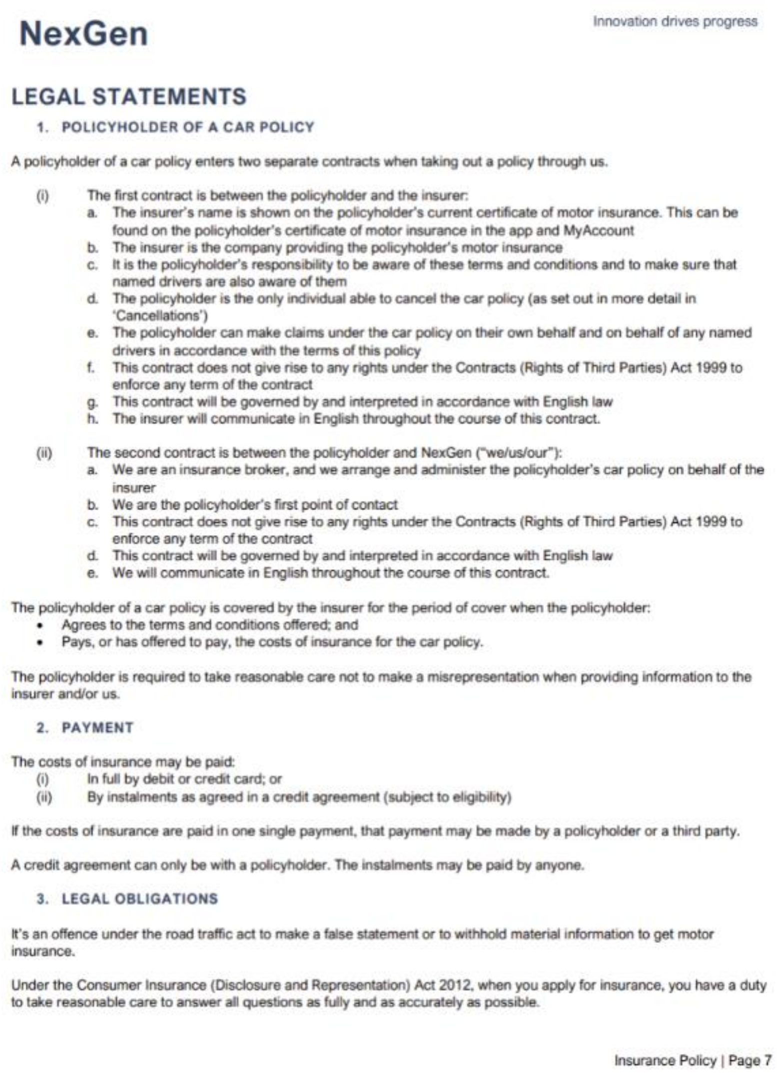
Company's Bank Details
Bank Name: THE KARAD URBAN CO-OPRATIVE BANK LTD - 20
A/c No. : 1033102000020
Branch & IFS Code : MARKET YARD SANGLI & KUCB0488033

for DILIP TYRES
Authorised Signatory

This is a Computer Generated Invoice

14/3/19

Faktury o prostych i złożonych układach, z odręcznymi podpisami, zakrytymi fragmentami i notatkami na marginesach.



NexGen

Innovation drives progress

LEGAL STATEMENTS

1. POLICYHOLDER OF A CAR POLICY

A policyholder of a car policy enters two separate contracts when taking out a policy through us.

(i) The first contract is between the policyholder and the insurer:
a. The insurer's name is shown on the policyholder's current certificate of motor insurance. This can be found on the policyholder's certificate of motor insurance in the app and MyAccount
b. The insurer is the company providing the policyholder's motor insurance
c. It is the policyholder's responsibility to be aware of these terms and conditions and to make sure that named drivers are also aware of them
d. The policyholder is the only individual able to cancel the car policy (as set out in more detail in 'Cancellations')
e. The policyholder can make claims under the car policy on their own behalf and on behalf of any named drivers in accordance with the terms of this policy
f. This contract does not give rise to any rights under the Contracts (Rights of Third Parties) Act 1999 to enforce any term of the contract
g. This contract will be governed by and interpreted in accordance with English law
h. The insurer will communicate in English throughout the course of this contract.

(ii) The second contract is between the policyholder and NexGen ("we/us/our"):
a. We are an insurance broker, and we arrange and administer the policyholder's car policy on behalf of the insurer
b. We are the policyholder's first point of contact
c. This contract does not give rise to any rights under the Contracts (Rights of Third Parties) Act 1999 to enforce any term of the contract
d. This contract will be governed by and interpreted in accordance with English law
e. We will communicate in English throughout the course of this contract.

The policyholder of a car policy is covered by the insurer for the period of cover when the policyholder:
- Agrees to the terms and conditions offered; and
- Pays, or has offered to pay, the costs of insurance for the car policy.

The policyholder is required to take reasonable care not to make a misrepresentation when providing information to the insurer and/or us.

2. PAYMENT

The costs of insurance may be paid:
(i) In full by debit or credit card; or
(ii) By instalments as agreed in a credit agreement (subject to eligibility)

If the costs of insurance are paid in one single payment, that payment may be made by a policyholder or a third party.

A credit agreement can only be with a policyholder. The instalments may be paid by anyone.

3. LEGAL OBLIGATIONS

It's an offence under the road traffic act to make a false statement or to withhold material information to get motor insurance.

Under the Consumer Insurance (Disclosure and Representation) Act 2012, when you apply for insurance, you have a duty to take reasonable care to answer all questions as fully and as accurately as possible.

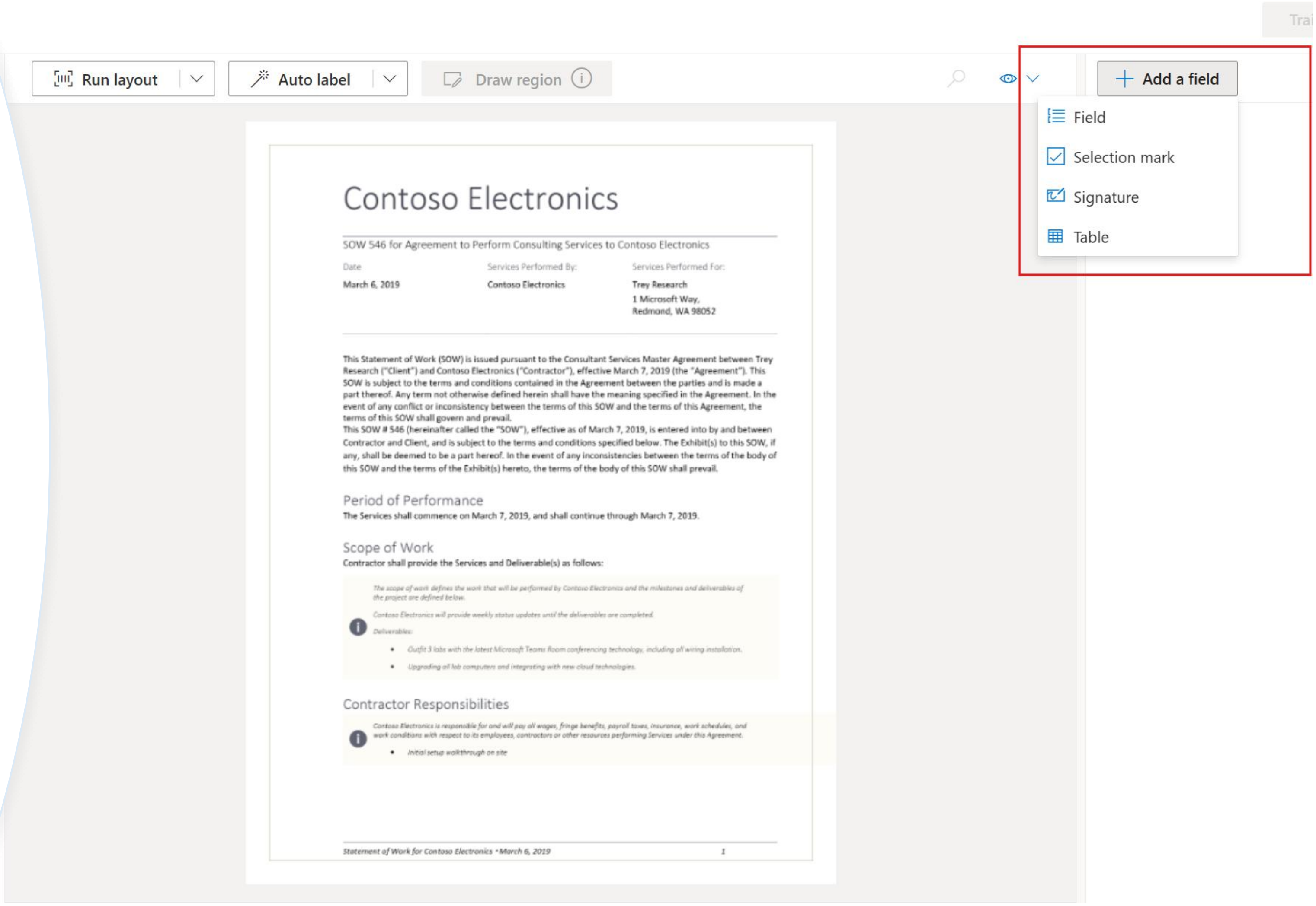
Insurance Policy | Page 7

Ponad 10-stronicowe polisy zawierająca dane strukturalne i niestrukturalne, w tym język branżowy i wywnioskowane informacje.

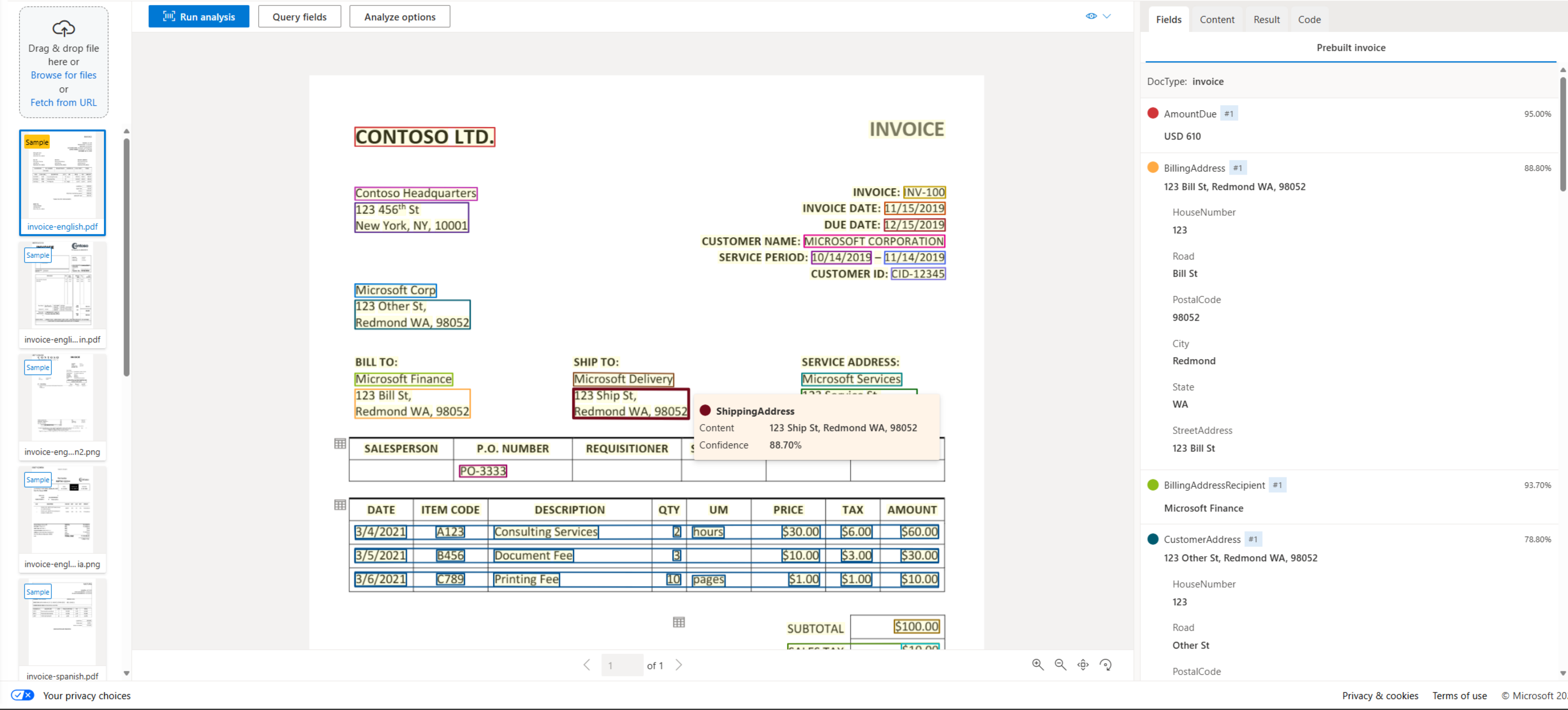
Obiekt naszego działania – oczekiwany wynik


```
{
  "customer_name": "ALD AUTOMOTIVE PVT LTD",
  "customer_tax_id": "27AAFCA0924K1ZR",
  "customer_address": {
    "street": null,
    "city": null,
    "state": "Maharashtra",
    "postal_code": null,
    "country": "India"
  },
  "shipping_address": null,
  "purchase_order": null,
  "invoice_id": "GT\\5153",
  "invoice_date": "2019-02-27",
  "due_date": null,
  "vendor_name": "DILIP TYRES",
  "vendor_address": {
    "street": "Pushparaj Chowk",
    "city": "Sangli",
    "state": "Maharashtra",
    "postal_code": null,
    "country": "India"
  }
  "vendor_tax_id": "27AAEPH8682F1ZI",
  "remittance_address": null,
  "subtotal": {
    "currency_code": "INR",
    "amount": 2287.5
  },
  "total_discount": null,
  "total_tax": {
    "currency_code": "INR",
    "amount": 640.5
  },
  "invoice_total": {
    "currency_code": "INR",
    "amount": 2928.0
  },
  "payment_term": null,
  "items": [
    {
      "product_code": "40111090",
      "description": "165.80R/14 ASSURANCE DP T/L TYRES",
      "quantity": 1,
      "tax": {
        "currency_code": "INR",
        "amount": 640.5
      },
      "unit_price": {
        "currency_code": "INR",
        "amount": 2287.5
      },
      "total": {
        "currency_code": "INR",
        "amount": 2928.0
      }
    }
  ],
  "customer_signature": null
}
```

The old Way


Run layout
Auto label
Draw region
Add a field
Field
Selection mark
Signature
Table
Contoso Electronics
SOW 546 for Agreement to Perform Consulting Services to Contoso Electronics
Date
Services Performed By:
Services Performed For:
March 6, 2019
Contoso Electronics
Trey Research
1 Microsoft Way,
Redmond, WA 98052
Period of Performance
The Services shall commence on March 7, 2019, and shall continue through March 7, 2019.
Scope of Work
Contractor shall provide the Services and Deliverable(s) as follows:
Contractor Responsibilities
Statement of Work for Contoso Electronics • March 6, 2019
1

A jak już się nauczy...



Fast & Furious

Deployment * + Create new deployment

gpt-4o (version:2024-11-20)

Give the model instructions and context

Extract data from the image in JSON format using the following schema:

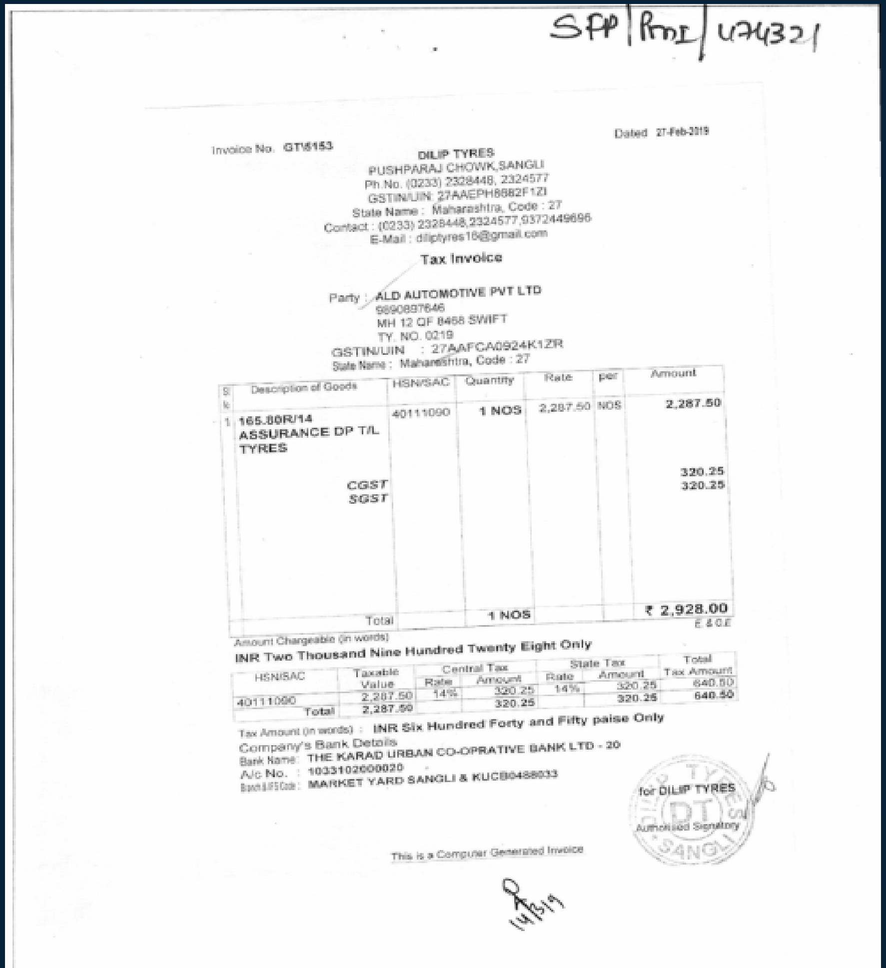

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Invoice",
  "type": "object",
  "properties": {
    "customer_name": {
      "type": "string"
    },
    "customer_tax_id": {
      "type": "string"
    },
```


Apply changes | Generate system prompt

+ Add section

> Add your data

> Parameters



SPP | RMI | 474321

Invoice No. GT\6153

Dated 27-Feb-2019

DILIP TYRES
PUSHPARAJ CHOWK,SANGLI
Ph.No. (0233) 2328448, 2324577
GSTIN/UIN: 27AAEPH8882F1ZI
State Name : Maharashtra, Code : 27
Contact : (0233) 2328448,2324577,9372449696
E-Mail : diliptyres16@gmail.com

Tax Invoice

Party : ALD AUTOMOTIVE PVT LTD
9890897646
MH 12 QF 8468 SWIFT
TY. NO. 0219
GSTIN/UIN : 27AAFCA0924K1ZR
State Name : Maharashtra, Code : 27

Sl. No.	Description of Goods	HSN/SAC	Quantity	Rate	per	Amount
1	165.80R/14 ASSURANCE DP T/L TYRES	40111090	1 NOS	2,287.50	NOS	2,287.50
	CGST					320.25
	SGST					320.25
	Total		1 NOS			₹ 2,928.00

E & O.E

Amount Chargeable (in words)
INR Two Thousand Nine Hundred Twenty Eight Only

HSN/SAC	Taxable Value	Central Tax Rate	Central Tax Amount	State Tax Rate	State Tax Amount	Total Tax Amount
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Company's Bank Details
Bank Name: THE KARAD URBAN CO-OPRATIVE BANK LTD - 20
A/c No. : 1033102000020
Branch & IFS Code: MARKET YARD SANGLI & KUCB0488033

for DILIP TYRES
Authorised Signatory

This is a Computer Generated Invoice

14/3/19

Azure AI Foundry | gpt-4o-2024-11-20 AI-generated content may be incorrect


```
{
  "customer_name": "ALD AUTOMOTIVE PVT LTD",
  "customer_tax_id": "27AAFCA0924K1ZR",
  "customer_address": {
    "street": "MH 12 DF 8468 SWIFT, TY. NO. 0219",
    "city": null,
    "state": "Maharashtra",
    "postal_code": null,
```

Tymczasem w prawdziwej aplikacji

- Wykorzystujemy Structured Outputs i definiujemy klasę oczekiwanego formatu, np. za pomocą pydantic.

```
class Invoice(BaseModel):
    """
    A class representing an invoice.
    """

    customer_name: Optional[str] = Field(
        description='Name of the customer being invoiced, e.g. Microsoft Corp'
    )
    customer_tax_id: Optional[str] = Field(
        description='Government tax ID of the customer, e.g. 765432-1'
    )        You, 2 days ago • initial commit
    customer_address: Optional[InvoiceAddress] = Field(
        description='Full mailing address of the customer, e.g. 123 Other St.,
    )
```

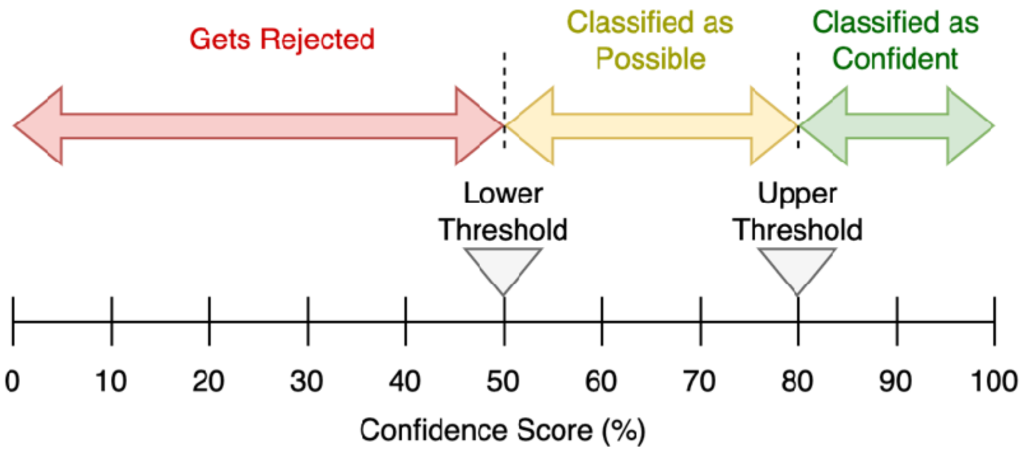
Tymczasem w prawdziwej aplikacji

- Wykorzystujemy Structured Outputs i definiujemy klasę oczekiwanego formatu, np. za pomocą pydantic.
- Konwertujemy strony do formatu PNG i tworzymy kolekcję messages

```
[
    {
        "role": "system",
        "content": "You are an AI assistant that extracts data from documents."
    },
    {
        "type": "image_url",
        "image_url": {
            "url": "data:image/jpeg;base64,BASE64_IMAGE_PLACEHOLDER",
        }
    }
]
```

Podstawowe wyzwanie AI: niezawodność

- Systemy sztucznej inteligencji nie gwarantują 100% dokładności wyników.
- Niektóre modele uczenia maszynowego oferują skalibrowane oceny wiarygodności swoich przewidywań.
- Wyniki ufności pozwalają ocenić, kiedy model jest godny zaufania.
- W praktycznych zastosowaniach AI wykorzystuje się poziomy ufności do podejmowania decyzji o akceptacji przewidywań.
- Jeśli wynik ufności jest niski, decyzja lub analiza przekazywana jest do weryfikacji przez człowieka.



Gdzie w LLM poziom ufności?

Dec 20, 2023

Using logprobs

James Hills (OpenAI), Shyamal Anadkat (OpenAI)

Open in Github

View as Markdown

Czym jest logprobs?

logprobs boolean or null Optional Defaults to false

Whether to return log probabilities of the output tokens or not. If true, returns the log probabilities of each output token returned in the `content` of `message`.

https://platform.openai.com/docs/api-reference/chat/create#chat-create-logprobs

logprobs array or null

The log probabilities of the transcription.

^ Hide properties

bytes array

The bytes that were used to generate the log probability.

logprob number

The log probability of the token.

token string

The token that was used to generate the log probability.

p = exp(logprob)

A jak to się ma do danych:

```
...
 "customer_tax_id": "GB34567890123",
...
```

A jak to się ma do danych:

```
…
 "customer_tax_id": "GB34567890123",
…
```

GB34567890123


```
[
 {
  "token": "GB",
  "logprob": -3.5313153e-06,
  "bytes": [71, 66]
 },
 {
  "token": "345",
  "logprob": -1.7432603e-06,
  "bytes": [51, 52, 53]
 },
 {
  "token": "678",
  "logprob": -3.1281633e-07,
  "bytes": [54, 55, 56]
 },
 {
  "token": "901",
  "logprob": -9.0883464e-07,
  "bytes": [57, 48, 49]
 },
 {
  "token": "23",
  "logprob": -2.6968896e-06,
  "bytes": [50, 51]
 }
]
```

A jak to się ma do danych:

…
"customer_tax_id": "GB34567890123",
…

GB34567890123


```
[
 {
  "token": "GB",
  "logprob": -3.5313153e-06,
  "bytes": [71, 66]
 },
 {
  "token": "345",
  "logprob": -1.7432603e-06,
  "bytes": [51, 52, 53]
 },
 {
  "token": "678",
  "logprob": -3.1281633e-07,
  "bytes": [54, 55, 56]
 },
 {
  "token": "901",
  "logprob": -9.0883464e-07,
  "bytes": [57, 48, 49]
 },
 {
  "token": "23",
  "logprob": -2.6968896e-06,
  "bytes": [50, 51]
 }
]
```


p = exp(SumOrAvg(logprob))

A jak to się ma do danych:

...
"customer_tax_id": {
  "confidence": 0.9999837894431287,
  "value": "GB34567890123"
},

...

GB34567890123

p = exp(SumOrAvg(logprob))

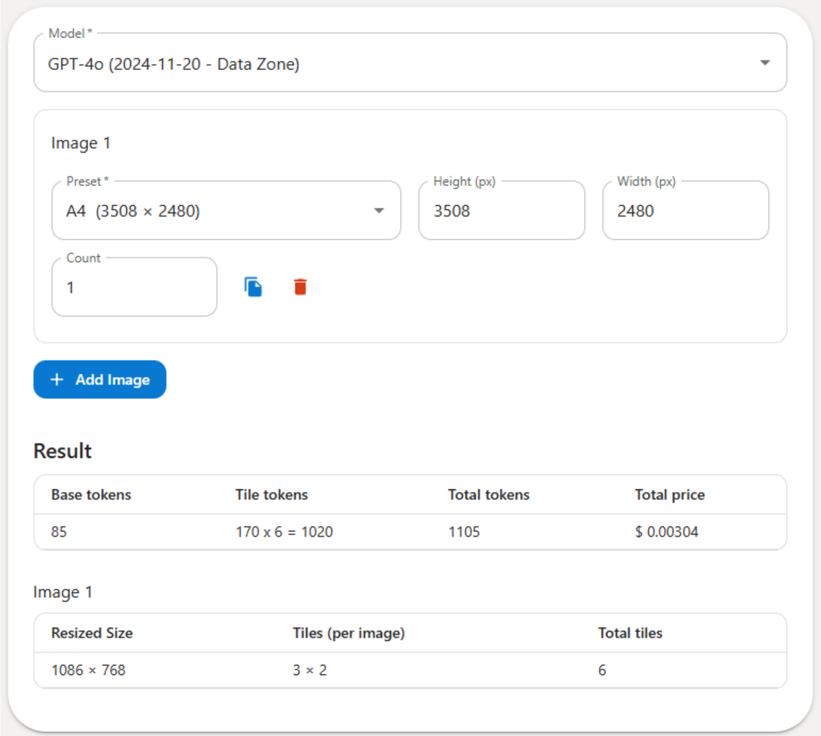
[
 {
  "token": "GB",
  "logprob": -3.5313153e-06,
  "bytes": [71, 66]
 },
 {
  "token": "345",
  "logprob": -1.7432603e-06,
  "bytes": [51, 52, 53]
 },
 {
  "token": "678",
  "logprob": -3.1281633e-07,
  "bytes": [54, 55, 56]
 },
 {
  "token": "901",
  "logprob": -9.0883464e-07,
  "bytes": [57, 48, 49]
 },
 {
  "token": "23",
  "logprob": -2.6968896e-06,
  "bytes": [50, 51]
 }
]

Wyniki

	Model	Accuracy (95th)	Confidence (95th)	Speed (95th)	Est. Cost (1,000 pages)
Faktury	GPT-4o	98.99%	99.85%	22.80s	$7.45
	GPT-4o Mini	79.31%	99.76%	56.71s	$8.02
Dokumenty ubezpieczeniowe	GPT-4o	97.04%	98.71%	66.24s	$2.31
	GPT-4o Mini	67.25%	98.73%	83.01s	$5.67

Evaluating the quality of AI document data extraction with small and large language models 10 dokumentów z każdej klasy https://jamesmcroft.github.io/openai-image-token-calculator/

Drobnym drukiem



- Model, jak to model może odjechać w….
- Maksymalnie 50 stron (ograniczenie API)
- Realnie lepiej <=10
- 1 obraz <20MB
- Koszty – zależne od rozdzielczości przesłanych plików
- Zgodnie z FAQ na stronie OpenAI:

4. **Rotation**: The model may misinterpret rotated / upside-down text or images.

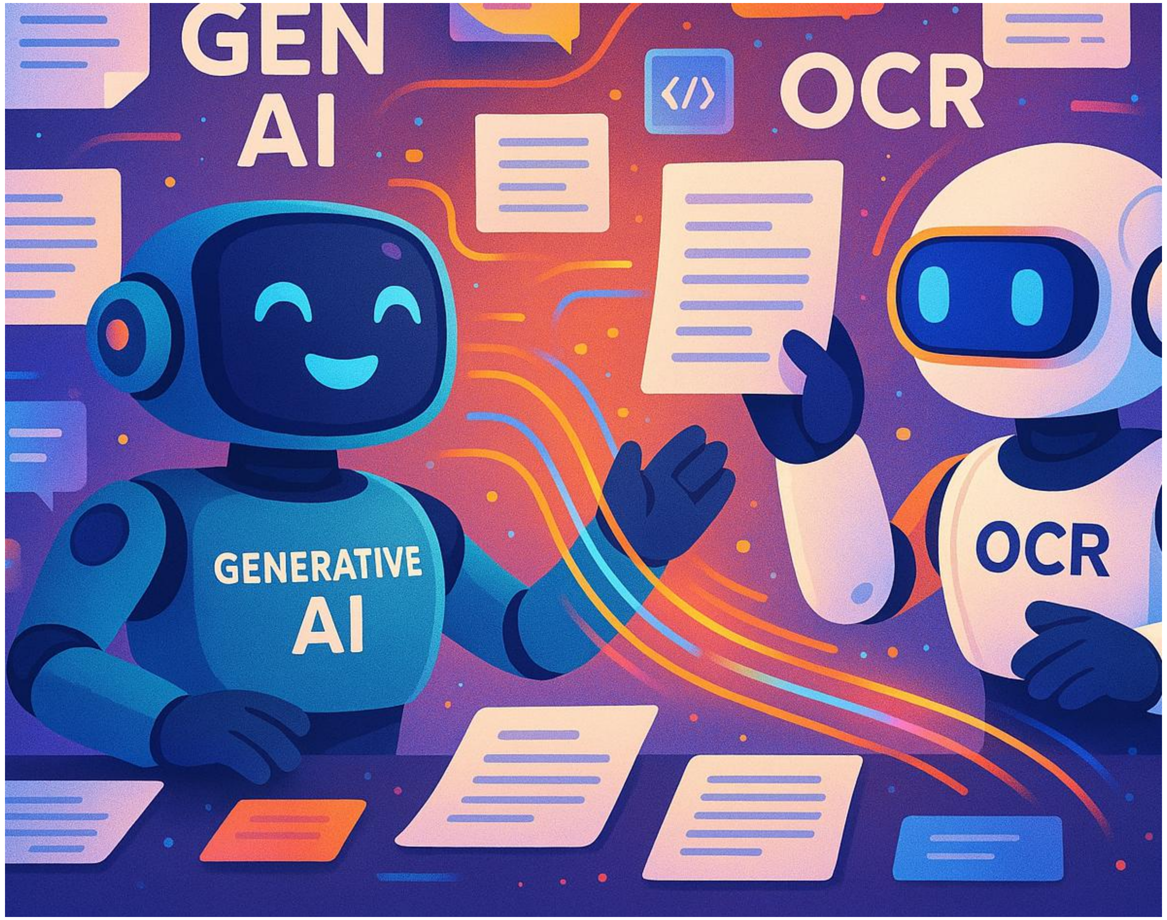
https://help.openai.com/en/articles/8400551-chatgpt-image-inputs-faq

One more thing...

Can we do better?

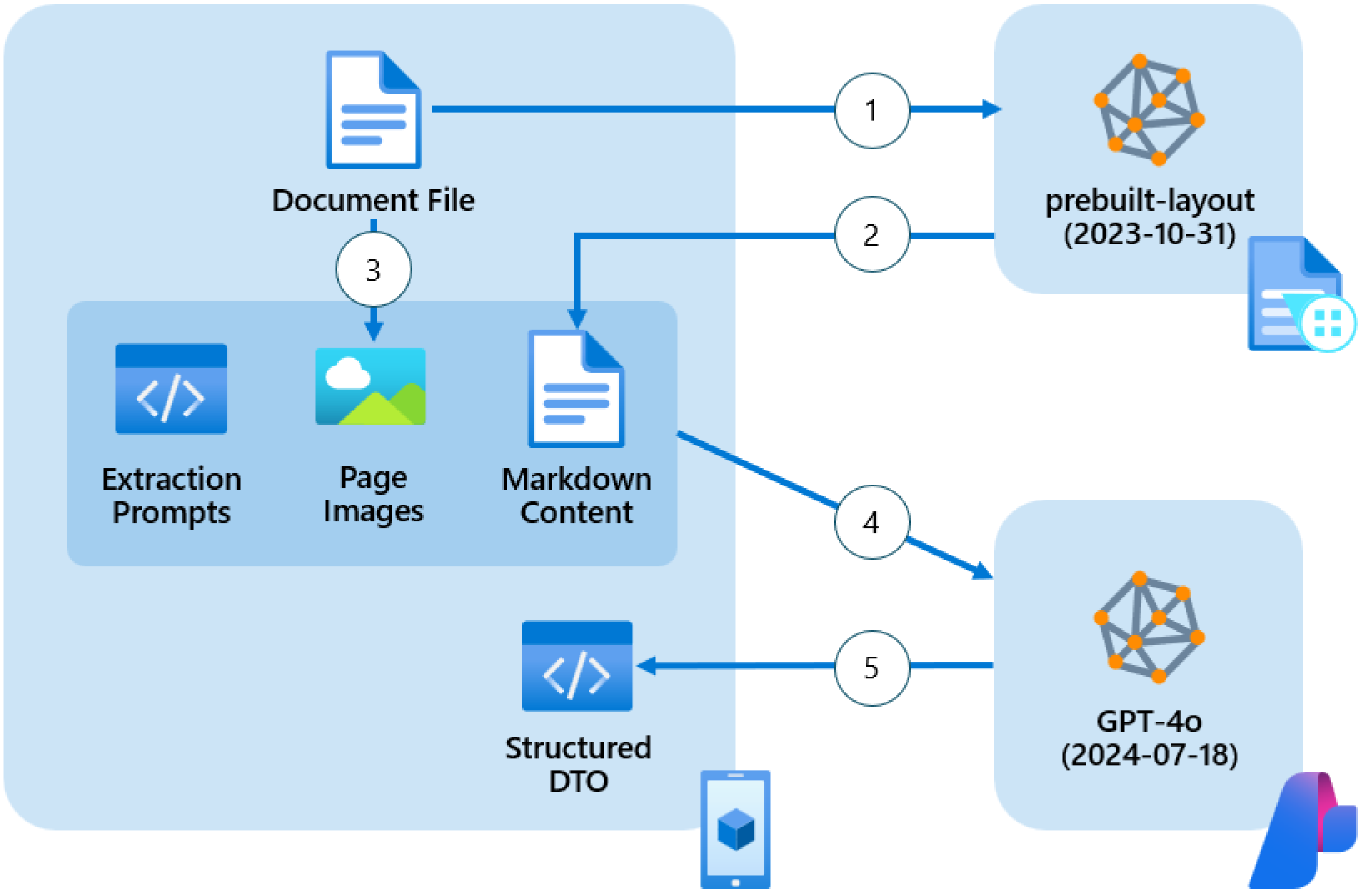
Połączmy siły obu rozwiązań

- Wysoka elastyczność, niezawodność w przypadku wstępnie przetworzonych danych wejściowych
- Programowalny za pomocą „jednego” promptu



- Dokładne, niezawodne i spójne przetwarzanie
- Tani i skalowalny do przetwarzania masowego

Połączmy siły obu rozwiązań



Co wyciągamy z AI Document Intelligence?

- Kąt obrotu strony!
- Treść dokumentu w formacie Markdown
- Ramki ograniczające tekst
- Poziom ufności dla tekstu

Co wysyłamy do LLM

- Konwertuj strony na format PNG
- Obróć jeśli:

  result.pages[page_num].angle>10

  or result.pages[page_num].angle<-10
- Utwórz kolekcję:

```json
[
    {
        "role": "system",
        "content": "You are an AI assistant that extracts data from documents."
    },
    {
        "type":"text",
        "content": "PAGE1_MARKDOWN_PLACEHOLDER"
    },
    {
        "type": "image_url",
        "image_url": {
            "url": "data:image/jpeg;base64,BASE64_IMAGE1_PLACEHOLDER"
        }
    },
    {
        "type":"text",
        "content": "PAGE2_MARKDOWN_PLACEHOLDER"
    },
    {
        "type": "image_url",
        "image_url": {
            "url": "data:image/jpeg;base64,BASE64_IMAGE2_PLACEHOLDER"
        }
    }
]
```

Wyniki

		Model	Accuracy (95th)	Confidence (95th)	Speed (95th)	Est. Cost (1,000 pages)
VISION	Faktury	GPT-4o	98.99%	99.85%	22.80s	$7.45
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		GPT-4o Mini	67.25%	98.73%	83.01s	$5.67
VISION + MARKDOWN	Faktury	GPT-4o	96.60%	99.82%	22.25s	$19.47
		GPT-4o Mini	91.84%	99.99%	56.69s	$18.14
	Dokumenty ubezpieczeniowe	GPT-4o	100%	99.35%	68.93s	$13.96
		GPT-4o Mini	82.99%	99.16%	101.89s	$15.71

Evaluating the quality of AI document data extraction with small and large language models

10 dokumentów z każdej klasy

One more thing…

Can we do better?

Dodatkowa weryfikacja danych


```
...
 "customer_tax_id": {
     "confidence": 0.99998378944431287,
     "value": "GB34567890123"
   },

...
```


'lines'

...

{'content': 'GB34567890123', 'polygon': [5.8191, 1.9572, 6.8896, 1.9567, 6.8894, 2.1057, 5.8192, 2.1065], 'confidence': 0.994, 'span': {'offset': 137, 'length': 13}}

...

Dodatkowa weryfikacja danych

```
...
 "customer_tax_id": {
     "confidence": 0.99998378944431287,
     "value": "GB34567890123"
   },

...
```

'lines'

```
...
{'content': 'GB34567890123', 'polygon': [5.8191, 1.9572, 6.8896, 1.9567,
6.8894, 2.1057, 5.8192, 2.1065], 'confidence': 0.994, 'span': {'offset': 137,
'length': 13}}
...
```

Dodatkowa weryfikacja danych

...
"customer_tax_id": {
    "confidence": 0.99998378944431287,
    "value": "GB34567890123"
  },

...

'lines'

...

{'content': 'GB34567890123', 'polygon': [5.8191, 1.9572, 6.8896, 1.9567, 6.8894, 2.1057, 5.8192, 2.1065], 'confidence': 0.994, 'span': {'offset': 137, 'length': 13}}

...

Dodatkowa weryfikacja danych

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...
```

'lines'

```
...
{'content': 'GB34567890123', 'polygon': [5.8191, 1.9572, 6.8896, 1.9567, 6.8894, 2.1057, 5.8192, 2.1065], 'confidence': 0.994, 'span': {'offset': 137, 'length': 13}}
...
```

Invoice No. GT\5153

Dated 27-Feb-2019

DILIP TYRES
PUSHPARAJ CHOWK,SANGLI
Ph.No. (0233) 2328448, 2324577
GSTIN/UIN: 27AAEPH8682F1ZI
State Name : Maharashtra, Code : 27
Contact : (0233) 2328448,2324577,9372449696
E-Mail : diliptyres16@gmail.com

Tax Invoice

Party : **ALD AUTOMOTIVE PVT LTD**
9890897646
MH 12 QF 8468 SWIFT
TY. NO. 0219
GSTIN/UIN : 27AAFCA0924K1ZR
State Name : Maharashtra, Code : 27

Sl No	Description of Goods	HSN/SAC	Quantity	Rate	per	Amount
1	**165.80R/14 ASSURANCE DP T/L TYRES**	40111090	**1 NOS**	2,287.50	NOS	**2,287.50**
	CGST					**320.25**
	SGST					**320.25**
	Total		**1 NOS**			**₹ 2,928.00**

E. & O.E

Amount Chargeable (in words)
INR Two Thousand Nine Hundred Twenty Eight Only

HSN/SAC	Taxable Value	Central Tax Rate	Central Tax Amount	State Tax Rate	State Tax Amount	Total Tax Amount
40111090	2,287.50	14%	320.25	14%	320.25	640.50
Total	**2,287.50**		**320.25**		**320.25**	**640.50**

Tax Amount (in words) : **INR Six Hundred Forty and Fifty paise Only**

Company's Bank Details
Bank Name: **THE KARAD URBAN CO-OPRATIVE BANK LTD - 20**
A/c No. : **1033102000020**
Branch & IFS Code : **MARKET YARD SANGLI & KUCB0488033**

for DILIP TYRES

DILIP TYRES DT SANGLI

Authorised Signatory

This is a Computer Generated Invoice

14/3/19

Invoice No. GT\5153

Dated 27-Feb-2019

DILIP TYRES
PUSHPARAJ CHOWK,SANGLI
Ph.No. (0233) 2328448, 2324577
GSTIN/UIN: 27AAEPH8682F1ZI
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Total	**2,287.50**		**320.25**		**320.25**	**640.50**

Tax Amount (in words) : **INR Six Hundred Forty and Fifty paise Only**

Company's Bank Details
Bank Name: **THE KARAD URBAN CO-OPRATIVE BANK LTD - 20**
A/c No. : **1033102000020**
Branch & IFS Code : **MARKET YARD SANGLI & KUCB0488033**

for DILIP TYRES

DILIP TYRES DT SANGLI

Authorised Signatory

This is a Computer Generated Invoice

14/3/19

ROUTE 78 Auto Mechanika

TAX INVOICE
PmI/47175.
Original Buyer's Copy

M/s. AUTOMECHANIKA
78, SHAMBHUNATH PANDIT STREET
KOLKATA - 700 020
PHONE : 2454-3734

GST. No. 19AAKFA7292C1ZB
Tax Invoice No. 338
Date 28/01/19

M/s. ALD AUTOMOTIVE Pvt LTD
611 Paratolla Road, KOLKATA

GST Registration No. 19 AAFCA0924K1ZO
Order No. Bookmy 28777/Appr 309078 Date : 28/01/19
Challan No. 11173 Date : 28/01/19

QUANTITY	PARTICULARS i20	HSN / SAC CODE	RATE	AMOUNT Rs.
	WB-08B/9070 kms: 56717			
One	Battery - 40B 20L —	8507	2968/75	2968=75
	101935-N T-WA 3800.01			
		CGST@ 14%		415=63
		SGST@ 14%		415=63
		CGST@ 9%		
		SGST@ 9%		
		Round Off	(–)	201
		TOTAL		3800=00
	GST No. : 19AAKFA7292C1ZB	E. & O. E.		

₹ Three thousand eight hundred Only

N. B. 1) Payment must be made by Cash or Crossed Cheque.
2) Goods sold are guaranteed against manufacturing defects only. The decision of the manufacturer in replacing and rejecting the goods in all respects will be final. The dealer in no way will be held responsible for the alleged defects.
3) All Transactions are Subject to Calcutta Jurisdiction.
4) No. Complaint Regarding this Bill will be Entertained if not made in writing within 6 days from the date of this Bill.
5) Interest @ 20% will be charged if payment is not made with 15 days from date of issue.

For M/s. AUTOMECHANIKA

{
"customer_name": "ALD AUTOMOTIVE PVT LTD",
"customer_tax_id": "19AAFCA0924K1ZO",
"customer_address": {
"street": "611 Paratolla Road",
"city": "KOLKATA",
"state": null,
"postal_code": null,
"country": null
},
"shipping_address": null,
"purchase_order": "Bookmy 28777/Appr 309078",
"invoice_id": "338",
"invoice_date": "28/01/19",
"due_date": null,
"vendor_name": "M/s. AUTOMECHANIKA",
"vendor_address": {
"street": "78, SHAMBHUNATH PANDIT STREET",
"city": "KOLKATA",
"state": null,
"postal_code": "700 020",
"country": null
},
"vendor_tax_id": "19AAKFA7292C1ZB",
"remittance_address": null,
"subtotal": {
"currency_code": "INR",
"amount": 2968.75 },
"total_discount": null,
"total_tax": {
"currency_code": "INR",
"amount": 831.26
},
"invoice_total": {
"currency_code": "INR",
"amount": 3800.0
},
"payment_term": null,
"items": [
{
"product_code": null,
"description": "Battery - 40B 20L -",
"quantity": null,
"tax": null,
"unit_price": {
"currency_code": "INR",
"amount": 2968.75
},
"total": {
"currency_code": "INR",
"amount": 2968.75
}
}
],
"customer_signature": null
}

ROUTE 78 Auto Mechanika

TAX INVOICE
PMI/47175
Original Buyer's Copy

M/s. AUTOMECHANIKA
78, SHAMBHUNATH PANDIT STREET
KOLKATA - 700 020
PHONE : 2454-3734

GST. No. 19AAKFA7292C1ZB
Tax Invoice No. 338
Date 28/01/19

M/s. ALD AUTOMOTIVE Pvt LTD
611 Paratolla Road KOLKATA

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QUANTITY	PARTICULARS i20	HSN / SAC CODE	RATE	AMOUNT Rs.
	WB-08B/9070			
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One	Battery - 40B 20L —	8507	2968/75	2968=75
	10193S-N			
	T-NA			
	3800.01			
		CGST@ 14%		415=63
		SGST@ 14%		415=63
		CGST@ 9%		
		SGST@ 9%		
		Round Off	(-)	=01
		TOTAL		3800=00

GST No. : 19AAKFA7292C1ZB

E. & O. E.

₹ Three thousand eight hundred Only

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3) All Transactions are Subject to Calcutta Jurisdiction.
4) No. Complaint Regarding this Bill will be Entertained if not made in writing within 6 days from the date of this Bill.
5) Interest @ 20% will be charged if payment is not made with 15 days from date of issue.

For M/s. AUTOMECHANIKA


```
{
"customer_name": "ALD AUTOMOTIVE PVT LTD",
"customer_tax_id": "19AAFCA0924K1ZO",
"customer_address": {
  "street": "611 Paratolla Road",
  "city": "KOLKATA",
  "state": null,
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"invoice_id": "338",
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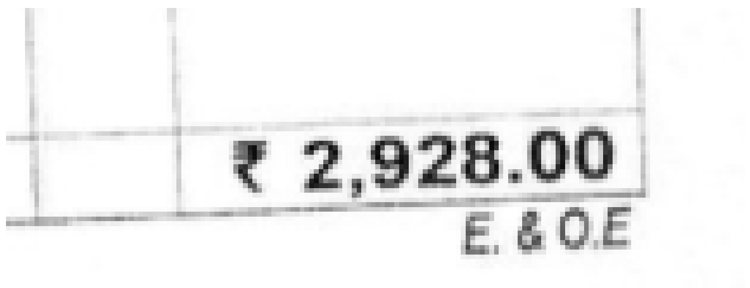
"vendor_address": {
  "street": "78, SHAMBHUNATH PANDIT STREET",
  "city": "KOLKATA",
  "state": null,
  "postal_code": "700 020",
  "country": null
},
"vendor_tax_id": "19AAKFA7292C1ZB",
"remittance_address": null,
"subtotal": {
  "currency_code": "INR",
  "amount": 2968.75   },
"total_discount": null,
"total_tax": {
  "currency_code": "INR",
  "amount": 831.26
},
"invoice_total": {
  "currency_code": "INR",
  "amount": 3800.0
},
"payment_term": null,
"items": [
  {
    "product_code": null,
    "description": "Battery - 40B 20L -",
    "quantity": null,
    "tax": null,
    "unit_price": {
      "currency_code": "INR",
      "amount": 2968.75
    },
    "total": {
      "currency_code": "INR",
      "amount": 2968.75
    }
  }
],
"customer_signature": null
}
```

Drobnym drukiem

LLM nie wyciąga tekstu z dokumentu on wyciąga INFORMACJĘ




```
<td></td>
<td></td>
<td>₹ 2,928.00</td>
</tr>
</table>
```

```
"invoice_total": {
    "currency_code": {
        "confidence": 0.933,
        "value": "INR"
    },
    "amount": {
        "confidence": 0.9978815356680525,
        "value": 2928.0
    }
},
```

Dobre praktyki

- Użyj promptu systemowego do zwiększenia kontekstu:

```
system_text_prompt = """You are an AI assistant that extracts data from documents. Extract the data from this invoice.
- If a value is missing, enter null.
- Do not make assumptions or modify any data.
- Format all data exactly as in the original document.
- Preserve date formats as shown.
- Extract numeric values as strings, exactly as they appear.
"""
```

- Zarządź typami danych – szczególnie datami:


```
"invoice_date": {
    "confidence": 0.992,
    "value": "27-Feb-2019"
},
```

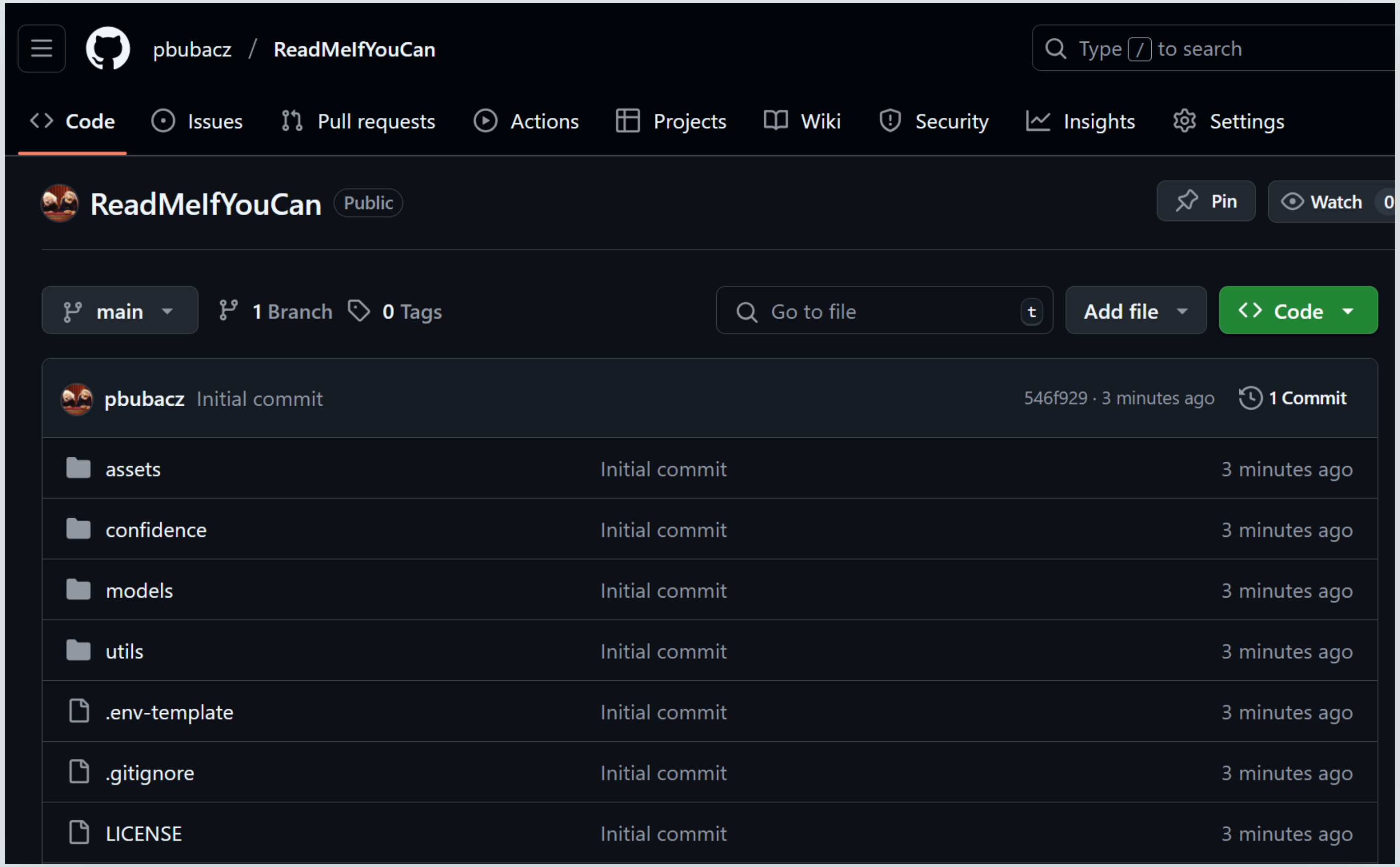
Dobre praktyki

- Testuj najlepsze rozwiązania dla Twoich danych
- Rozpocznij od obiektu wygenerowanego przez AI na podstawie listy pól wyjściowych, a następnie edytuj opisy i prompt

		Model	Accuracy (95th)
VISION	Faktury	GPT-4o	98.99%
		GPT-4o Mini	79.31%
	Dokumenty ubezpieczeniowe	GPT-4o	97.04%
		GPT-4o Mini	67.25%
VISION + MARKDOWN	Faktury	GPT-4o	96.60%
		GPT-4o Mini	91.84%
	Dokumenty ubezpieczeniowe	GPT-4o	100%
		GPT-4o Mini	82.99%

Próba	Zmiana	Accuracy [%]
0	Początkowa lista pól	75
1	Dodano podstawowe informacje do promptu	86
2	Dodano dodatkowe informacje dla pól z problemami	92

Github repo


pbubacz / ReadMeIfYouCan
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
ReadMeIfYouCan
Public
Pin
Watch 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
pbubacz Initial commit
546f929 · 3 minutes ago
1 Commit
assets Initial commit 3 minutes ago
confidence Initial commit 3 minutes ago
models Initial commit 3 minutes ago
utils Initial commit 3 minutes ago
.env-template Initial commit 3 minutes ago
.gitignore Initial commit 3 minutes ago
LICENSE Initial commit 3 minutes ago